

Abstract

Investigations into Instabilities of the Fermi surface This thesis reports (i) the development of a new auxiliary model method for studying translation-invariant models of strongly-interacting electrons, and (ii) an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with emphasis on the signatures at a critical Fermi surface. Together, these results shed light on how a system of electrons behaves as it undergoes a metal-insulator transition.

The auxiliary model method allows us to create a mapping between a lattice model and an auxiliary impurity model in terms of the Hamiltonian, eigenstates, Greens functions and entanglement measures, among other things. This mapping ensures that such lattice model quantities can be computed indirectly from similar computations within the auxiliary impurity model. In terms of phenomenology, this mapping connects various renormalisation group (RG) fixed points of the impurity model to those of the lattice model. As examples, local Fermi liquid and local moment phases of the impurity site get mapped to Fermi liquid and Mott insulating phases, respectively, on the lattice model. The mapping makes use of translation operators and a manybody version of Bloch’s theorem to recreate the lattice model via repeated “tiling” operations on the impurity model. In contrast to other auxiliary model methods such as dynamical mean-field theory (DMFT) and its cluster variants (C-DMFT), this approach does not involve a self-consistency loop, making it more transparent. It also provides significantly more k -space resolution which is necessary for studying phenomena such as the pseudogap observed in the copper-oxide high- T_c superconductors.

A short description of the protocol for applying our method is as follows. We first identify the appropriate quantum impurity model corresponding to the lattice model at hand. For example, the impurity model must have a localisation-delocalisation transition on the impurity site in order to capture a metal-insulator transition on the lattice model. As another example, topological features of the band structure can be encoded by working with a topologically non-trivial conduction bath within the impurity model. Having decided upon a quantum impurity model, we solve the model using an impurity solver. In the works described in this thesis, we use the unitary renormalisation group because of its (i) analytical transparency, (ii) ability to deal with weak and strong correlations, (iii) ability to work at zero temperature, and (iv) reduced computational cost. Using the URG, we obtain a hierarchy of Hamiltonians that describe the high-energy and low-energy behaviour of the auxiliary model in various parameter regimes. This requires numerically solving the URG flow equations for various Hamiltonian couplings.

The phases of the impurity model are then characterised by (a) effective Hamiltonians, (b) static equal-time correlations, (c) dynamical correlations such as Greens functions and self-energy, and

(d) entanglement measures. Calculating some of these objects requires carrying out an iterative diagonalisation procedure on the hierarchy of Hamiltonians. Our tiling method then “periodises” these objects into the lattice model, generating required k -dependence in the process. Since the conduction bath of the impurity model already inherits the lattice geometry of the parent model, the appropriate symmetries are baked into the quantities. As an example, the lattice k -space Greens function G_k obtains contributions from the impurity site Greens function as well as impurity-bath off-diagonal Greens functions. While the former captures the local component of G_k , the latter captures the non-local components. With similar mappings existing for the dynamical correlations and entanglement measures, our method allows us to characterise various phases of the lattice model purely through impurity model computations.

Within our work on the holographic principle, we demonstrate the emergence of a holographic dimension in a system of 2D non-interacting Dirac fermions placed on a torus, by studying the scaling of multipartite entanglement measures under a sequence of renormalisation group (RG) transformations applied in momentum space. Geometric measures defined in this emergent space can be related to the RG beta function of the spectral gap, hence establishing a holographic connection between the spatial geometry of the emergent spatial dimension and the entanglement properties of the boundary quantum theory. We prove, analytically, that changing the boundedness of the holographic space involves a topological transition accompanied by a critical Fermi surface in the boundary theory. We go on to show that this results in the formation of a quantum wormhole geometry that connects the UV and the IR of the emergent dimension. The additional conformal symmetry at the transition also supports a relation between the emergent metric and the stress-energy tensor. In the presence of an Aharonov-Bohm flux, the entanglement gains a geometry-independent piece which is shown to be topological, sensitive to changes in boundary conditions, and related to the Luttinger volume of the system. Upon the insertion of a strong transverse magnetic field, we show that the Luttinger volume is linked to the Chern number of the occupied single-particle Landau levels.

Some Important Questions In this thesis, we have used the auxiliary model tiling method to study three fundamental problems in quantum condensed matter physics: (i) Mott metal-insulator transition on the Bethe lattice in infinite dimensions, (ii) Mott metal-insulator transition on the 2D square lattice, and (iii) quantum criticality in heavy fermion materials in 2D. Towards this, we have applied the method to two paradigmatic models: (a) an extended Hubbard model, and (ii) a bilayer extended Hubbard model. In preparation for applying the tiling approach, we have first analysed the corresponding quantum impurity models.

1. The Mott transition on the Bethe lattice is studied through an extended Anderson impurity model (eSIAM) that hosts local electronic correlation in the conduction bath proximate to the impurity site. Due to the lack of non-local correlations in this limit, it is sufficient to treat the conduction electron in the continuum limit.
2. The Mott transition in 2D is highly sensitive to details of the lattice. This requires us to study a lattice-embedded impurity model, where the impurity site is embedded in a 2D lattice of conduction electrons, and its hybridisation with the bath has the symmetries of the underlying lattice.

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3. The heavy fermions problem necessitates a two layer impurity model, each layer acting as a lattice-embedded impurity model and coupled via inter-layer hybridisation.

The tiling approach then translates results from the auxiliary models into appropriate lattice models. In this thesis, we have attempted to answer a number of important open questions in the field:

- *Is there a minimal but effective quantum impurity model Hamiltonian that describes the Mott MIT of the $1/2$ -filled Hubbard model on the Bethe lattice in infinite dimensions?* Finding such an impurity model would also reduce the considerable computational effort that is presently required in self-consistent approaches. In the same context, is it possible to obtain a low-energy theory for the local gapless excitations precisely at the transition, where the metal is on the brink of destruction?
- For strongly-correlated electronic models, *is it possible to map the renormalisation group fixed points of such lattice models to those of quantum impurity models?* This would provide a firmer theoretical backing to various quantum impurity models, as well as reduce computational costs.
- The phenomenon of Mott insulation involves the localization of itinerant electrons due to strong local repulsion. Upon doping, a pseudogap (PG) phase emerges - marked by selective gapping of the Fermi surface without conventional symmetry breaking in spin or charge channels. *A key challenge is understanding how quasiparticle breakdown in the Fermi liquid gives rise to this enigmatic state, and how it connects to both the Mott insulating and superconducting phases.*

Summary Of The Chapters. For the readers' convenience, we now briefly describe the content of the various chapters.

The *Introduction* ...

After the introduction, the *Methods and Preliminaries* chapter describes the various techniques and some preliminary ideas employed throughout the thesis. The chapter begins by deriving the unitary renormalisation group formalism and describes its various properties and a practical protocol for implementing it. That is followed by demonstrations of the unitary RG on three simple models - a star graph model of spins in a magnetic field, the two-channel Kondo model, and the problem of a particle in a periodic potential. Through these problems, various aspects of the method are clarified. In order to provide more insights, the chapter also details how the unitary RG method is related to other RG-based methods that rely on canonical transformations, such as Poor Man's scaling, Schrieffer-Wolff transformation and CUT (continuous unitary transformation) RG. Apart from the URG, the chapter also describes an iterative diagonalisation method that has been used within this thesis to compute various correlation functions and entanglement measures from the hierarchy of Hamiltonians obtained from the URG. Since various entanglement measures have been used throughout the later chapters, we have also added a section on various entanglement measures here - what they mean, what are their inter-relations, why they are important and how one can compute some of them. The chapter concludes by describing some of the other strategies that have been employed to reduce computational costs in various projects.

In Chapter 3, *Kondo frustration via charge fluctuations: A route to Mott localisation*, we study our first impurity model, the extended Anderson model with local correlation U_b in the conduction bath. We demonstrate that this model hosts a local version of the Mott metal-insulator transition on the Bethe lattice in infinite dimensions, as seen from DMFT. At a critical value of the ratio of U_b and the impurity-bath hybridisation, the effective impurity model shows a transition from a Kondo screened phase into an unscreened local moment phase. For the case of attractive local bath correlations, the model sheds new light on several aspects of the DMFT phase diagram. For example, the $T = 0$ metal-to-insulator quantum phase transition (QPT) is preceded by an excited state QPT (ESQPT) where the local moment eigenstates are emergent in the low-lying spectrum. Long-ranged fluctuations are observed near both the QPT and ESQPT, suggesting that they are the origin of the quantum critical scaling observed recently at high temperatures in DMFT simulations. The $T = 0$ gapless excitations at the quantum critical point display particle-hole interconversion processes, and exhibit power-law behaviour in self-energies and two-particle correlations. These are signatures of non-Fermi liquid behaviour that emerge from the partial breakdown of the Kondo screening. A many-body perturbation theoretic treatment of the Hubbard sidebands reveals that they are comprised of holon-doublon excitations created by the hybridisation of the impurity site with the conduction bath. These excitations consist of decoupled local Fermi liquids for the holons and doublons at the lowest order, which, at higher orders, become coupled via correlated holon-doublon scattering between impurity and bath.

In Chapter 4, *Mapping From Auxiliary Model To Lattice: The Tiling Approach*, we lay out the details of the auxiliary model method developed to map impurity models to lattice models. We first motivate the general idea behind auxiliary model methods. This is followed by a construction of the manybody translation operators. We use this, along with a manybody Bloch's theorem, to construct lattice models from quantum impurity models and relate their eigenstates. This then allows us to map correlations functions, self-energies and entanglement measures across models. In particular, we demonstrate how the presence of the conduction bath introduces k -space resolution in various quantities.

In Chapter 5, *Mott Criticality as the Confinement Transition of a Pseudogap-Mott Metal: Insights from a lattice-embedded impurity model*, we apply the auxiliary model tiling method to a lattice-embedded extended SIAM. Applying a many-body tiling scheme to the fixed-point impurity model uncovers a lattice model with electron interactions and Kondo physics. At half-filling, the interplay between Kondo screening and bath charge fluctuations in the impurity model leads to Fermi liquid breakdown. This reveals a pseudogap phase characterized by a non-Fermi liquid (the Mott metal) residing on nodal arcs, gapped antinodal regions of the Fermi surface (Luttinger surfaces), and an anomalous scaling of the electronic scattering rate with frequency. The eventual confinement of holon-doublon excitations of this exotic metal obtains a continuous transition into the Mott insulator. Our results identify the pseudogap as a distinct long-range entangled quantum phase, and offer a new route to Mott criticality beyond the paradigm of local quantum criticality. The phase appears to encode multipartite entanglement, as seen from a large value of the quantum Fisher information. The critical point at the pseudogap-Mott insulator boundary is found to be described by the exactly-solvable Hatsugai-Kohmoto model. The emergence of Luttinger surfaces alters the anomaly structure associated with the generalized symmetry of the FS [44–46]. The corresponding topological charge is defined via the Luttinger–Ward functional. This charge is modified by Green's

function zeros, signalling a shift in the underlying anomaly. Within our framework, this reorganization grants topological protection to the RG flows terminating in the PG regime, ensuring the stability of its NFL excitations.

In Chapter 7, *Holographic entanglement renormalisation for fermionic quantum matter*, we provide an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

We conclude in the final chapter by summarising the main results obtained within the thesis, in terms of the methods developed as well as insights obtained into the behaviour of various models. We describe the impact of these results on the field, the broad conclusions emerging from them and some open questions resulting from the work.

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Chapter 1

Introduction: Mott Transitions and Kondo Breakdown

The rich physics of Mott metal-insulator transitions in strongly correlated systems has been an active subject of study for quite some time [1–4]. Their appearance in various materials such as the nickelates, cuprates, iridates and manganites, among others, is marked by complex phase diagrams that display dramatic changes in properties across changes in filling and/or bandwidth, even at low temperatures. The physics of Mott transitions is driven by the competition between two energy scales in the system: (i) the scale of itineracy, quantified by the bandwidth, and (ii) the scale of electron-electron repulsion. A Mott insulator arises when the electronic correlations are strong enough that low-energy electronic excitations become gapped, and the metallic valence band gets split into lower and upper bands; for an initially half-filled valence band, the lower band is filled at $T = 0$ while the upper band is vacant. Because of strong electronic repulsion, the electrons in the lower band form local moments which can show a variety of behaviour. This includes various symmetry-broken ground states such as an antiferromagnet [5], or symmetry-preserved states such as quantum spin liquids [6]. The physics of Mottness is also responsible for several very interesting phenomena, such as high-temperature superconductivity [5], non-Fermi liquids [7] and the various phases in magic-angle twisted bilayer graphene [8, 9].

1.1 Metal-insulator transition arising from strong correlation

Initial attempts at explaining metal-insulator transitions was based on a non-interacting picture of electrons (through the work of Bethe, Sommerfeld, Bloch and Wilson among others); in such a picture, electronic bands can arise only from the scattering of electrons off the periodic lattice potential. One then distinguishes between metals and insulators at zero temperature by the position of the Fermi level E_F with respect to the bands: if E_F lies within a band gap, an integer number of bands will be completely filled and the system will be an insulator, whereas if E_F lies within one of the bands, the system will be a metal because there are vacant gapless excitations within the band.

It was first pointed out by Boer and Verwey [10] that certain insulators and semiconductors (such as nickel oxides) have a particle filling that, purely from band theory, would lead to partially filled

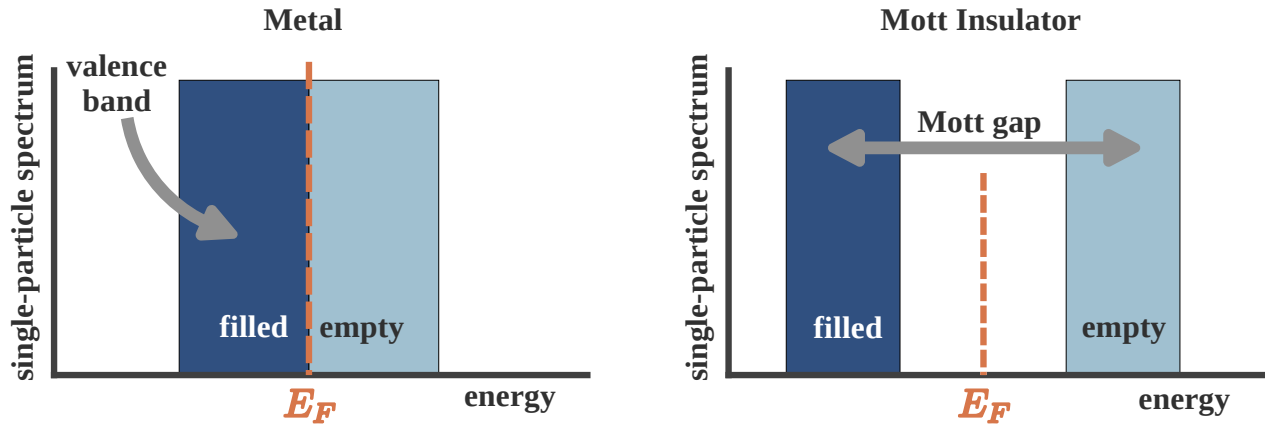


Figure 1.1

bands. This poses a challenge because a partially filled band necessarily leads to a metallic phase (see Fig. 1.2 (left)). The insulating behaviour of these materials therefore signals a fundamental limitation of single-particle band theory and points to a qualitatively different mechanism for electron localization—one driven by strong electron–electron interactions rather than by scattering from the lattice potential alone. This realization led to the concept of a Mott insulator, in which charge localization emerges as a genuine many-body effect despite the absence of a band gap in the non-interacting electronic structure. A very simple mechanism for this was put forth by Mott [11–13]: in the presence of a large local Coulomb repulsion cost U on a lattice site, the quantum states $|\Psi_1\rangle$ in which the site is singly occupied get separated in energy from the states $|\Psi_2\rangle$ where the site is doubly occupied by a gap of size U ; the former class of states $|\Psi_1\rangle$ form the lower band (LB) while the latter $|\Psi_2\rangle$ form the upper band (UB). If we now have a half-filled system (one electron per site), this electron will completely fill LB and leave UB vacant, leading to an insulating state (see Fig. 1.2 (right)).

Given the above discussion, it is clear that a Mott insulator must involve a gap to charge excitations (excitations that add or remove an electron). It does not however determine the fate of the spin, orbital or other degrees of freedom. One scenario is when the spin or orbital degree of freedom undergoes spontaneous symmetry breaking into an ordered state; in the case of spin, this usually results in a superexchange interaction being generated and leads to a long-range ordered antiferromagnetic Neel state coexisting with the Mott state. Examples of such systems include vanadium sesquioxide V_2O_3 and its derivatives and the high- T_c cuprates (such as La_2CuO_4) [4]. The other scenario is one where the spin degree of freedom retains its symmetry, and we end up with a quantum mechanically non-trivial entangled state (in order to quench the entropy). Examples of such states include quantum spin liquids [14]. Mott insulators without long-range order are of particular interest because of the possibility of topological order, fractional excitations, and novel metallic parent states.

A prototypical model for gaining theoretical understanding of Mott metal-insulator transitions

on a lattice is the single-band Hubbard model [1, 15–17]

$$H = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) - \frac{U}{2} \sum_i (n_{i,\uparrow} - n_{i,\downarrow})^2 - \mu N. \quad (1.1)$$

The first term incorporates nearest-neighbour hopping on the lattice, while the second term introduces inter-electron repulsion when two electrons site on the same lattice site; in this way, the model incorporates the two competing tendencies of delocalisation and localisation that are responsible for the Mott MIT. The third term fixes the chemical potential and determines the filling of the lattice. Many of the materials that are studied for Mott MIT involve d -electrons that are five-fold degenerate, and these orbitals can hybridise with ligand atoms. The Hubbard model avoids these complexities by focusing on only a single band that lies closest to the Fermi energy, and it is the states of this band that are described by the Hubbard model.

This leaves us with two channels for observing Mott MIT [4]: (i) tuning the bandwidth, and (ii) tuning the filling. The former can be done, in experiments, by applying pressure. Filling is tuned by doping specific elements of a compound with elements of a different valence. The former is referred to as bandwidth-control metal-insulator transition, while the latter is called filling-control metal-insulator transition.

1.2 Features of Mott metal-insulator transition

Two complementary pictures of the Mott transition: Brinkman-Rice vs Mott-Hubbard. Initial insights into the Mott MIT were obtained by Hubbard [1], thinking along lines similar to Mott. Within this picture, one starts from the insulating large U limit of the Hubbard model (eq. 1.1) ($U \gg t$). We stay at half-filling for simplicity ($\mu = 0$). The large on-site repulsion leads to the formation of two bands at each site, the lower Hubbard band (LHB) associated with singly-occupied states and the upper Hubbard band (UHB) associated with doubles and holes (states that incur a cost U). In the large U limit, these bands are well-separated, and the dynamics within the bands is a result of the motion of tightly bound holons and doubles (more on this later) that are generated because of the small but non-zero hopping probability t . As U/t is lowered, the bands come closer and a metal-insulator transition occurs at the critical value of U at which the gap vanishes. For lower U/t , the bands overlap and we get a metal. This picture fails to provide an accurate description of the metal resulting from the melting of such a Mott insulator [18].

An alternative approach to the Mott MIT is due to Brinkman and Rice [19], where the authors applied Gutzwiller's variational method [20] to the half-filled Hubbard model. Gutzwiller's approach involves modifying, through variational factors, the hopping matrix elements that start from doubly occupied states, in order to reflect the electronic correlation at each site. By requiring that the number of doubly-occupied states vanish in the Mott insulating ground state, they obtained a critical value of the interaction strength. In the Brinkman-Rice scenario, therefore, the transition from the metal to the insulating phase occurs when the hopping matrix elements from the doubly-occupied to the singly-occupied states strictly vanishes. Note that in contrast to the Mott-Hubbard picture, this approach starts from a correlated Fermi liquid (renormalised due to the Gutzwiller factors). While it gives a better low-energy description of the correlated metal compared to the Mott-Hubbard picture,

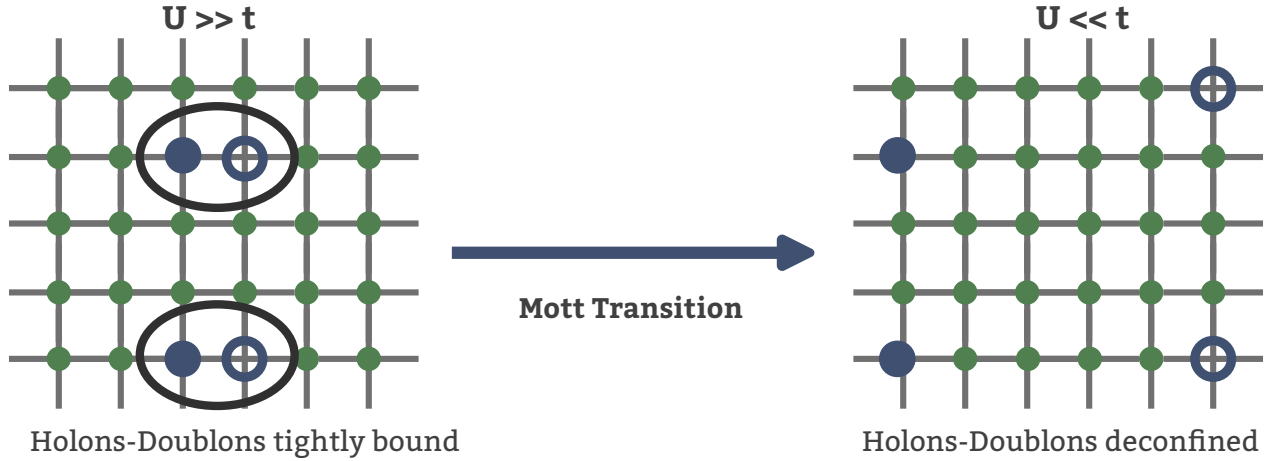


Figure 1.2

it fails to account for the high-energy physics, and treats the Mott insulator as a direct product state of local moments at each state (on account of the strictly zero hopping). This of course means that this method cannot explain antiferromagnetic superexchange processes in the Mott insulator [18].

Holon-doublon unbinding picture of the Mott MIT. In 1949, Mott proposed a zero temperature explanation for a correlation-driven insulator and a mechanism for its transition into a metal in terms of the motion of holes and doubles in the background of a half-filled lattice. For $U \ll t$ (strong correlation), the ground state predominantly involves lattice sites with a single electron. Hopping processes between two nearest-neighbour sites transfer an electron between them and create a pair of hole and double, but the pair remains tightly bound to each other. This is because moving the hole independently involves scattering processes that cross the Mott gap, and these processes are not allowed at $T = 0$. But without the independent motion of the hole (or double), there is no current, and the system is an insulator.

As U/t is lowered, the gap closes and processes that transport the hole without transporting the double become gapless. This essentially means that the pair becomes deconfined, and the hole can be moved far away from the double. Current can then be passed by having the hole move through the lattice, and we get a metallic state. This leads to a picture of the Mott transition into a metal in terms of an unbinding of the hole-double bound states (excitons) that were present in the insulator.

There have been more quantitative studies of this mechanism. I list some of the relatively recent ones here. In [21], Prelovšek et al. studied the motion of holon-doublon pairs in a spin polarised background by rewriting the repulsive Hubbard model in terms of electrons as an attractive Hubbard model of holes and doubles. The model was treated using a BCS-like treatment, and the Mott transition was obtained when the binding energy for the hole-double condensate vanishes. To mark the deconfinement, they show that the approach to the transition from the insulating side is concomitant with an increase in the size of the pairs. Following a slave-boson treatment of holon-doublon unbinding, Zhou et al. [22] show that the Mott transition from a paramagnetic metal to a Mott insulator on the half-filled honeycomb and 2D square lattices occur through an intervening Slater insulator with coherent quasiparticles. In particular, their approach sheds light on the nature of the inco-

herent excitations in the Hubbard sidebands of the local spectral function. Finally, an older work by Watanabe et al. [23] studies the half-filled Hubbard model on an anisotropic triangular lattice using a Monte carlo method with a variational binding parameter between holons and doublons. Their approach reveals a first order Mott transition, along with robust d -wave superconductivity and short-range magnetic order.

Spectral features of the Mott transition. One of the important diagnostics of the Mott metal-insulator transition is the spectrum for one-particle electron-like excitations, as expressed through the one-particle retarded Greens function $G(\mathbf{k}, \sigma; t) = -i\theta(t)\langle\{c_{\mathbf{k},\sigma}(t), c_{\mathbf{k},\sigma}^\dagger\}\rangle$. In general, the presence of poles in the Green function at the Fermi energy and in its neighbourhood imply the existence of gapless excitations and hence a metallic phase. It turns out that this statement is equivalent to Luttinger's theorem,

$$\frac{N}{V} = 2 \int_{G(0,\mathbf{k})>0} \frac{d\mathbf{k}}{(2\pi)^3}, \quad (2.1)$$

which states that the particle density in a metal is equal to the number of states in the Fermi volume. As clarified in the equation, the Fermi volume is defined as the region of k -space where the Greens function is positive at $\omega = 0$.

In a Mott insulator, strong correlations lead to the destruction of electronic excitations around $\omega = 0$. This results in the Greens function poles transmuting into zeros. This is also tracked by analogous changes in the electronic self-energy $\Sigma(\mathbf{k}; \omega)$; Greens function zeros leads to a Mott pole at $\omega = 0$ in $\Sigma(\mathbf{k}; \omega)$. The points in k -space where $\Sigma(\mathbf{k}; 0)$ vanishes is called a Luttinger surface. The Mott metal-insulator transition is therefore about the topological change of a Fermi surface (surface of Greens function poles) into a Luttinger surface (surface of self-energy poles).

As discussed earlier, the appearance of Greens function zeros at $\omega = 0$ leads to the formation of a charge gap that freezes one-particle low-energy excitations. The spectrum gets redistributed into two bands on either side of the gap, the upper and lower Hubbard bands. In the ground state, the lower band is obviously filled while the upper band is empty. Bands of a Mott insulator formed from electronic correlation differ from those formed due to one-particle scattering in a particular way: dynamical spectral weight transfer. Unlike a band insulator, the bands of a Mott insulator are not rigid; tuning the lattice filling results in a continuous transfer of states from one band to another.

The non-rigidity of the bands can be seen most easily from the atomic limit ($t = 0$) of the Hubbard model (eq. 1.1). The local retarded Greens function at an arbitrary site i can be obtained using the equation of motion approach. The final result is [24]

$$G_R(i; \omega) = \frac{1+x}{\omega + \mu + U/2} + \frac{1-x}{\omega + \mu - U/2}, \quad (2.2)$$

where $x = 1 - \sum_\sigma \langle n_{i\sigma} \rangle$ is the average number of holes at any site. The spectral weight for the two poles makes it clear that the lower band has $1+x$ number of states while the upper band has $1-x$ states. Changing the number of states in the lower band (by tuning μ and hence x) also changes those in the upper band. For example, by adding 0.1 holes per site to the lower band, the spectral weight in the upper band decreases by the same amount. This is the non-rigidity of the bands mentioned previously.

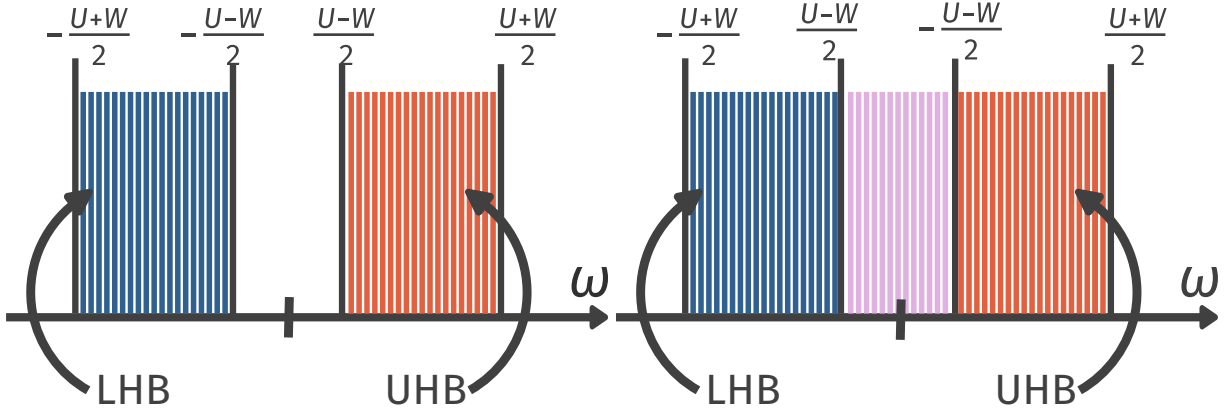


Figure 1.3

1.3 Preliminary Insights Into Mottness: The Baskaran-Hatsugai-Kohmoto model

The Baskaran-Hatsugai-Kohmoto model [25, 26] introduced independently in 1991 and 1992 is an exactly solvable model of electronic correlation that displays a transition from a metallic phase to a Mott insulator. While its solvability offers an instructive method for studying Mott transition, the infinitely long-ranged nature of its interaction makes it tricky to apply these conclusions to realistic materials.

At half-filling, the model is defined by the following Hamiltonian:

$$H_{\text{BHK}} = \sum_{\mathbf{k}, \sigma} \epsilon(\mathbf{k}) n_{\mathbf{k}, \sigma} - \frac{U}{2} \sum_{\mathbf{k}} (n_{\mathbf{k}, \uparrow} - n_{\mathbf{k}, \downarrow})^2. \quad (3.1)$$

Clearly the model is diagonal in momentum-space; the Hamiltonian splits up into commuting 4×4 blocks for every point $\mathbf{k}$:

$$H_{\text{BHK}} = \sum_{\mathbf{k}} H(\mathbf{k}), \quad H(\mathbf{k}) = \epsilon(\mathbf{k}) \sum_{\sigma} n_{\mathbf{k}, \sigma} - \frac{U}{2} (n_{\mathbf{k}, \uparrow} - n_{\mathbf{k}, \downarrow})^2. \quad (3.2)$$

The vacant eigenstate $|0\rangle$ of $H(\mathbf{k})$ has zero energy ($E_0 = 0$), the two singly-occupied states $|\uparrow\rangle$ and $|\downarrow\rangle$ have energy $E_1 = \epsilon(\mathbf{k}) - U/2$ and the doubly-occupied state $|\uparrow, \downarrow\rangle$ has energy $E_2 = 2\epsilon(\mathbf{k})$. This leads to two possible single-particle excitations – one for adding an electron on the vacant state: ($\Delta_1 = E_1 - E_0 = \epsilon(\mathbf{k}) - U/2$), the other for adding an additional electron on a singly-occupied state: ($\Delta_2 = E_2 - E_1 = \epsilon(\mathbf{k}) + U/2$).

Upon combining states from other k -modes, these isolated excitations broaden into bands around $\pm U/2$ because of the continuum of values available to $\epsilon(\mathbf{k})$. For large enough U (compared to the non-interacting bandwidth W), these bands remain well-separated in energy (Fig. 1.3 left), forming lower and upper Hubbard bands. The half-filled system results in the lower band being completely filled and the upper band empty, leading to a Mott insulator. The states within this band are all singly-occupied (per k -state). Upon lowering the correlation strength U , the bands

touch each other at the critical value of $U_c = W$ (obtained by equating the top of the lower band to the bottom of the upper band). For lower values of U , the system is a metal due to the presence of gapless excitations within the combined band (Fig. 1.3 right). This therefore provides a simple demonstration of a correlation-driven Mott metal-insulator transition.

1.4 Strong local correlations and auxiliary models

1.4.1 Introduction to the auxiliary model approach

One of the main results obtained within this thesis is the development of a formalism to study the more realistic case of finite-dimensional interacting models by using appropriately chosen impurity models as auxiliary models. The key to an *auxiliary model method* is to construct an appropriate auxiliary model that while being tractable captures some essential property of the full lattice model [27]. This involves separating the complete degrees of freedom into a system part (H_S) which one treats exactly and the rest of the system (H_R) that one might simplify in order to be able to solve the problem. The non-trivial nature of the problem arises of course from the hybridisation H_{SR} between the system and the bath:

$$H = \begin{bmatrix} H_R & H_{RS} \\ H_{RS}^* & H_S \end{bmatrix}. \quad (4.1)$$

This separation can always be done exactly, if one does not care about tractability. For example, one can take the Hubbard model that describes electrons hopping on a lattice and interacting when two electrons reside on the same lattice site:

$$H_{\text{hub}} = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (4.2)$$

and separate this into parts, the system being the local orbitals of a specific site l and the rest of the system being all the other sites:

$$H_S = U n_{l\uparrow} n_{l\downarrow}, \quad H_R = -t \sum'_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad H_{SR} + H_{RS} = -t \sum_{i,\sigma} (c_{l,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}), \quad (4.3)$$

where the primed summation is carried out over all nearest-neighbour pairs that do not involve l and the summation in $H_{SR} + H_{RS}$ is over all sites adjacent to l .

The smaller system H_S (often called the *cluster*) is typically chosen such that its eigenstates are known exactly. Progress is then made by choosing a simpler form for H_R and its coupling H_{RS} with the smaller system. This combination of the cluster and the simpler bath is then called the *auxiliary system*. A tractable auxiliary system for the Hubbard model is the single-impurity Anderson model (SIAM); the correlated impurity site represents the set of local orbitals of any given site, while the conduction bath represents the rest of the lattice sites in an approximate fashion. Such a construction is shown in fig. 1.4.

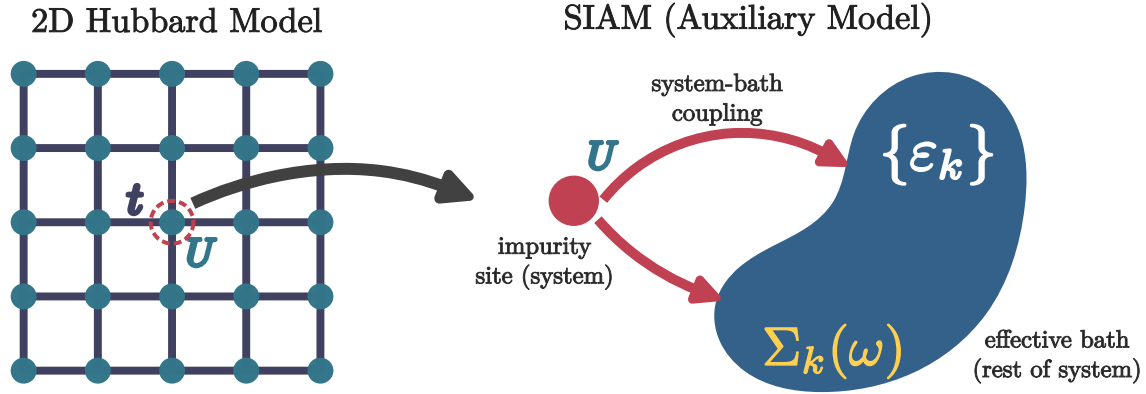


Figure 1.4: Left: Full Hubbard model lattice with onsite repulsion U^H on all sites and hopping between nearest neighbour sites with strength t^H . Right: Extraction of the auxiliary (cluster+bath) system from the full lattice. The central site on left becomes the impurity site (red) on the right (with an onsite repulsion ϵ_d), while the rest of the $N - 1$ sites on the left form a conduction bath (green circles) (with dispersion ϵ_k and correlation modelled by the self-energy $\Sigma_k(\omega)$) that hybridizes with the impurity through the coupling V .

It should be noted that any reasonable choice of the cluster and bath would break the translational symmetry of the full model: To allow computing quantities, one would need to make the bath (which is a much larger system) simpler than the cluster (which is a single site). This distinction breaks the translational symmetry of the Hubbard model. As a result, calculations that make use of the translation symmetry of the full model (such as k -space objects) will require a periodisation operation of an auxiliary model quantity.

1.4.2 Correspondence between impurity and lattice models in $d = \infty$

Dynamical Mean-Field Theory (DMFT) provides a nonperturbative framework for treating strong local electronic correlations in lattice models by mapping them onto a self-consistent quantum impurity problem [18, 28]. The central idea of DMFT is that in the limit of infinite spatial dimension, or equivalently infinite coordination number, the electronic self-energy becomes purely local while retaining its full dynamical (frequency-dependent) structure. As a result, the many-body lattice problem reduces to an effective single-site problem embedded in a self-consistently determined bath, capturing local quantum fluctuations exactly while neglecting nonlocal correlations. DMFT has become a cornerstone of modern correlated-electron theory, successfully describing Mott metal-insulator transitions, quasiparticle formation, and spectral weight transfer in Hubbard-like models [18, 29]

In order to motivate the impurity model-lattice model mapping inherited by DMFT, we first recall the textbook example of classical mean-field theory. Mean-field theory provides one of the earliest and most instructive examples of controlled approximations in many-body physics. In its

classical incarnation, exemplified by the Ising model,

$$H_{\text{Ising}} = - \sum_{i \neq j} J_{ij} S_i S_j - h \sum_i S_i = - \sum_i S_i (h + \sum_{j \neq i} J_{ij} S_j), \quad (4.4)$$

mean-field theory proceeds by replacing the interaction between a given spin and its neighbors by an *effective static field* proportional to the average magnetization. Writing $S_j = \langle S_j \rangle + \delta S_j$ and neglecting quadratic fluctuations and constant parts, one obtains an *auxiliary problem* in the form of a mean-field Hamiltonian

$$H_{\text{Ising}}^{(\text{MF})} = - \sum_i (h + h_{\text{MF}}(i)) S_i, \quad (4.5)$$

where the mean-field h_{MF} comes out to be

$$h_{\text{MF}}(i) = \sum_{j \neq i} 2 \langle S_j \rangle J_{ij}. \quad (4.6)$$

The above approach assumes two things: (i) the presence of a non-zero *order parameter* $m_j = \langle S_j \rangle$ (in this case, a non-zero magnetisation), and (ii) the smallness of the fluctuations $S_j - \langle S_j \rangle$. The mean-field Hamiltonian can be used to compute the partition function (at inverse temperature β) and hence the magnetisation at a site i of the lattice:

$$m_i = \tanh(\beta(h_{\text{MF}}(i) + h)) = \tanh(\beta \sum_{j \neq i} 2m_j J_{ij} + \beta h). \quad (4.7)$$

The problem hasn't yet been solved of course; one needs to determine the appropriate value of m_j . We therefore need an additional equation; this is provided by the so-called *self-consistency condition*. The translation invariance of the lattice ensures that the magnetisation must be uniform across sites i and j . This allows us to replace m_j with m_i in the above equation:

$$m_i = \tanh(\beta m_i \sum_{j \neq i} 2J_{ij} + \beta h). \quad (4.8)$$

This equation can be solved numerically to obtain the uniform magnetisation $m_i = m$. The mean-field approximation becomes exact in the limit of large coordination, where fluctuations are suppressed and spatial correlations beyond a single site vanish.

An equivalent way of framing the self-consistency equation is to equate the mean-fields on different lattice sites. Combining eqs. 4.7 and 4.6, we have

$$h_{\text{MF}}(i) = \sum_{j \neq i} 2m_j J_{ij} = \sum_{j \neq i} 2 \tanh(\beta(h_{\text{MF}}(j) + h)) J_{ij}. \quad (4.9)$$

The equation in terms of $h_{\text{MF}}(i)$ and $h_{\text{MF}}(j)$ can again be solved numerically by claiming that the mean-field has to be uniform across all the sites owing to translation invariance. This form of the self-consistency requirement (in terms of the field) will be useful to us in the upcoming discussion.

The essential feature of classical mean-field theory is therefore the replacement of a many-body problem by an effective *single-site* problem embedded in a self-consistently determined environment. Quantum lattice models admit a closely analogous construction, but with an important conceptual difference: while the classical mean field is static, quantum fluctuations necessarily introduce a tower of excited states that allow for temporal dynamics. This observation underlies the formulation of dynamical mean-field theory (DMFT), which may be viewed as the natural quantum generalization of classical mean-field theory [18, 28].

Similar to the previous case of the Ising model, we will now derive the self-consistency equation for a quantum model. Consider an interacting fermionic lattice Hamiltonian, such as the Hubbard model:

$$H = - \sum_{ij\sigma} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (4.10)$$

defined on a lattice with coordination number z .

The point of this exercise is to calculate the interacting self-energy $\Sigma_k(\omega)$ for the fermions. This therefore acts as an “order parameter” and plays the same role as the local magnetisation in the previous exercise. The self-energy defines the interacting k -space Green’s function for a translation-invariant problem:

$$G_{\mathbf{k}}(\omega) = \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}, \quad (4.11)$$

where $\varepsilon_{\mathbf{k}}$ denotes the noninteracting dispersion, and a real-space local Green’s function, obtained by summing over momentum:

$$G_{\text{loc}}(\omega) = \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}. \quad (4.12)$$

Similar to the earlier exercise, we need to apply a mean-field decoupling to drop certain fluctuations and obtain a similar effective problem. In the context of interacting electrons, such a decoupling is obtained in the limit of infinite dimensionality or coordination number, where the self-energy becomes k -independent [28]:

$$\Sigma_{\mathbf{k}}(\omega) = \Sigma_{\text{loc}}(\omega), \text{ or (equivalently) } \Sigma_{ij}(\omega) = \delta_{ij} \Sigma_{\text{loc}}(\omega). \quad (4.13)$$

The mean-field decoupling here therefore corresponds to ignoring spatial fluctuations of the one-particle self-energy, and leads to a local Greens function that is directly connected to the local self-energy:

$$G_{\text{loc}}(\omega) = \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\text{loc}}(\omega)}. \quad (4.14)$$

It turns out that a local Greens function connected to a local self-energy also defines a quantum impurity problem describing the dynamics of a single correlated site with the other environment sites integrated out:

$$S_{\text{imp}} = - \int_0^\beta d\tau d\tau' c^\dagger(\tau) G_{\text{MF}}^{-1}(\tau - \tau') c(\tau') + U \int_0^\beta d\tau n_\uparrow(\tau) n_\downarrow(\tau). \quad (4.15)$$

This acts as an auxiliary problem for the more complete problem locally interacting electrons on a lattice. The non-interacting Greens function G_{MF} is as of now unknown, and defines the environment of the impurity site and comprises the dynamics of the sites that have been integrated out. Such an impurity problem is solvable, and the impurity Green's function is given by

$$G_{\text{imp}}(\omega) = \frac{1}{G_{\text{MF}}^{-1}(\omega) - \Sigma_{\text{imp}}(\omega)}, \quad (4.16)$$

The first term in eq. 4.15 is therefore what makes this problem dynamical – it represents the probability of electrons hopping into and out of the impurity site, and leads to fluctuations of its occupancy. We can make the identification of the Greens function G_{MF} as the *mean-field* encountered in the previous example. More specifically, G_{MF} acts as a “field” because it is obtained by integrating out the other sites on the lattice, and it acts on the impurity site by affecting its dynamics.

The complete set of equations for the solving such a dynamical mean-field problem is finally obtained by equating the impurity local Greens function G_{imp} to the lattice local Greens function G_{loc} . As discussed earlier, such a solution exists in an exact fashion only in the limit of infinite dimensions, as long as the correct G_{MF} can be found. In order to find the appropriate G_{MF} that allows equating the impurity model Greens function to a lattice mode Greens function, we require an additional equation: the self-consistency condition. Similar to the case of the Ising model, we require that G_{MF} be equal to G_{loc} , owing to translation invariance. The full set of coupled DMFT equations is

$$\begin{aligned} G_{\text{MF}} &= G_{\text{loc}}, \\ \Sigma_{\text{imp}}(\omega) &= G_{\text{MF}}^{-1}(\omega) - G_{\text{imp}}^{-1}(\omega), \\ \Sigma_{\text{loc}}(\omega) &= \Sigma_{\text{imp}}(\omega), \\ G_{\text{loc}}(\omega) &= \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\text{loc}}(\omega)}. \end{aligned} \quad (4.17)$$

- The first equation is equivalent to the feature in the previous exercise where we equated the mean-fields on i and j .
- The second equation is equivalent to computing, in the previous exercise, the magnetisation on the j^{th} site from the partition function.
- The third equation is the essential mean-field approximation, where we equate the self-energy of the simpler decoupled problem to that of the more complete lattice problem and is justified in the limit of infinite dimensions.
- The fourth equation is equivalent, in the previous exercise, to using the computed “magnetisation” of site j in the second step to further compute the mean-field at site i .

In direct analogy with classical mean-field theory, DMFT replaces a lattice model by an effective single-site problem embedded in a self-consistent environment. The crucial distinction is that the mean field is now dynamical rather than static, encoding quantum fluctuations exactly while neglecting spatial correlations. Importantly, this demonstrates the importance of quantum impurity

models in investigating correlated lattice models, particularly in the presence of strong local interactions. This is because the effects of local correlations are captured exactly by such an auxiliary model mapping.

1.4.3 Examples of materials with strong local correlations

As discussed in the previous section, the route to studying lattice problems via quantum impurity models makes most sense for materials with strong local correlations. Such systems are often referred to as strongly correlated quantum materials, and their phenomenology often lies beyond conventional approaches like density functional theory or perturbative weak-coupling methods. In practise, such materials often involve transition metal elements or rare earth/actinide elements, due to the presence of partially filled $3d$ - or $4f$ -orbitals in such elements. This is because these orbitals are more more localised around the nucleus than the s - or p -orbitals of comparable energy, and hence lead to a reduced itinerant energy scale and amplify the effects of local correlations [29].

The localised nature of these orbitals can be understood as follows. For the $3s$ -orbital, the $1s$ and $2s$ orbitals have same orbital angular momentum l but lower principal quantum number n ; since all such wavefunctions must be orthogonal to each other, the $3s$ -orbital must develop nodes in its radial form and extend farther away from the nucleus in order to be orthogonal to these other orbitals. However, for the $3d$ -orbital, there is no orbital with same l but lower n ; this means that the orthogonality is already enforced by the behaviour of the angular part (the spherical harmonics), and the radial part can remain localised around the nucleus without having to spread out. A similar argument holds for the localised nature of the $4f$ -orbital.

We now discuss the phenomenology of some of these quantum materials in order to identify the challenges in trying to explain them and the kind of models that are likely to be useful in this endeavour.

Transition Metal Oxides (TMOs) Transition Metal Oxides (TMOs) are defined chemically by the presence of partially filled d -shells in transition metals (e.g., Cu^{2+} , Mn^{3+} , V^{4+}) coordinated by oxygen ligands. The physics of TMOs is governed by the competition between the kinetic energy hopping amplitude t and the on-site Coulomb repulsion U ; due to the small overlap between the localised d -orbitals, the d -electrons can delocalise only by hopping onto the ligand atoms. According to the Zaanen-Sawatzky-Allen (ZSA) classification scheme [30], these materials are categorized as either Mott-Hubbard insulators, where the charge gap is determined by the d - d Coulomb interaction ($E_g \approx U$), or charge-transfer insulators, where the gap is dictated by the energy difference between the oxygen p -band and the metal d -band. These derive from considerations of several factors, such as crystal field splitting, arrangement of atoms and lattice symmetries.

For example, in the case of octahedral symmetry, repulsion from the surrounding atoms leads to the raising of energy of the $d_{x^2-y^2}$ and $d_{3z^2-r^2}$ orbitals forming the e_g doublet, while the other three orbitals (d_{xy} , d_{yz} and d_{zx}) form the t_{2g} multiplet. For the heavier TMOs (Ti, V, Cr...), the Fermi level lies within e_g . The increased positive nuclear charge (compared to the lighter elements) lowers the energy of the d -orbitals and brings them closer to the p -orbitals, reducing the charge transfer energy $\Delta = \varepsilon_d - \varepsilon_p$ of transporting an electron from a d -state to a p -state, compared to the d -level repulsion U ($|\Delta| < U$). Geometric reasons (the e_g orbitals point towards the p -orbitals)

also enhance the hybridisation between d - and p -orbitals in such materials. If Δ becomes larger than the bandwidth, we obtain a *charge-transfer insulator*, where the charge gap is decided by Δ . In the lighter TMOs, the Fermi level passes through t_{2g} , and because they point away from the p -orbitals, the hybridisation is weaker. Moreover, the reduced positive nuclear charge leads to an increased $\Delta > U$; a U greater than the bandwidth leads to what's called a *Mott-Hubbard insulator*.

Experimental studies of transition metal oxides (TMOs) have established metal-insulator transitions (MITs) as a defining consequence of strong electronic correlations in systems with partially filled (d)-electron shells [31, 32]. Early experiments on vanadates such as V_2O_3 characterised pressure-driven MITs through the pressure- and temperature-dependent conductivity as well as an anomalous enhancement of the optical conductivity [33–35]. Copper oxides constitute the most extensively studied class of correlated TMOs and provide a canonical example of a doping-driven MIT. Transport measurements on parent compounds such as La_2CuO_4 establish a robust insulating state, while carrier doping leads to anomalous metallic behavior characterized by linear-in-temperature resistivity and unconventional superconductivity [36–38]. Such metal-insulator transition is also reported from angle-resolved photoemission spectroscopy (ARPES) measurements in doped La_2CuO_4 and $Bi_2Sr_2CaCu_2O_8$, along with the emergence of pseudogap near the antinodes at high temperatures and a hole-to-electronlike Fermi surface reconstruction [39–41]. Nickelates ($LaNiO_2$, $La_3Ni_2O_7$, $La_4Ni_3O_{10}$, etc) exhibit related but distinct experimental signatures. With Ni^{2+} sharing the same $3d^9$ valence configuration as Cu^{2+} in the cuprate superconductors, these materials also display superconductivity, although the mechanism in the nickelates is different due to the strong coupling between the layers [42, 43]. The optimal doped regime in thin films shows strange metal behaviour, while infinite-layer nickelates show the absence of long-range antiferromagnetic order [44].

Transition Metal Dichalcogenides Transition Metal Dichalcogenides (TMDs) are layered van der Waals materials with the stoichiometry MX_2 , where M is a transition metal (e.g., Mo, W) and X is a chalcogen (e.g. S, Se, Te) [45, 46]. Recent interest has focused on the interplay between strong spin-orbit coupling (SOC), electronic correlations and direct bandgap – properties that make them particularly useful for electronic applications, spintronics and DNA sequencing. Layered TMDs have the structure $X - M - X$, with the transition metal sandwiched between two hexagonal planes of chalcogenides. Different combinations of the metals with the chalcogens lead to widely varying structures and electronic properties in TMDs. Experimental studies show that metal-insulator transitions (MITs) in TMDs are commonly intertwined with charge density wave (CDW) formation rather than arising from purely on-site Coulomb localization [47, 48]. In prototypical compounds such as 1T-TaS₂, ARPES and transport measurements reveal fluctuating magnetic order parameters in the Mott state and a transition to superconductivity driven by pressure [49–51]. Scanning tunnelling microscope (STM) pulses have been successfully used to manipulate the metal-insulator transition of 1T-TaS₂, demonstrating the macroscopic controllability of the Mott phase [52].

ARPES measurements in 2H-NbSe₂ and 2H-TaSe₂ partial gapping of the Fermi surface restricted to nested momentum regions and a temperature-dependent pseudogap similar to the cuprates [53, 54]. This is further supported by optical conductivity experiments that show spectral weight in the midinfrared region along with a low-frequency Drude contribution [55]. Transport measurements on single crystals of TaSe₂ reveal that disorder enhances superconductivity by suppressing

other competing orders such as CDW [56]. Collectively, these experimental results establish TMDs as a distinct class of correlated materials in which reduced dimensionality, band narrowing, and lattice reconstruction cooperate to produce metal–insulator behavior beyond conventional band theory.

Iron-Based Superconductors The iron-based superconductors (FeSCs), discovered in 2008, constitute a broad family of layered materials that share a common structure: a layer of iron atoms joined by tetrahedrally coordinated pnictogen (P, As) or chalcogen (S, Se, Te) anions arranged in a stacked sequence; the layers are separated by alkali, alkaline-earth or rare-earth and oxygen/fluorine ‘blocking layers’ [57–60]. Doping or applying external pressure reveals coupled antiferromagnetic and structural transitions along with superconducting and nematic orders [61–63]. Electrical resistivity measurements in the neighbourhood of the quantum critical point show a crossover from a bad-metal behavior (T –linear resistivity) to a Fermi liquid behaviour (T^2 –resistivity) [64]. Optical conductivity measurements indicate point to intermediate-to-strong electronic correlations [65].

ARPES measurements reveal a k –dependent superconducting gap and de Haas-van Alphen measurements indicate substantial quasiparticle mass enhancements, shedding more light on the correlated nature of the material [66, 67]. Strain-tuning measurements of the nematic susceptibility, and differential elastoresistance measurements have provided compelling evidence for the nematic transition and its surrounding fluctuations being the cause of the structural transition as well as the superconductivity [68, 69]. Neutron scattering and NMR measurements further reveal persistent low-energy spin fluctuations across wide regions of the phase diagram, supporting scenarios in which unconventional superconductivity emerges from the interplay of magnetic, orbital, and nematic degrees of freedom [70, 71].

Heavy-Fermion / Kondo-Lattice Systems Heavy-fermion materials are a class of strongly correlated electron systems typically realized in intermetallic compounds of rare earth metals (such as Ce, Sm and Yb) and actinides (such as U, Np and Pu), where localized f electrons hybridize with itinerant conduction bands [72, 73]. At high temperatures, these systems exhibit local-moment behavior governed by Curie–Weiss susceptibilities and incoherent transport, while at low temperatures the Kondo effect leads to the formation of a coherent many-body state with extremely large quasiparticle effective masses [74]. Experimentally, this crossover is most clearly manifested in transport and thermodynamic measurements: the electrical resistivity shows a characteristic maximum followed by a rapid drop signaling the onset of coherence, while the linear specific heat coefficient $\gamma = C/T$ reaches values orders of magnitude larger than in conventional metals, directly reflecting the heavy quasiparticle mass [75].

A defining feature of heavy-fermion materials is their proximity to magnetic quantum phase transitions, which can be accessed experimentally by pressure, chemical substitution, or magnetic field [76]. Near such quantum critical points (QCPs), transport measurements reveal pronounced non-Fermi-liquid behavior (T –linear resistivity) and a vanishing Hall response, along with a divergent effective quasiparticle mass as the transition is approached [77, 78]; these are all indications of the breakdown of electronic excitations near the transition. Neutron scattering experiments provide direct evidence for spin fluctuations down to very low temperatures, interplaying with the itinerant electrons at the QCP and potentially driving the superconducting transition [79, 80]. Quan-

tum oscillation measurements through de Haas–van Alphen experiments further demonstrate Fermi surface reconstruction across the quantum critical point, highlighting the drastic change in Fermi volume [81]. Together, these experimental observations establish heavy-fermion compounds as paradigmatic systems in which strong local correlations, hybridization-driven coherence, and quantum critical fluctuations conspire to produce emergent low-energy quasiparticles and unconventional metallic behavior [82].

1.4.4 Generator of strong local correlations: Quantum impurity models

Within this thesis, we have developed an auxiliary model approach that combines the physics of quantum impurity models with a manybody version of Bloch’s theorem to investigate interacting fermionic lattice models. This is motivated by the relative ease in solving impurity models and transparency afforded by such solutions. We now review the phenomenology of some of these impurity models and point out how the various impurity features affect the physics of the lattice models.

The single-impurity Anderson model describes the dynamics of a system of non-interacting itinerant electrons ($c_{k,\sigma}$) hybridising with a localised correlated impurity orbital (d_σ) [15]:

$$H_{\text{AIM}} = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + \sum_{k,\sigma} V_k (c_{k,\sigma}^\dagger d_\sigma + \text{h.c.}) + \epsilon_d \sum_{\sigma} n_{d\sigma} + U n_{d\uparrow} n_{d\downarrow} . \quad (4.18)$$

V_k controls the hybridisation strength, and ϵ_d and U control the onsite energy and local correlation respectively of the impurity site. In the limit of large U , the impurity dynamics gets projected onto its spin states, and the model then describes the scattering of non-interacting electrons off magnetic impurities, through the Kondo model [83]. Formally, the Kondo model is obtained by carrying out a Schrieffer-Wolff transformation on the Anderson impurity model and then using the condition to $V \ll U$ to drop all contributions beyond second order in the ratio V^2/U [84]:

$$H_{\text{Kondo}} = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + J \sum_{k_1,k_2} \sum_{\alpha,\beta} \vec{S}_d \cdot \vec{\sigma}_{\alpha,\beta} c_{k_1,\alpha}^\dagger c_{k_2,\beta} , \quad (4.19)$$

where $\vec{S}_d$ is the impurity spin formed by projecting the Hilbert space of the impurity to the singly-occupied states.

Even though the conduction bath is non-interacting, the presence of local repulsion on the impurity site makes the problem *many-electron* in nature: it makes it impossible to study the electrons in isolation. This can be seen as follows: imagine two conduction electrons $|k_1 \uparrow\rangle$ and $|k_2 \downarrow\rangle$ trying to hybridise with an empty impurity site. The first electron moves onto the impurity site through a scattering process of the form $c_{d\uparrow}^\dagger c_{k_1\uparrow}$ and makes the impurity site single occupied ($|\uparrow\rangle_d$), but now the second electron k_2 *must pay a cost* of the order of U in order to hybridise with this singly-occupied impurity site through a process of the kind $c_{d\downarrow}^\dagger c_{k_1\downarrow}$. This shows that the conduction electrons *are aware* of the other electrons, and all of them must be treated together. Another way of seeing this collective behaviour is from the Kondo model. Again imagine two spin-up electrons k_1, k_2 that are about to scatter off the impurity spin (which is presently down) via a *spin-flip* process. The first

electron scatters and leaves the impurity spin up while it itself becomes down, but now the second conduction electron cannot spin-flip scatter because the spins cannot be exchanged (both are down!). This is another example of how the conduction electrons can *feel* the presence of the other electrons, and this is what makes the problem many-electron in nature.

Kondo screening Despite the interacting nature of the problem, the problem can be solved in an effectively exact fashion, via Wilson’s numerical renormalisation group approach [85, 86], density-matrix renormalisation group approach [87] or the method of *Bethe ansatz* [88, 89]. The former solution is more transparent: for any non-zero value of the hybridisation V (or Kondo coupling J), the low energy physics of the impurity is described by a strong-coupling fixed point ($V = \infty$, or $J = \infty$), where the impurity forms a maximally-entangled singlet state with the local spin density of the conduction bath. The low-energy excitations about such a fixed point can be calculated from renormalised perturbation theory, and describe a local Fermi liquid – one particle excitations in the neighbourhood of the singlet renormalised by a local repulsion term [90, 91]. The crossover from the free local moment at high temperatures to the screened non-magnetic state at low temperatures occurs at an emergent energy scale T_K , the Kondo temperature, which depends exponentially on the strength of the Kondo coupling, and is the only energy scale remaining at low temperatures, so that all thermodynamic quantities must go as a function of T/T_K [85].

At half-filling (attained for $\epsilon_d = -U/2$), the only other fixed points are the free orbital fixed point at $V = 0, U = \epsilon_d = 0$ where the impurity site is freely occupied by two independent spins, and the local moment fixed point at $V = 0, U = \infty$, where the impurity site is occupied by a local moment that is decoupled from the bath [74, 86]. Away from half-filling, two other fixed points become available [74, 86]: a valence fluctuation fixed point at $V = 0, \epsilon_d = 0, U = \infty$ where the doubly-occupied state has been projected out from the symmetric free orbital fixed point, and a frozen impurity fixed point at $V = U = 0, \epsilon_d = \infty$ where the impurity site is left completely empty.

The crossover from local moment physics to Kondo screening as the system is tuned from high to low temperatures can also be seen from the impurity spectral function [92–95]. At low frequencies, the spectral function exhibits a sharp resonance around $\omega = 0$ that indicates the presence of gapless single-particle local Fermi liquid excitations. At higher frequencies, one sees satellite peaks that involve impurity excitations that cost energy of the order of U . The height of the central Kondo resonance is a topological invariant; it is fixed by the impurity occupancy owing to an application of the Friedel sum rule to impurity models [96], and remains unchanged as the interaction U is increased adiabatically while remaining at particle-hole symmetry. The topological nature can also be ascertained by identifying such a sum rule as the impurity model analogue of Luttinger’s theorem [97, 98] that relates the total number density of electrons to the number of Greens function poles within the Fermi volume, and which is itself topological [99, 100].

Breakdown of Kondo screening While the vanilla Anderson model (or the Kondo model derived from it) only display a strongly-coupled screened phase, it is possible to tweak it in order to introduce Kondo frustration which can eventually lead to a breakdown in the Kondo screening process. Such models are interesting because they provide a means of studying zero temperature *continuous* phase transitions and address experimental signatures of quantum criticality in correlated systems. The

canonical example of this is the multichannel Kondo model, involving a single impurity hybridising with multiple independent conduction channels [101]:

$$H_{\text{MCK}} = \sum_{k,\sigma,\alpha} \epsilon_{k,\lambda} n_{k,\sigma,\lambda} + \sum_{k_1,k_2,\lambda} J_\lambda \sum_{\alpha,\beta} \vec{S}_d \cdot \vec{\sigma}_{\alpha,\beta} c_{k_1,\alpha,\lambda}^\dagger c_{k_2,\beta,\lambda}, \quad (4.20)$$

here λ is the channel index. In the case of an isotropic Kondo coupling across the channels ($J_\lambda = J$), the strong-coupling $J = \infty$ fixed-point of the single-channel Kondo model is replaced by an intermediate-coupling fixed-point at a finite value J^* [101, 102]. Powerful methods like boundary conformal field theory [103–108], bosonization [109–113], numerical renormalization group [105, 114, 115] and Bethe ansatz [116–120] have shown that the low energy theory of such models exhibit non-Fermi liquid excitations. Further, the system displays anomalous behaviour in thermodynamic properties near $T = 0$ and a fractional zero temperature entropy in the thermodynamic limit (upon ensuring that temperature is taken to zero after the system size is taken to infinity [112, 121]). This anomalous divergent behaviour is a signature of the fact that the MCK system with a single channel-symmetric exchange coupling is quantum critical: the ground state is susceptible to perturbations that introduce channel anisotropy in the exchange couplings [101, 105, 111, 119, 122].

Another class of models that shows Kondo breakdown is that of pseudogap Anderson and Kondo models, where the impurity hybridises with a conduction bath whose density of states (DOS) varies as $\rho(\omega) \sim |\omega|^r$ with a parameter $r > 0$ that controls how fast the DOS vanishes as the Fermi level is approached. Such behaviour can arise in semi-metals and in long-range ordered systems where the order parameter vanishes at certain points on the Fermi surface (such as d -wave superconductors). The pseudogap impurity models exhibit a boundary quantum phase transition between a Kondo-screened strong-coupling phase and an unscreened local-moment phase as a function of interaction strength or exchange coupling [123–126]. At the critical Kondo coupling, the impurity spectral function and spin susceptibility exhibit nontrivial power-law scaling with anomalous exponents determined by r [124, 127–130]. For $0 < r < 1/2$ (in the particle-hole symmetric case), both the local-moment and strong-coupling phases are stable, separated by a critical fixed point characterized by fractional residual entropy and nontrivial hyperscaling relations [124, 131]. For larger exponents, the strong-coupling phase may disappear entirely, and the system remains in the local-moment phase for all couplings [126].

The multi-impurity Anderson and Kondo models furnish another canonical realization of Kondo breakdown, driven by the competition between Kondo screening and inter-impurity interaction. The paradigmatic example is the two-impurity Kondo model (TIKM), in which two localized spins couple both to conduction electrons and to each other via an antiferromagnetic exchange K [132, 133]. In the limit of large $K > 0$, the impurities form a non-magnetic singlet decoupled from the conduction electrons and suppressing the Kondo resonance. This fixed point is separated from the strong-coupling Kondo screened fixed point through a quantum critical point [132–134] displaying logarithmic or power-law scaling in thermodynamic quantities and fractional residual entropy [135]. A modified version of this model that uses conduction bath-mediated inter-impurity interaction leads to a metallic spin liquid displaying nonlocal impurity-spin entanglement and fractionalized charge excitations [136]. More intricate geometries can be constructed by arranging antiferromagnetically coupled Kondo impurities along a triangle, each hybridising with its own conduction bath; such a model displays a topological phase transition from a symmetry-preserved gapless spin liquid phase

with “topological order” to a screened Fermi liquid phase [137]. Indeed, recent experiments have validated such phase diagrams for the case of two impurities by studying pairs of copper atoms in the presence of strong magnetic fields [138].

1.5 Mott transition in Hubbard-like models

The Hubbard model (introduced in eq. 1.1) presents one of the most fruitful playgrounds for investigating correlation-driven metal-insulator transitions and its various facets. This model, despite its conceptual simplicity, exhibits a rich variety of phenomena, including antiferromagnetism, superconductivity, and, most notably for our purposes, the correlation-driven Mott metal–insulator transition (MIT) [4, 139–141]. While exact solutions exist in the limits of $d = 1$ [142] and $d = \infty$ [143], the problem remains largely open in more realistic situations such as two and three spatial dimensions.

As discussed earlier, the limit of infinite dimensions presents a drastic simplification for the manybody scattering processes within the problem: the one-particle k -space self-energy becomes momentum-independent due to the vanishing of all non-local contributions (they scale as negative powers of the coordination number, which diverges in the limit of $d = \infty$). This simplification lies at the heart of dynamical mean-field theory and its ability to capture the Mott metal–insulator transition at both zero and finite temperatures for this limiting case [18, 143–147]. At half-filling and zero temperature, DMFT predicts a continuous transition as a function of U wherein the quasiparticle weight Z tends to zero at a critical interaction strength U_{c2} , signaling the vanishing of coherent Fermi liquid behavior and the opening of a Mott gap in the single-particle spectral function [18, 145]. The three-peak structure of the interacting spectral function, with a central quasiparticle resonance flanked by incoherent lower and upper Hubbard bands, indicates the presence of highly renormalised Landau quasiparticles [145]. A doping-driven continuous metal-insulator transition also exists for this model at zero temperature, marked by the vanishing of electronic compressibility at a critical value of the chemical potential [148].

While the DMFT solution captures the zero temperature transition accurately, the self-consistently determined nature of the conduction bath renders the solution somewhat opaque and makes it difficult to write down an effective Hamiltonian for the low-energy theories. Since the vanilla Anderson impurity model does not contain the mechanism to stabilise a localised phase over a finite-sized parameter regime, an important question arising from the DMFT solution is: what are the minimal promotions that one must apply to the Anderson impurity in order to capture such a metal-insulator transition (1/2-filled Hubbard model on the Bethe lattice in $d = \infty$), in sufficient accuracy and precision? Finding such an impurity model would also reduce the considerable computational effort that is presently required in self-consistent approaches.

At finite temperatures, the interaction-driven as well as doping-driven Mott transition become first order, and is characterized by a region of phase coexistence and hysteresis [146, 149]. Two spinodal lines, $U_{c1}(T)$ and $U_{c2}(T)$, emerge; within this region metallic and insulating DMFT solutions coexist, and responses such as resistivity show a significant discontinuity [150]. These lines meet at a finite-temperature critical endpoint (U_c, T_c) , above which there is no sharp distinction between metallic and insulating solutions and only a crossover exists between the Mott insulator and

the Fermi liquid through a bad-metal regime, as the chemical potential is tuned [146, 148, 151–153]. The coexistence of metallic and insulating phases shows that the insulating solution is present within the many-body spectrum of the metallic phase. This naturally leads to the following question: since a quantum impurity lies at the heart of the self-consistency procedure, what does the presence of coexistence region mean for the Hamiltonian dynamics and the spectrum of this impurity model? Can the impurity model Hamiltonian display the emergence of the insulating state prior to the transition, through changes in the eigenstates? Analytic insight of this kind is particularly important because approaching a coexistence region numerically is often fairly tricky.

The bad metal regime displays anomalous linear-in-temperature resistivity which corresponds to quantum critical ω/T scaling, along the quantum widom line [154, 155]. This is accompanied by the disappearance of the Drude peak in optical conductivity, signalling the loss of quasiparticle transport [154]. Similar quantum critical scaling is also obtained from DMFT solutions at very low temperatures, hinting at a “hidden” quantum criticality that is continuously connected to and responsible for the critical behaviour above the finite temperature endpoint [156]. Luttinger’s theorem, which relates the poles of the Greens function within the Fermi volume to the number of electrons in the system at zero temperature, is satisfied within the Fermi liquid phase but is violated in the particle-hole asymmetric Mott insulator by a universal magnitude [157].

The above studies of single band models describe the Mott transition to be of the Mott-Hubbard kind (described in previous sections), where the size of the gap is determined by the double-occupancy cost U . In order to obtain transition of the charge-transfer kind, one needs to consider at least a two-band model. Indeed, in a non-degenerate two-band Hubbard model with a charge-transfer energy Δ for hopping from a d -orbital to a p -orbital, one sees a Mott-Hubbard transition when $U < \Delta$ and a charge-transfer transition when $U > \Delta$ [158]. The degenerate two-band model with inter-band local correlation shows successive Mott-Hubbard transitions at low temperatures whenever the filling passes through integer values and the interaction strength is intermediate [159].

The emergence of a quantum critical regime at high temperatures and its connection to similar signatures at low temperatures suggests the presence of quantum fluctuations in the underlying impurity model that modify the impurity-bath scattering processes and the associated impurity responses. One can therefore ask: What are the fluctuations that destroy the metal and lead to the insulating phase? Can we obtain a universal theory for these competing tendencies? The quantum critical point itself (at $T = 0$) is likely to be an exotic state. Is it possible to obtain a low-energy theory for the local gapless excitations precisely at the MIT, where the metal is on the brink of destruction? How do these excitations compare with those of the local Fermi liquid, e.g., in terms of self-energies and two-particle correlation functions?

1.6 Effect of low dimensionality: Mott MIT in two dimensions

1.6.1 DCA, CDMFT results (among other methods)

1.6.2 Open Questions

1.7 Effect of multiple bands: Orbital selective Mott transition

1.7.1 results on heavy fermions

1.7.2 Open Questions

1.8 Brief introduction to the tiling auxiliary model approach

1.8.1 Why do we need another auxiliary model method? Limitations of existing approaches

1.8.2 Overview

1.8.3 Features

1.9 Entanglement holography and fermionic criticality

Role of entanglement in fermionic systems The present millennium has seen a rapid surge in the use of quantum entanglement towards the study of quantum and condensed matter systems [160–165]. Being non-local by nature, it is helpful in tracking correlations among particles across both small and large distances, giving valuable insight into the nature of the ground state and the low-energy excitations. Entanglement, in the form of various measures like entanglement entropy and mutual information, is the engine that gives rise to new low-energy degrees of freedom from a renormalisation group (RG) perspective, and is therefore key to understanding and characterising phase transitions and criticality in quantum matter. For instance, entanglement entropy is at the heart of understanding the so-called topological states of matter that can be described in terms of a purely topological effective Lagrangian and have a robust ground state degeneracy and gapless edge states [166–186]. In contrast to systems with local interactions whose subsystem entanglement entropy scales with the area of the subsystem [162, 187, 188], topologically ordered systems have an additional subdominant topological term dubbed the topological entanglement entropy [189–207] that quantifies the long-ranged nature of the entanglement in such systems. The non-local nature arises from the fact that the ground state of such a system cannot be made separable (non-entangled) by the application of purely local unitary transformations [164, 208, 209].

More pertinent to the work at hand, however, is the entanglement in free fermionic systems. In this thesis, we will consider relativistic Dirac fermions in two dimensions, both massless and massive. They emerge in several condensed matter systems [210], appearing at the surfaces of three-dimensional topological insulators [211–214], quasi-2D organic materials [215–217] and in artificial

materials like cold atoms, molecular and microwave crystals [218–224]. For massless critical systems (or conformal field theories (CFTs)) living in one spatial dimension, the entanglement entropy of a single subsystem of length l scales as $\sim \frac{c}{3} \ln l$ [225–234], where c is the conformal central charge of the CFT. The case of multiple subsystems is more involved, but has been solved [232, 235, 236]. For massive fermions, the entanglement entropy (or rather the entropic c -function, which is the derivative of the entropy along the distance) can be expressed in the form of differential equations which need to be solved numerically [235, 237–239]. The solutions lend themselves to small and large mass expansions, reducing to $\frac{1}{3} \ln \frac{l}{\epsilon} - \frac{1}{6} (ml \ln ml)^2 + O((ml)^2)$ for the case of small ml , and to $\sim (ml)^{-1/2} e^{-2ml}$ for that of large ml , both for a single interval. The $\ln l$ behaviour is a specific case of the more general $l^{d-1} \ln l$ violation of the area law, as observed in d -dimensional fermionic critical systems [235, 240–254]. This violation arises from the non-local nature of the quantum fluctuations that exist in quantum critical fermionic systems.

While there is no geometry-independent subdominant term in the entanglement entropy of free fermionic systems [255, 256], it is possible to generate such a term by placing the system on a manifold with a non-trivial topology, say a cylinder, and inserting a magnetic flux through the cylinder [255, 257–260]. For a massless Dirac fermion, if there are ϕ number of electronic flux quanta piercing the system, the additional term that is generated in the entanglement entropy of a subsystem of sufficiently (as defined in Ref. [255]) large length is of the form $\frac{1}{6} \ln |2 \sin \frac{\phi}{2}|$ [255, 258]. There is evidence from numerical computations on discrete models that such a term is a signature of the specific field theory and encodes universality [260–262]. In this thesis, we put this idea on a stronger analytical footing by relating the flux-dependent term to a topological invariant of the field theory, its Luttinger volume [97–100, 263–265]. Given by the number of k -states enclosed by the Fermi surface, Luttinger volume V_L has proved to be one of the most important indicators that distinguish metals from insulating states of matter. In metals, V_L is constrained by the number of particles in the system, following Luttinger’s theorem [97–99]. As shown most prominently in Ref. [99], Luttinger’s theorem depends crucially on the presence of extended states in the metal that can respond to flux insertion experiments. We demonstrate that indeed it is this subdominant term in the entanglement entropy that carries this dependence and hence encodes Luttinger’s theorem. It is known that for strongly-interacting states like Mott insulators or quantum hall states, V_L is modified due to changes in the topology of the Fermi surface [100, 263–265]. In this context, we relate the topological invariant on the metallic side (Luttinger volume) to the topological invariants (the Chern numbers) in the quantum hall system that the metal would lead to when placed in a perpendicular magnetic field. Such a relation makes explicit the transmutation of the extended states into the localised states of the insulating phase.

Entanglement hierarchy and the holographic principle The leading geometry-dependent piece is also found to have intriguing properties of its own. The fact that the free fermion Lagrangian is diagonal in momentum space means that the entanglement arises purely from real space quantum fluctuations, and this allows us to study the entanglement of subsystems of varying sizes in momentum space. We find that the entanglement entropy hence obtained has a Russian doll-like nested structure, such that increasing the size of the subsystem in momentum space gradually reveals layers of entanglement. In fact, we argue that such transformations between the subsystems amount to

a renormalisation group (RG) flow in the space of the field theories with varying spectral gaps, and the above-mentioned hierarchy in the entanglement entropy then stretches across the RG transformations. The precise prescription for choosing the subsystems is described in Sec. 3.3. We also show, simultaneously, that the hierarchy holds across multipartite entanglement measures, leading to an onion-like structure where increasing the number of parties in the entanglement measure (mutual information, tripartite information, 4-partite information and so on) leads to additional entanglement that did not exist before. More importantly, we interpret the hierarchy of entanglement along the RG flow as the emergence of an additional dimension [266–268], allowing us to define a measure of distance using some non-negative measure of entanglement, like the mutual information [269–272]: the more the entanglement, the less will be the distance between those points. Generation of an additional dimension is, of course, the essence of the AdS/CFT conjecture or, more generally, the holographic principle.

The holographic principle [273–281], also known as the gauge-gravity duality, posits that there exists a duality relation between a quantum field theory with a gauge field, and a gravity theory having one additional spatial dimension [282–285]. The additional dimension can be visualised as a renormalisation group flow that starts from the boundary field theory and extends into the bulk [286–295]. The most popular realisation of the holographic principle is the AdS/CFT correspondence that links a super-symmetric gauge theory with a string theory [296–298]. The essence of holography is that all the information about the bulk gravity theory is already contained within the boundary gauge theory; this idea is very similar to the discovery by Bekenstein and Hawking [299–301] that the entropy of a black hole is determined by its surface area. In fact, Ryu and Takayanagi proposed a generalisation of this result where the entanglement entropy of subsystem region a CFT is determined by the area of the minimal surface that bounds the region in the bulk [302, 303]. The AdS/CFT correspondence happens to be a strong-weak duality - it relates a strongly coupled CFT to a theory of weak classical gravity. This allows physicists to use the holographic principle and convert a problem of strong quantum correlation to one of classical gravity which can be solved using well-known methods [276, 277]. This has led to new insights in various topics of high energy and condensed matter physics, such as the viscosity of the quark-gluon plasma [304], the response functions of the Bose-Hubbard model [305, 306], and the phenomenology of non-Fermi liquids [307–310], holographic superconductors [311–316], striped phases [317–319] and holographic Fermi liquids [320–324]. An overview of the metallic states (phases that satisfy Luttinger’s theorem) and their connections to the gauge-gravity duality have been provided in Refs. [325–327].

Of course, on the flip side, the holographic principle also implies that studying the boundary conformal theory can provide insights into the nature of the bulk quantum gravity theory. This however requires constructing, *ab initio*, the emergent dimension by starting from the field theory, and has proved to be more difficult than the converse, mostly because the field theories are often strongly coupled. Nevertheless, there exist some examples of constructive attempts towards a bulk gravity theory [264, 265, 271, 272, 328–361]. Some well-known approaches include the momentum shell renormalisation approach of Sung-Sik Lee [341–344] and the multi-scale entanglement renormalisation ansatz (MERA) of Vidal [345–349]; both use certain forms of RG flow to generate the extra spatial dimension. The former integrates out high momentum degrees of freedom in favour of generating dynamics for the boundary gauge field, with the RG trajectory variable giving

rise to the additional spatial dimension [275]. On the other hand, the MERA works by disentangling degrees of freedom in real space, creating a tensor network in the process [350–352] and leading to a renormalisation in the entanglement. The renormalised entanglement is interpreted as the additional dimension [347]. There exist other approaches that also proceed via entanglement renormalisation, such as the exact holographic mapping [271, 272] (EHM) and the unitary renormalisation group [264, 265, 353] (URG) that operate purely through unitary transformations and have been explicitly shown to lead to a holographic tensor network along the RG flow. Apart from these, there is a body of work due largely to Ki-Seok Kim that provides a geometric description of Wilson’s numerical renormalisation group [86, 354–356, 362], and creates a more general RG framework in terms of an order-parameter field that extends the Ads/CFT correspondence to non-critical theories [357–361].

Questions on entanglement and holography become particularly interesting in systems of critical fermions. As mentioned earlier, such systems differ from bosonic and gapped systems through the presence of longer-ranged entanglement that follows a modified area-law [246, 248] as well as topological features arising from the incompressibility of the Fermi volume [99, 100]. Fermi surface gapping phase transitions in such systems are often driven by changes in topological quantum numbers and the nature of the inter-particle entanglement [76, 264, 265]. Constructing a framework for fermionic criticality is therefore an active area of research with consequences for important systems such as unconventional superconductors, strange metals and heavy fermions [77, 363–365]. Still less is known about the holographic implications of such systems. We extend this body of work by providing an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

Some open questions

Chapter 2

Methods and Preliminaries

2.1 The Unitary renormalisation group method

The URG method was introduced and formalised in refs. [264,265,363,364]. This section is adapted from those references and expanded wherever required.

Description of the problem

We are given a Hamiltonian $\mathcal{H}$ which is not completely diagonal in the occupation number basis of the electrons, $\hat{n}_k$: $[\mathcal{H}, n_k] \neq 0$. k labels any set of quantum numbers depending on the system. For spin-less Fermions it can be the momentum of the particle, while for spin-full Fermions it can be the set of momentum and spin. There are terms that scatter electrons from one quantum number k to another quantum number k' .

We take a general Hamiltonian,

$$\mathcal{H} = H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta}) + c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \quad (1.1)$$

Formally, we can decompose the entire Hamiltonian in the subspace of the electron we want to decouple ($q\beta$).

$$\mathcal{H} = \begin{pmatrix} |1\rangle & |0\rangle \\ H_1 & T \\ T^\dagger & H_0 \end{pmatrix} \quad (1.2)$$

The basis in which this matrix is written is $\{|1\rangle, |0\rangle\}$ where $|i\rangle$ is the set of all states where $\hat{n}_{q\beta} = i$. The aim of one step of the URG is to find a unitary transformation U such that the new Hamiltonian $U\mathcal{H}U^\dagger$ is diagonal in this already-chosen basis.

$$\tilde{\mathcal{H}} \equiv U\mathcal{H}U^\dagger = \begin{pmatrix} |1\rangle & |0\rangle \\ \tilde{H}_1 & 0 \\ 0 & \tilde{H}_0 \end{pmatrix} \quad (1.3)$$

U_q is defined by

$$\tilde{\mathcal{H}} = U_q \mathcal{H} U_q^\dagger \text{ such that } [\tilde{\mathcal{H}}, n_q] = 0 \quad (1.4)$$

It is clear that U is the diagonalizing matrix for $\mathcal{H}$. Hence we can frame this problem as an eigenvalue equation as well. Let $|\psi_1\rangle, |\psi_0\rangle$ be the basis in which the original Hamiltonian $\mathcal{H}$ has no off-diagonal terms corresponding to $q\beta$. Hence, we can write

$$\mathcal{H} |\psi_i\rangle = \tilde{H}_i |\psi_i\rangle, i \in \{0, 1\} \quad (1.5)$$

Since $|\psi_i\rangle$ is the set of eigenstates of $\mathcal{H}$ and $|i\rangle$ is the set of eigenstates in which $U\mathcal{H}U^\dagger$ has no off-diagonal terms corresponding to $q\beta$, we can relate $|\psi_i\rangle$ and $|i\rangle$ by the same transformation : $|\psi_i\rangle = U^\dagger |i\rangle$. We can expand the state $|\psi_i\rangle$ in the subspace of $q\beta$:

$$|\psi_i\rangle = \sum_{j=0,1} |j\rangle \langle j | \psi_i \rangle \equiv |1\rangle |\phi_1^i\rangle + |0\rangle |\phi_0^i\rangle \quad (1.6)$$

where $|\phi_j^i\rangle = \langle j | \psi_i \rangle$. If we substitute the expansion 1.2 into the eigenvalue equation 1.5, we get

$$\left[H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta}) + c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \right] |\psi_i\rangle = \tilde{H}_i |\psi_i\rangle \quad (1.7)$$

The diagonal parts $H_e = \text{tr} [\mathcal{H} \hat{n}_{q\beta}]$ and $H_h = \text{tr} [\mathcal{H} (1 - \hat{n}_{q\beta})]$ can be separated into a purely diagonal part $\mathcal{H}^d$ that contains the single-particle energies and the multi-particle correlation energies or Hartree-like contributions, and an off-diagonal part $\mathcal{H}^i$ that scatters between the remaining degrees of freedom $k\sigma \neq q\beta$. That is,

$$H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta}) = \mathcal{H}^d + \mathcal{H}^i$$

This gives

$$\left[c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \right] |\psi_i\rangle = \left(\tilde{H}_i - \mathcal{H}^i - \mathcal{H}^d \right) |\psi_i\rangle \quad (1.8)$$

Obtaining the decoupling transformation

We now define a new operator $\hat{\omega}_i = \tilde{H}_i - \mathcal{H}^i$, such that

$$\left[c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \right] |\psi_i\rangle = \left(\hat{\omega}_i - \mathcal{H}^d \right) |\psi_i\rangle \quad (1.9)$$

From the definition of $\hat{\omega}_i$, we can see that it is Hermitian and has no term that scatters in the subspace of $q\beta$, so it is diagonal in $q\beta$ and we can expand it as $\hat{\omega}_i = \hat{\omega}_i^1 \hat{n}_{q\beta} + \hat{\omega}_i^0 (1 - \hat{n}_{q\beta})$. Using the expansion 1.6, we can write

$$\hat{\omega}_i |\psi_i\rangle = \hat{\omega}_i^1 |1\rangle |\phi_1^i\rangle + \hat{\omega}_i^0 |0\rangle |\phi_0^i\rangle \quad (1.10)$$

Since the only requirement on $|\psi_i\rangle$ is that it diagonalize the Hamiltonian in the subspace of $q\beta$, there is freedom in the choice of this state. We can exploit this freedom and choose the $|\phi_0^i\rangle$ to be an eigenstates of $\hat{\omega}_i^{1,0}$ corresponding to real eigenvalues $\omega_i^{1,0}$:

$$\left[\mathcal{H}^d + c_{q\beta}^\dagger T + T^\dagger c_{q\beta} \right] |\psi_i(\omega_i)\rangle = \left(\omega_i^1 - \mathcal{H}^d \right) |1\rangle |\phi_1^i\rangle + \left(\omega_i^0 - \mathcal{H}^d \right) |0\rangle |\phi_0^i\rangle \quad (1.11)$$

If we now substitute the expansion 1.6 and gather the terms that result in $\hat{n}_{q\beta} = 1$, we get

$$c_{q\beta}^\dagger T |0\rangle |\phi_0^i\rangle = (\omega_i^1 - \mathcal{H}^d) |1\rangle |\phi_1^i\rangle \quad (1.12)$$

Similarly, gathering the terms that result in $\hat{n}_{q\beta} = 0$ gives

$$T^\dagger c_{q\beta} |1\rangle |\phi_1^i\rangle = (\omega_i^0 - \mathcal{H}^d) |0\rangle |\phi_0^i\rangle \quad (1.13)$$

We now define two many-particle transition operators:

$$\begin{aligned} \eta^\dagger(\omega_i^1) &= \frac{1}{\omega_i^1 - \mathcal{H}^d} c_{q\beta}^\dagger T \equiv G_1 c_{q\beta}^\dagger T \\ \eta(\omega_i^0) &= \frac{1}{\omega_i^0 - \mathcal{H}^d} T^\dagger c_{q\beta} \equiv G_0 T^\dagger c_{q\beta} \end{aligned} \quad (1.14)$$

where G_j is the propagator $\frac{1}{\omega_i^j - \mathcal{H}^d}$. We can write this compactly as

$$\eta(\hat{\omega}) = G T^\dagger c_{q\beta} = \frac{1}{\hat{\omega} - \mathcal{H}^d} T^\dagger c_{q\beta} \quad (1.15)$$

where $\hat{\omega}_i = \omega_i^0 (1 - \hat{n}_{q\beta}) + \omega_i^1 \hat{n}_{q\beta} = \begin{pmatrix} \omega_i^1 & \\ & \omega_i^0 \end{pmatrix}$ is a 2x2 matrix and $\mathcal{H}^d = \mathcal{H}_0^d (1 - \hat{n}_{q\beta}) + \mathcal{H}_1^d \hat{n}_{q\beta}$ and $G = (\hat{\omega} - \mathcal{H}^d)^{-1}$. It is easy to check that this reproduces the previous forms of η_0 and $\eta_1^\dagger$. We will later find that it is important to demand that these two be Hermitian conjugates of each other; that constraint is imposed on the denominators:

$$\eta^\dagger(\omega_i^0) = \eta^\dagger(\omega_i^1) \implies \frac{1}{\omega_i^1 - \mathcal{H}^d} c_{q\beta}^\dagger T = c_{q\beta}^\dagger T \frac{1}{\omega_i^0 - \mathcal{H}^d} \quad (1.16)$$

Henceforth we will assume that this constraint has been imposed.

In terms of these operators, eq. 1.13 becomes

$$|1\rangle |\phi_1^i\rangle = \eta^\dagger |0\rangle |\phi_0^i\rangle, \quad |0\rangle |\phi_0^i\rangle = \eta |1\rangle |\phi_1^i\rangle \quad (1.17)$$

These allow us to write

$$|\psi_1\rangle = |1\rangle |\phi_1^i\rangle + |0\rangle |\phi_0^i\rangle = (1 + \eta) |1\rangle |\phi_1^i\rangle, \quad |\psi_0\rangle = (1 + \eta^\dagger) |0\rangle |\phi_0^i\rangle \quad (1.18)$$

Recalling that $|\psi_i\rangle = U^\dagger |i\rangle$, we can read off the required transformation:

$$U_1 = 1 + \eta \quad (1.19)$$

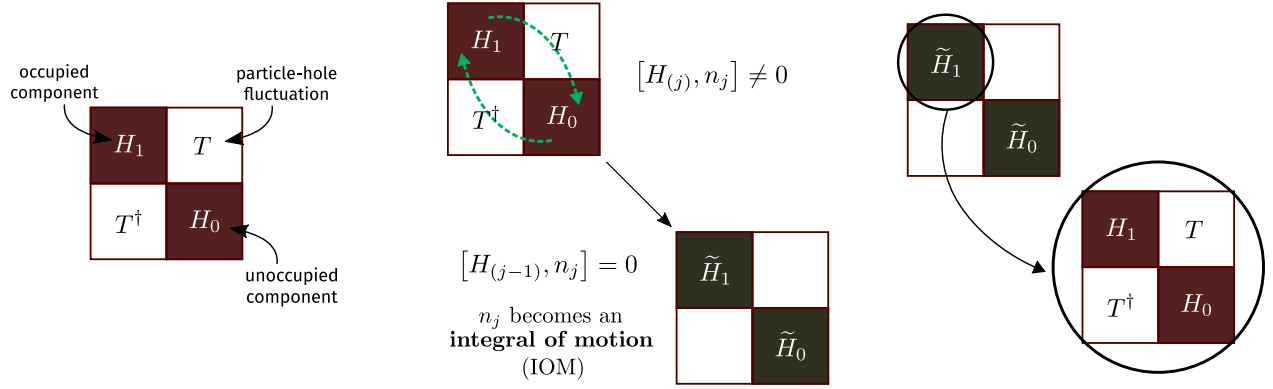


Figure 2.1: Three steps of the URG: Decompose the Hamiltonian in a 2×2 matrix, apply the unitary operator to rotate it, then repeat these steps with one of the rotated blocks.

Properties of the many-body transition operators

The operators η have some important properties. First is the Fermionic nature:

$$\eta^2 = \eta^{\dagger 2} = 0 \quad [c^{\dagger 2} = c^2 = 0] \quad (1.20)$$

Second is:

$$\begin{aligned} |1\rangle |\phi_1^i\rangle &= \eta^\dagger |0\rangle |\phi_0^i\rangle = \eta^\dagger \eta |1\rangle |\phi_1^i\rangle \implies \eta^\dagger \eta = \hat{n}_{q\beta} \\ |0\rangle |\phi_0^i\rangle &= \eta |1\rangle |\phi_1^i\rangle = \eta \eta^\dagger |0\rangle |\phi_0^i\rangle \implies \eta \eta^\dagger = 1 - \hat{n}_{q\beta} \end{aligned} \quad (1.21)$$

and hence the anticommutator

$$\implies \{\eta, \eta^\dagger\} = 1 \quad (1.22)$$

Note that the three equations in 1.21 work only when applied on the eigenstate $|\psi_i\rangle$ and not any arbitrary state.

$$\begin{aligned} \eta^\dagger \eta |\psi_i\rangle &= |1\rangle |\phi_1^i\rangle = \hat{n}_{q\beta} |\psi_i\rangle \\ \eta \eta^\dagger |\psi_i\rangle &= |0\rangle |\phi_0^i\rangle = (1 - \hat{n}_{q\beta}) |\psi_i\rangle \\ \{\eta^\dagger, \eta\} |\psi_i\rangle &= |\psi_i\rangle \end{aligned}$$

Form of the unitary operators

Although we have found the correct similarity transformations U_i (eqs. 1.19), we need to convert them into a unitary transformation. Say we are trying to rotate the eigenstate $|\psi_1\rangle$ into the state $|1\rangle$. We can then work with the transformation

$$U_1 = 1 + \eta \quad (1.23)$$

In this form, this transformation is not unitary. It can however be written in an exponential form:

$$U_1 = e^\eta \quad (1.24)$$

using the fact that $\eta^2 = 0$. It is shown in ref. [366] that corresponding to a similarity transformation e^ω , there exists a unitary transformation e^G where

$$G = \tanh^{-1} (\omega - \omega^\dagger) \quad (1.25)$$

Applying that to the problem at hand gives

$$U_1^\dagger = \exp \left\{ \tanh^{-1} (\eta - \eta^\dagger) \right\} \quad (1.26)$$

Let $x = \tanh y$. Then,

$$x = \frac{e^{2y} + 1}{e^{2y} - 1} \implies y = \frac{1}{2} \log \frac{1+x}{1-x} \implies e^y = e^{\tanh^{-1} x} = \sqrt{\frac{1+x}{1-x}} \quad (1.27)$$

Therefore,

$$\exp \left\{ \tanh^{-1} (\eta - \eta^\dagger) \right\} = \frac{1 + \eta - \eta^\dagger}{\sqrt{(1 + \eta^\dagger - \eta)(1 - \eta^\dagger + \eta)}} = \frac{1 + \eta - \eta^\dagger}{\sqrt{1 + \{\eta, \eta^\dagger\}}} = \frac{1}{\sqrt{2}} (1 + \eta - \eta^\dagger) \quad (1.28)$$

The *unitary* operator that transforms the entangled eigenstate $|\psi_1\rangle$ to the state $|1\rangle$ is thus

$$U_1 = \frac{1}{\sqrt{2}} (1 + \eta^\dagger - \eta) \quad (1.29)$$

It can also be written as $\exp \left\{ \frac{\pi}{4} (\eta^\dagger - \eta) \right\}$ because

$$\begin{aligned} \exp \left\{ \frac{\pi}{4} (\eta^\dagger - \eta) \right\} &= 1 + (\eta^\dagger - \eta) \frac{\pi}{4} + \frac{1}{2!} (\eta^\dagger - \eta)^2 \left(\frac{\pi}{4} \right)^2 + \frac{1}{3!} (\eta^\dagger - \eta)^3 \left(\frac{\pi}{4} \right)^3 + \dots \\ &= 1 + (\eta^\dagger - \eta) \frac{\pi}{4} - \frac{1}{2!} \left(\frac{\pi}{4} \right)^2 - \frac{1}{3!} (\eta^\dagger - \eta) \left(\frac{\pi}{4} \right)^3 + \frac{1}{4!} \left(\frac{\pi}{4} \right)^4 + \dots \\ &= \cos \frac{\pi}{4} + (\eta^\dagger - \eta) \sin \frac{\pi}{4} \\ &= \frac{1}{\sqrt{2}} (1 + \eta^\dagger - \eta) \end{aligned} \quad (1.30)$$

There we used

$$(\eta^\dagger - \eta)^2 = \eta^{\dagger 2} + \eta^2 - \{\eta^\dagger, \eta\} = -1 \quad \left[\because \eta^2 = \eta^{\dagger 2} = 0 \right] \quad (1.31)$$

and hence

$$(\eta^\dagger - \eta)^3 = -1 (\eta^\dagger - \eta) \quad (1.32)$$

and so on.

Effective Hamiltonian

We can now compute the form of the effective Hamiltonian that comes about when we apply U_1 - that is - when we rotate one exact eigenstate $|\psi_1\rangle$ into the occupied Fock space basis $|1\rangle$. From eq. 1.29,

$$\begin{aligned}
 U_1 \mathcal{H} U_1^\dagger &= \frac{1}{2} (1 + \eta^\dagger - \eta) \mathcal{H} (1 + \eta - \eta^\dagger) \\
 &= \frac{1}{2} (1 + \eta^\dagger - \eta) (\mathcal{H} + \mathcal{H}\eta - \mathcal{H}\eta^\dagger) \\
 &= \frac{1}{2} (\mathcal{H} + \mathcal{H}\eta - \mathcal{H}\eta^\dagger + \eta^\dagger \mathcal{H} + \eta^\dagger \mathcal{H}\eta - \eta^\dagger \mathcal{H}\eta^\dagger - \eta \mathcal{H} - \eta \mathcal{H}\eta + \eta \mathcal{H}\eta^\dagger) \\
 &= \frac{1}{2} (\mathcal{H}^d + \mathcal{H}^i + \mathcal{H}^l + \mathcal{H}\eta - \mathcal{H}\eta^\dagger + \eta^\dagger \mathcal{H} + \eta^\dagger \mathcal{H}\eta - \eta^\dagger \mathcal{H}\eta^\dagger - \eta \mathcal{H} - \eta \mathcal{H}\eta + \eta \mathcal{H}\eta^\dagger) \\
 &= \frac{1}{2} (\mathcal{H}^d + \mathcal{H}^i + \mathcal{H}^l + [\eta^\dagger - \eta, \mathcal{H}] + \eta^\dagger \mathcal{H}\eta - \eta^\dagger \mathcal{H}\eta^\dagger - \eta \mathcal{H}\eta + \eta \mathcal{H}\eta^\dagger)
 \end{aligned} \tag{1.33}$$

In the last two lines, we expanded the Hamiltonian into the three parts $\mathcal{H}^d$, $\mathcal{H}^i$ and a third piece $\mathcal{H}^l \equiv c_{q\beta}^\dagger T + T^\dagger c_{q\beta}$.

For reasons that will become apparent, we will split the terms into two groups:

$$\tilde{\mathcal{H}} = \frac{1}{2} \left(\underbrace{\mathcal{H}^d + \mathcal{H}^i + [\eta^\dagger - \eta, \mathcal{H}] + \eta^\dagger \mathcal{H}\eta + \eta \mathcal{H}\eta^\dagger}_{\text{group 1}} + \overbrace{\mathcal{H}^l - \eta^\dagger \mathcal{H}\eta^\dagger - \eta \mathcal{H}\eta}^{\text{group 2}} \right) \tag{1.34}$$

Group 2 can be easily shown to be 0. Note that terms that have two η or two $\eta^\dagger$ sandwiching a $\mathcal{H}$ can only be nonzero if the intervening $\mathcal{H}$ has an odd number of creation or destruction operators.

$$\eta \mathcal{H} \eta = \eta c_q^\dagger T \eta \tag{1.35}$$

and

$$\eta^\dagger \mathcal{H} \eta^\dagger = \eta^\dagger T^\dagger c_q \eta^\dagger \tag{1.36}$$

Group 2 becomes

$$\text{group 2} = \mathcal{H}^l - \eta^\dagger T^\dagger c_q \eta^\dagger - \eta c_q^\dagger T \eta = c_q^\dagger T + T^\dagger c_q - \eta^\dagger T^\dagger c_q \eta^\dagger - \eta c_q^\dagger T \eta \tag{1.37}$$

To simplify this, we use the relation

$$\begin{aligned}
 \eta c_q^\dagger T \eta &= \frac{1}{\omega_i^0 - \mathcal{H}^d} T^\dagger c_q c_q^\dagger T \eta \\
 &= T^\dagger c_q \frac{1}{\omega_i^1 - \mathcal{H}^d} c_q^\dagger T \eta \quad [\text{eq. 1.16}] \\
 &= T^\dagger c_q \eta^\dagger \eta \quad [\text{eq. 1.15}] \\
 &= T^\dagger c_q \hat{n}_q \quad [\text{eq. 1.21}]
 \end{aligned} \tag{1.38}$$

which gives

$$\eta c_q^\dagger T \eta = T^\dagger c_q \quad (1.39)$$

Taking the Hermitian conjugate of eq. 1.39 gives

$$\eta^\dagger T^\dagger c_q \eta^\dagger = c_q^\dagger T \quad (1.40)$$

Substituting the expressions 1.39 and 1.40 into the expression for group 2, 1.37, shows that it vanishes. This leaves us only with group 1:

$$\tilde{\mathcal{H}} = \frac{1}{2} \left(\mathcal{H}^d + \mathcal{H}^i + \overbrace{\eta^\dagger \mathcal{H} \eta + \eta \mathcal{H} \eta^\dagger}^{\text{group A}} + \underbrace{[\eta^\dagger - \eta, \mathcal{H}]}_{\text{group B}} \right) \quad (1.41)$$

Group A simplifies in the following way. First note that $\eta^\dagger \mathcal{H}^I \eta = \eta^\dagger \mathcal{H}^I \eta = 0$ must be 0 because it will involve consecutive $c_{q\beta}$ or consecutive $c_{q\beta}^\dagger$. We are therefore left with the diagonal part of $\mathcal{H}$, which is $H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta})$.

$$\eta^\dagger [H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta})] \eta + \eta [H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta})] \eta^\dagger = \eta^\dagger H_h \eta + \eta H_e \eta^\dagger \quad (1.42)$$

This can be shown to be equal to the diagonal part:

$$\text{group A} = \eta^\dagger H_h \eta + \eta H_e \eta^\dagger = H_e \hat{n}_{q\beta} + H_h (1 - \hat{n}_{q\beta}) = \mathcal{H}^d + \mathcal{H}^i \quad (1.43)$$

It can also be shown that

$$\text{group B} = [\eta^\dagger - \eta, \mathcal{H}] = 2 [c_{q\beta}^\dagger T, \eta] \quad (1.44)$$

Putting it all together,

$$\tilde{\mathcal{H}} = \mathcal{H}^d + \mathcal{H}^i + [c_{q\beta}^\dagger T, \eta] \quad (1.45)$$

The renormalizing in the Hamiltonian is

$$\Delta \mathcal{H} = \tilde{\mathcal{H}} - \mathcal{H}^d - \mathcal{H}^i = [c_{q\beta}^\dagger T, \eta] \quad (1.46)$$

Because of eq. 1.44, it can also be written as

$$\Delta \mathcal{H} = \frac{1}{2} [\eta^\dagger - \eta, \mathcal{H}_X] = \frac{1}{2} [\eta^\dagger - \eta, \mathcal{H}] \quad (1.47)$$

This form will be useful later when we make the connection with one-shot Schrieffer-Wolff transformation and CUT RG.

To check that the renormalised Hamiltonian indeed commutes with $\hat{n}_{q\beta}$,

$$\begin{aligned} [\tilde{\mathcal{H}}, \hat{n}_{q\beta}] &= [[c_{q\beta}^\dagger T, \eta], \hat{n}_{q\beta}] = [c_{q\beta}^\dagger T \eta, \hat{n}_{q\beta}] - [\eta c_{q\beta}^\dagger T, \hat{n}_{q\beta}] \\ &= c_{q\beta}^\dagger T \eta \hat{n}_{q\beta} - \hat{n}_{q\beta} c_{q\beta}^\dagger T \eta \quad \left[2^{\text{nd}} [\cdot] \text{ is 0, } \because c_{q\beta}^\dagger \hat{n}_{q\beta} = \hat{n}_{q\beta} \eta = 0 \right] \\ &= c_{q\beta}^\dagger T \eta - c_{q\beta}^\dagger T \eta = 0 \end{aligned} \quad (1.48)$$

Fixed point condition

Within the URG, it is a prescription that the fixed point is reached when the denominator of the RG equation vanishes. This is equivalent to either $\omega_i^1 = \mathcal{H}_1^d$ or $\omega_i^0 = \mathcal{H}_0^d$. This shows that at the fixed point, one of the eigenvalues of $\hat{\omega}_i$ matches the corresponding eigenvalue of the diagonal blocks. This also leads to the vanishing of the off-diagonal block, because eqs. 1.12 and 1.13 gives

$$c_{q\beta}^\dagger T |0\rangle |\phi_0^i\rangle = \left(\omega_i^1 - \mathcal{H}_1^d \right) |1\rangle |\phi_1^i\rangle = 0 \implies c_{q\beta}^\dagger T = 0 \quad (1.49)$$

Multiple off-diagonal terms

There is a subtle assumption in the definitions eq. 1.14. In order for η to be the Hermitian conjugate of $\eta^\dagger$, $\mathcal{H}_d$ cannot have any information that relates to the structure of T . To see why, say the total off-diagonal term is composed of two parts: $T = T_1 + T_2$.

$$\begin{aligned} \eta &= \frac{1}{\omega_0 - \mathcal{H}_d} \left(T_1^\dagger + T_2^\dagger \right) c = \left[\frac{1}{\omega^0 - E_1^0} T_1^\dagger c + \frac{1}{\omega^0 - E_2^0} T_2^\dagger c \right] \\ \eta^\dagger &= \frac{1}{\omega^1 - \mathcal{H}_d} c^\dagger (T_1 + T_2) = \left[\frac{1}{\omega^1 - E_1^1} c^\dagger T_1 + \frac{1}{\omega^1 - E_2^1} c^\dagger T_2 \right] \end{aligned} \quad (1.50)$$

where $\mathcal{H}_d T_i^\dagger c = E_i^0 T_i^\dagger c$ and $\mathcal{H}_d c^\dagger T_i = E_i^1 c^\dagger T_i$. We can now see that in order for $\eta = (\eta^\dagger)^\dagger$ to hold, two conditions must be met:

$$\omega^0 - E_1^0 = \omega^1 - E_1^1, \quad \omega^0 - E_2^0 = \omega^1 - E_2^1 \quad (1.51)$$

This will not hold generally. The correct solution is to realize that each such off-diagonal term T_i will come with its own quantum fluctuation scale ω_i .

$$\begin{aligned} \eta &= \sum_i \frac{1}{\omega_i^0 - E_i^0} T_i^\dagger c \\ \eta^\dagger &= \sum_i \frac{1}{\omega_i^1 - E_i^1} c^\dagger T_i \end{aligned} \quad (1.52)$$

If we now impose the condition that $\eta = (\eta^\dagger)^\dagger$, we get the relations

$$\omega_i^0 - \omega_i^1 = E_i^0 - E_i^1 \quad (1.53)$$

and so

$$\eta^\dagger - \eta = \sum_i \frac{1}{\omega_i^0 - E_i^0} \left(c^\dagger T_i - T_i^\dagger c \right) \quad (1.54)$$

The expression for the renormalization will not be just $[c^\dagger T, \eta]$ in this case. That form will be non-Hermitian. The correct form is obtained from the more general form $[\eta^\dagger - \eta, \mathcal{H}_X]$:

$$\begin{aligned}\Delta\mathcal{H} &= \frac{1}{2} [\eta^\dagger - \eta, c^\dagger T + T^\dagger c] = \frac{1}{2} \sum_{ij} \frac{1}{\omega_i^0 - E_i^0} [c^\dagger T_i - T_i^\dagger c, c^\dagger T_j + T_j^\dagger c] \\ &= \frac{1}{2} \sum_{ij} \frac{1}{\omega_i^0 - E_i^0} \left[\hat{n} (T_i T_j^\dagger + T_j T_i^\dagger) - (1 - \hat{n}) (T_i^\dagger T_j + T_j^\dagger T_i) \right] \\ &= \frac{1}{2} \sum_{ij} \left(\frac{1}{\omega_i^0 - E_i^0} + \frac{1}{\omega_j^0 - E_j^0} \right) [\hat{n} T_i T_j^\dagger - (1 - \hat{n}) T_i^\dagger T_j]\end{aligned}\tag{1.55}$$

Prescription

Given a Hamiltonian

$$\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_0 + c^\dagger T + T^\dagger c\tag{1.56}$$

the goal is to look at the renormalization of the various couplings in the Hamiltonian as we decouple high energy electron states. Typically we have a shell of electrons at some energy D . During the process, we make one simplification. We assume that there is only one electron on that shell at a time, say with quantum numbers q, σ , and calculate the renormalization of the various couplings due to this electron. We then sum the momentum q over the shell and the spin β , and this gives the total renormalization due to decoupling the entire shell.

From eq. 1.45, the first two terms in the rotated Hamiltonian are just the diagonal parts of the bare Hamiltonian; they are unchanged in that part. The renormalization comes from the third term. For one electron $q\beta$ on the shell, the renormalization is

$$\Delta\mathcal{H} = [c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}), \eta] = c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \eta - \eta c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta})\tag{1.57}$$

Since this assumes we have obtained this from U_1 , it is fair to tag the η with a suitable label:

$$\Delta\mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \eta_1 - \eta_1 c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta})\tag{1.58}$$

It is clear that the first term takes into account virtual excitations that start from a filled state ($\hat{n}_{q\beta} = 1$ initially) - such a term is said to be a part of the *particle sector*.

$$\Delta_1\mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \eta_1\tag{1.59}$$

The second term, on the other hand, considers excitations that start from an empty state. They constitute the *hole sector*.

$$\Delta_0\mathcal{H} = -\eta_1 c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta})\tag{1.60}$$

To write the total renormalization in a particle-hole symmetric form, we can use the relation $\eta_0 = -\eta_1$, such that both the terms will now come with a positive sign:

$$\Delta\mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta}) \eta_1 + \eta_0 c_{q\beta}^\dagger \text{Tr}(\mathcal{H} c_{q\beta})\tag{1.61}$$

We can make one more manipulation: using eq. 1.16, we get

$$\Delta\mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}) \eta_1 + \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} \eta_0^\dagger \quad (1.62)$$

This form of the total renormalization is identical to the one we use in the "Poor Man's scaling"-type of renormalization that was used to get the scaling equations in the Kondo and Anderson models [367, 368]. Writing down the forms of η and $\eta^\dagger$ explicitly, we get

$$\Delta\mathcal{H} = c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}) \frac{1}{\omega_1^0 - \mathcal{H}_0^d} \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} + \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} \frac{1}{\omega_0^1 - \mathcal{H}_1^d} c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}) \quad (1.63)$$

The renormalization due to the entire shell is obtained by summing over all states on the shell.

$$\Delta\mathcal{H} = \sum_{q\beta} \left[c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}) \frac{1}{\omega_1^0 - \mathcal{H}_0^d} \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} + \text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta} \frac{1}{\omega_0^1 - \mathcal{H}_1^d} c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta}) \right] \quad (1.64)$$

These equations will now need to be simplified. For example, in the particle sector, we can set $\hat{n}_{q\beta} = 0$ in the numerator, because there is no such excitation in the initial state. Similarly, in the hole sector, we can set $\hat{n}_{q\beta} = 1$ because that state was occupied in the initial state. Another simplification we typically employ is that $\mathcal{H}_{0,1}^d$ will, in general, have the energies of all the electrons. But we consider only the energy of the on-shell electrons in the denominator. After integrating out these electrons, we can rearrange the remaining operators to determine which term in the Hamiltonian it renormalizes and what is the renormalization.

At first sight, one might think that we must evaluate lots of traces to obtain the terms in $\Delta\mathcal{H}$. A little thought reveals that the terms in the numerator are simply the off-diagonal terms in the Hamiltonian; $\text{Tr}(c_{q\beta}^\dagger \mathcal{H}) c_{q\beta}$ is the off-diagonal term that has $c_{q\beta}$ in it, and $c_{q\beta}^\dagger \text{Tr}(\mathcal{H}c_{q\beta})$ is the off-diagonal term that has $c_{q\beta}^\dagger$ in it. $\mathcal{H}^D$ is just the diagonal part of the Hamiltonian.

2.2 Example I: URG analysis of the star graph model

The star graph problem has already been analyzed using URG and an extensive study of its entanglement properties has already been carried out, in ref. [369]. Here we focus on just deriving the RG equations. The system consists of N spin-like degrees of freedom (labeled 1 through N) individually talking to a spin at the center (labeled 0). Each spin i ($i \in [0, N]$) has an on-site energy ϵ_i . The coupling strength between 0 and i ($i \in [1, N]$) is J_i . We choose the on-site energies such that $\epsilon_{i+1} > \epsilon_i, i \in [N-1, 1]$. In this way, ϵ_1 is the infrared limit and ϵ_N is the ultraviolet limit.

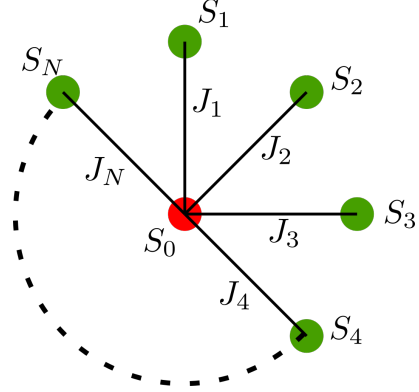


Figure 2.2: Star Graph model

$$\mathcal{H} = \epsilon_0 S_0^z + \sum_{i=1}^N \left[\epsilon_i S_i^z + J_i \vec{S}_0 \cdot \vec{S}_i \right] \quad (2.1)$$

By converting the last term into S^z and $S^\pm$, we can write the Hamiltonian as

$$\mathcal{H} = \epsilon_0 S_0^z + \sum_{i=1}^N \left[\epsilon_i S_i^z + J_i \left(S_0^z S_i^z + \frac{1}{2} (S_0^+ S_i^- + S_0^- S_i^+) \right) \right] \quad (2.2)$$

Calculation of Renormalization

The RG involves decoupling the nodes N through 1, and looking at the resultant renormalization in ϵ_i and J_i . As a simplification, we will ignore the lower nodes in the denominator and keep only the node currently being decoupled, ie node N . Since node 0 is connected to node N , we will keep node 0 in the denominator as well. Making this simplification gives

$$\mathcal{H}^D = \epsilon_0 S_0^z + \epsilon_N S_N^z + J_N S_0^z S_N^z \quad (2.3)$$

The off-diagonal part in the subspace of the node N is

$$\mathcal{H}_X = S_N^+ T + T^\dagger S_N^- = \frac{1}{2} J_N (S_N^+ S_0^- + S_N^- S_0^+) \quad (2.4)$$

The renormalization on doing one step of the URG is given by

$$\begin{aligned} \Delta \mathcal{H} &= S_N^+ T \frac{1}{\omega_0^1 - \mathcal{H}^D} T^\dagger S_N^- + T^\dagger S_N^- \frac{1}{\omega_1^0 - \mathcal{H}^D} S_N^+ T \\ &= \frac{J_N^2}{4} S_N^+ S_0^- \frac{1}{\omega_0^1 - \epsilon_0 S_0^z - \epsilon_N S_N^z - J_N S_0^z S_N^z} S_0^+ S_N^- + \frac{J_N^2}{4} S_0^+ S_N^- \frac{1}{\omega_1^0 - \epsilon_0 S_0^z - \epsilon_N S_N^z - J_N S_0^z S_N^z} S_N^+ S_0^- \end{aligned} \quad (2.5)$$

There, N refers to the spin being decoupled. The first Greens function has S_0^+ and S_N^- in front of it, so we substitute $S_0^z = \frac{1}{2}$, $S_N^z = -\frac{1}{2}$ in that Greens function. For the other Greens function we do the opposite.

$$\Delta\mathcal{H} = \frac{J_N^2}{4} S_N^+ S_0^- \frac{1}{\omega_1^0 - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} S_0^+ S_N^- + \frac{J_N^2}{4} S_0^+ S_N^- \frac{1}{\omega_0^1 + \frac{1}{2}\epsilon_0 - \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} S_N^+ S_0^- \quad (2.6)$$

To relate ω_1^0 and ω_0^1 , we use eq. ??:

$$\omega_1^0 + \omega_0^1 = \mathcal{H}_0^D + \mathcal{H}_1^D = -\frac{1}{2}J_N \implies \omega_1^0 \equiv \omega, \quad \omega_0^1 \equiv \omega' = -\frac{1}{2}J_N - \omega \quad (2.7)$$

So, the renormalization becomes

$$\Delta\mathcal{H} = \frac{J_N^2}{4} \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} (S_N^+ S_0^- S_0^+ S_N^- - S_0^+ S_N^- S_N^+ S_0^-) = \frac{J_N^2}{4} \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} (S_N^z - S_0^z) \quad (2.8)$$

There we used $S^+ S^- = \frac{1}{2} + S^z$ and $S^- S^+ = \frac{1}{2} - S^z$.

We can now read off the renormalizations in ϵ_N and ϵ_0 .

$$\begin{aligned} \Delta\epsilon_N &= \frac{1}{4} J_N^2 \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} \\ \Delta\epsilon_0 &= -\frac{1}{4} J_N^2 \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} \end{aligned} \quad (2.9)$$

Nature of flows

We are interested in looking at the renormalization of the central node energy ϵ_0 , upon removing the nodes N through 1. We will hence concentrate on the second RG equation. We first make some simplifying assumptions: $J_i = J$, $\epsilon_i = \epsilon$ for all $i \in \{1, N\}$.

$$\Delta\epsilon_0 = -\frac{1}{4} J^2 \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon + \frac{1}{4}J} \quad (2.10)$$

Define $\tilde{\omega} = \omega + \frac{1}{2}\epsilon + \frac{1}{4}J$.

$$\Delta\epsilon_0 = -\frac{1}{4} J^2 \frac{1}{\tilde{\omega} - \frac{1}{2}\epsilon_0} \quad (2.11)$$

Our goal here is to look for a fixed-point condition such that the denominator vanishes at some point of the RG. If we start with a bare of ϵ_0 such that $\tilde{\omega} - \frac{1}{2}\epsilon_0 > 0$, the denominator will be positive and the RG equation will be irrelevant. This means that ϵ_0 will keep on decreasing, and the denominator will keep on becoming more and more positive, meaning there cannot be a fixed point in this situation.

If, on other hand, we start with a bare of ϵ_0 such that $\tilde{\omega} - \frac{1}{2}\epsilon_0 < 0$, the denominator will be negative and the RG equation will be relevant. This means that ϵ_0 will keep on increasing, and the

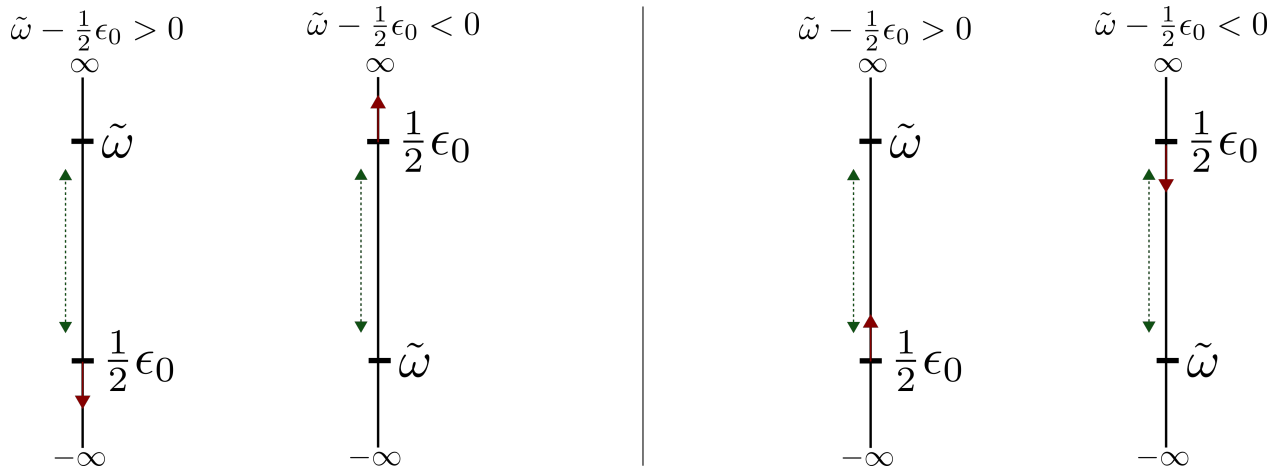


Figure 2.3: Left: RG flow for the two cases. The green line is the distance between the bare values of the two couplings, and hence also the magnitude of the denominator. The red arrow denotes the direction in which ϵ_0 will flow. Upward flow is increase. In both cases, the flow is such that the distance between the two quantities (and hence the magnitude of the denominator) increases. The RG fixed point occurs when the magnitude of the denominator goes to 0. This happens if the distance vanishes. Since the distance necessarily increases, we cannot get a fixed point in this way. Right: RG flow for the two cases with the new $-\tilde{\omega} = \omega' - \frac{1}{2}\epsilon + \frac{1}{4}J$. Now we can see that in both cases, the flow is such that the distance (green dotted line) between the couplings decreases. A fixed point is reached when this distance vanishes.

denominator will keep on becoming more and more negative, meaning there cannot be a fixed point in this situation either. These situations are depicted in figure 2.3.

Since we cannot find a fixed point, we will use a different ω . Instead of ω_1^0 , we will use ω_1^1 . From eq. ??, we have

$$\omega_1^0 - \omega_1^1 = \mathcal{H}_0^D - \mathcal{H}_1^D = \epsilon_0 - \epsilon_N = \epsilon_0 - \epsilon \quad (2.12)$$

Defining $\omega_1^1 = \omega'$ and substituting this in eq. 2.10 gives

$$\Delta\epsilon_0 = -\frac{1}{4}J^2 \frac{1}{\omega' - \frac{1}{2}\epsilon + \frac{1}{2}\epsilon_0 + \frac{1}{4}J} \quad (2.13)$$

We again define $-\tilde{\omega} = \omega' - \frac{1}{2}\epsilon + \frac{1}{4}J$.

$$\Delta\epsilon_0 = \frac{1}{4}J^2 \frac{1}{\tilde{\omega} - \frac{1}{2}\epsilon_0} \quad (2.14)$$

We now repeat the exercise of determining the relevance of the flows under various regime. If we start with a bare ϵ_0 such that $\tilde{\omega} + \frac{1}{2}\epsilon_0 > 0$, then the denominator is positive so the renormalization will be irrelevant. ϵ_0 will decrease until we reach $\tilde{\omega} + \frac{1}{2}\epsilon_0 = 0$. This will be a fixed point. However, if we start with a bare ϵ_0 such that $\tilde{\omega} + \frac{1}{2}\epsilon_0 < 0$, then the denominator is negative so the renormalization will be relevant. ϵ_0 will increase until we reach $\tilde{\omega} + \frac{1}{2}\epsilon_0 = 0$. This will again be a fixed point. This new situation is depicted in right panel of figure 2.3.

Effective Hamiltonians

If $\tilde{\omega}$ and ϵ_0 are of the same sign at the bare level, then it is easy to see that since the fixed point is defined by $\tilde{\omega} = \frac{1}{2}\epsilon_0^*$ (* denotes value at fixed point), the effective Hamiltonian at the fixed point will be

$$\mathcal{H}^* = 2\tilde{\omega}S_0^z + \epsilon \sum_i S_i^z + J \sum_i \vec{S}_i \cdot \vec{S}_0, \quad \text{if } \tilde{\omega}\epsilon_0 > 0 \quad (2.15)$$

If, at the bare level, ϵ_0 and $\tilde{\omega}$ are of opposite signs, then ϵ_0 would undergo a change in sign at some point as it flows towards $\tilde{\omega}$. Since we do not expect a coupling to change sign under RG, we will restrict it to 0 in such cases.

$$\mathcal{H}^* = \epsilon \sum_i S_i^z + J \sum_i \vec{S}_i \cdot \vec{S}_0, \quad \text{if } \tilde{\omega}\epsilon_0 < 0 \quad (2.16)$$

Things get much more simpler if we assume the onsite energies of the surrounding nodes are zero.

$$\begin{aligned} \mathcal{H}^* &= 2\tilde{\omega}S_0^z + J \sum_i \vec{S}_i \cdot \vec{S}_0, & \text{if } \tilde{\omega}\epsilon_0 > 0 \\ \mathcal{H}^* &= J \sum_i \vec{S}_i \cdot \vec{S}_0, & \text{if } \tilde{\omega}\epsilon_0 < 0 \end{aligned} \quad (2.17)$$

Fixed points

The fixed points are obtained numerically by solving the RG equation. As mentioned before, there are two types of solutions: The first kind is those in which ϵ_0 and $\tilde{\omega}$ are of the same sign, and the former flows to the latter without crossing the 0 axis. These flows are shown (obtained numerically) in fig. 2.4. The second kind are those where the two couplings have different signs, and so ϵ_0 flows to 0. These are shown in fig. 2.5.

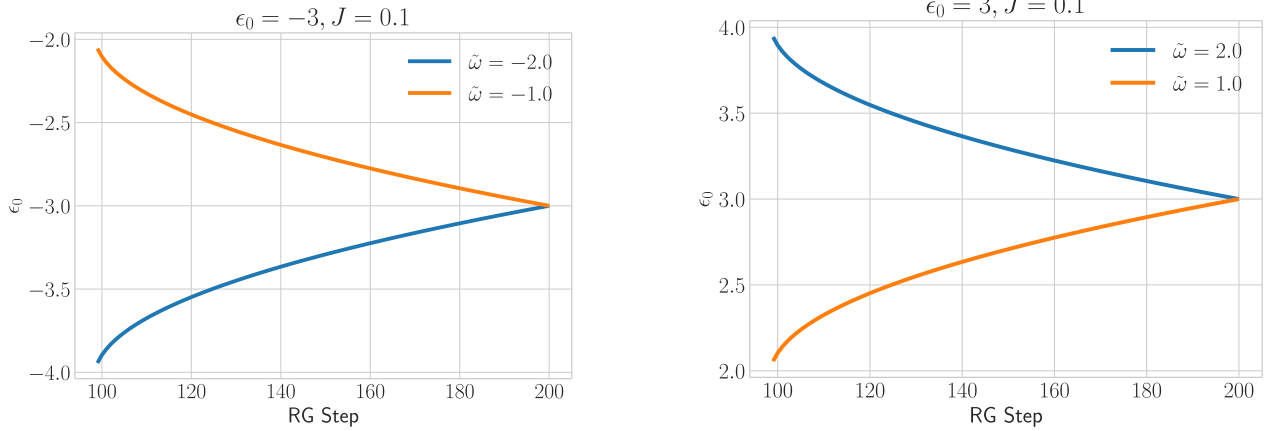


Figure 2.4: Flows where ϵ_0 and $\tilde{\omega}$ have same sign. The left and right panels show flows starting from negative and positive values respectively. The two plots in each panel correspond to different values of $\tilde{\omega}$, one greater than the bare ϵ_0 , the other less than that. The fixed point value is $2\tilde{\omega}$.

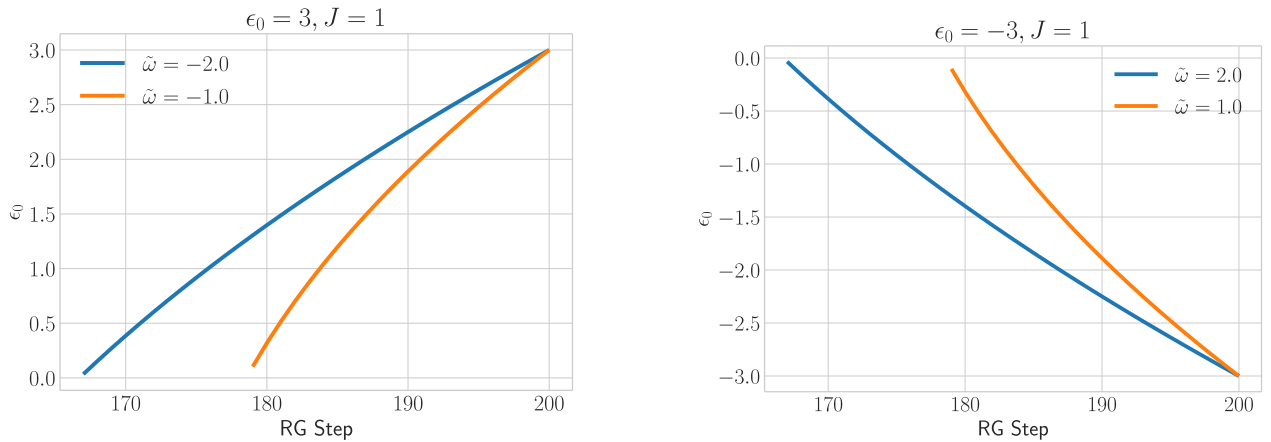


Figure 2.5: Flows where ϵ_0 and $\tilde{\omega}$ have opposite sign. The left and right panels show flows starting from negative and positive values respectively. The two plots in each panel correspond to different values of $\tilde{\omega}$, one greater than the bare ϵ_0 , the other less than that. The fixed point value is 0.

2.3 Example II: URG analysis of the two-channel Kondo problem

The Hamiltonian for the channel-isotropic MCK model is:

$$H = \sum_l \left[\sum_{\substack{k \\ \alpha=\uparrow,\downarrow}} \epsilon_{k,l} \hat{n}_{k\alpha,l} + \frac{\mathcal{J}}{2} \sum_{\substack{kk' \\ \alpha,\alpha'=\uparrow,\downarrow}} \vec{S}_d \cdot \vec{\sigma}_{\alpha\alpha'} c_{k\alpha,l}^\dagger c_{k'\alpha',l} \right]. \quad (3.1)$$

The URG equation for the single-channel Kondo model [?] shows a stable strong coupling fixed point. Ferromagnetic interactions are irrelevant. Strictly speaking, that RG equation already encodes, in principle, the multi-channel behaviour, through a modified $\hat{\omega}$. To extract this information, we consider the strong coupling fixed-point $J \gg D$ as a fixed point and analyze its stability from the star graph perspective. For the exactly-screened case, the star graph decouples from the conduction bath, leaving behind a local Fermi liquid interaction on the first site. Similarly, in the under-screened regime, the ground state is composed of states where the impurity spin is only partially screened by the conduction channels. If a particular configuration of the bath-impurity system has the total conduction bath spin down, the impurity will have a residual up spin. This induces a ferromagnetic super-exchange coupling that is irrelevant under RG, so this fixed point is stable as well.

We now come to the over-screened case, where there is a residual spin on the conduction channel site. The neighbouring electrons will now hop in with spins opposite to that of the impurity, so an antiferromagnetic interaction will be induced, and such an interaction is relevant under the RG. This shows that the over-screened regime cannot have a stable strong coupling fixed point, and we need to search for an intermediate coupling fixed point. We therefore need the generator of the unitary transformation that incorporates third order scattering scatterings explicitly. We should take account of all possible processes that render the set of states $\{|\hat{n}_{q\beta} = 1\rangle, |\hat{n}_{q\beta} = 0\rangle\}$ diagonal. The higher order generator itself has two scattering processes, such that the entire renormalisation term $c_{q\beta}^\dagger T \eta$ has in total three coherent processes. The complete generator up to third order can be written as

$$\eta = \frac{1}{\hat{\omega} - H_D} T^\dagger c \simeq \frac{1}{\omega' - H_D} T^\dagger c + \frac{1}{\omega' - H_D} H_X \frac{1}{\omega' - H_D} T^\dagger c + \frac{1}{\omega' - H_D} T^\dagger c \frac{1}{\omega' - H_D} H_X, \quad (3.2)$$

where $H_X = J \sum_{k,k' < \Lambda_j, \alpha, \alpha'} \vec{S}_d \cdot \vec{\sigma}_{\alpha\alpha'} c_{k\alpha}^\dagger c_{k'\alpha'}$ is scattering between the entangled electrons. There are two third order terms in the above equation corresponding to the two possible sequences in which the processes can occur while keeping the total renormalisation $c_{q\beta}^\dagger T \eta$ diagonal in $q\beta$. The second order processes remain unchanged. The total renormalisation takes the form:

$$\begin{aligned} \Delta H_{(j)} = & \underbrace{c^\dagger T \frac{1}{\omega' - H_D} T^\dagger c + (c^\dagger \leftrightarrow c)}_{\Delta H_{(j)}^{(2)}} \\ & + \underbrace{c^\dagger T \frac{1}{\omega' - H_D} H_X \frac{1}{\omega' - H_D} T^\dagger c + c^\dagger T \frac{1}{\omega' - H_D} T^\dagger c \frac{1}{\omega' - H_D} H_X + (c^\dagger \leftrightarrow c)}_{\Delta H_{(j)}^{(3)}}. \end{aligned} \quad (3.3)$$

$\Delta H_{(j)}^{(2)}$ and $\Delta H_{(j)}^{(3)}$ are the renormalisation arising from the second and third order processes respectively.

It is easier to see the RG flow of the couplings if we write the Hamiltonian in terms of the eigenstates of S_d^z . These eigenstates are defined by $S_d^z |m_d\rangle = m_d |m_d\rangle$, $m_d \in [-S, S]$. In terms of these eigenstates, the Hamiltonian becomes

$$\mathcal{H} = \sum_{k\sigma} \epsilon_k \tau_{k\sigma} + \sum_{m_d=-S}^S \sum_{\substack{kl, \\ \sigma=\uparrow, \downarrow}} J_{m_d}^\sigma |m_d\rangle \langle m_d| c_{k\sigma}^\dagger c_{l\sigma} + \sum_{kl} \sum_{m_d=-S}^{S-1} J_{m_d}^t (|m_d+1\rangle \langle m_d| s_{kl}^- + \text{h.c.}) \quad (3.4)$$

where k, l sum over the momentum states, σ sums over the spin indices, $J_m^\sigma = \frac{1}{2}\sigma m J$ in the UV Hamiltonian, and $J_m^t = J \frac{1}{2} \sqrt{S(S+1) - m(m+1)}$ is the coupling that connects $|m\rangle$ and $|m+1\rangle$. We first calculate $\Delta H_{(j)}^{(2)}$. There will be two types of processes - those processes that start from an occupied state (particle sector) and those that start from a vacant state (hole sector). Due to particle hole symmetry of the Hamiltonian, they will be equal to each other and we will only calculate the particle sector contribution.

In the particle sector, we have ($\hat{n}_{q\beta} = 1$), so we will work at a negative energy shell $\epsilon_q = -D$. The renormalisation can schematically represented as $H_0^I \frac{1}{\omega - H_{q\beta}^D} H_1^I$. Both H_0^I and H_1^I have all three operators $S_d^z, S_d^\pm$. We first consider specifically the case of spin- $\frac{1}{2}$ impurity. Those terms that have identical operators on both sides can be ignored because $S_d^{z2} = \text{constant}$ and $S^\pm{}^2 = 0$. All the six terms that *will* renormalise the Hamiltonian have a spin flip operator on at least one side of the Greens function. This means that in the denominator of the Greens function, S_d^z and s_{qq}^z have to be anti-parallel in order to produce a non-zero result for that term. This means we can identically replace $S_d^z s_{qq}^z = -\frac{1}{4}$. Also, in the particle sector, the Greens function always has $c_{q\beta}$ in front of it, so $\epsilon_q \tau_{q\beta} = \frac{D}{2}$. The upshot of all this is that the denominator of all scattering processes for the spin- $\frac{1}{2}$ impurity Hamiltonian will be $\omega - \frac{D}{2} + \frac{J}{4}$.

We now come to the general case of spin- S impurity. The various terms that renormalise the Hamiltonian can be described in terms of the bath spin operators that come into them. For example, the term that has s^z on both sides of the intervening Greens function can be represented as $z|z$. There are 7 such terms: $z|z, \pm|\mp, z|\pm, \pm|z$. Each of these terms occurs both in the particle and the hole sectors. We will demonstrate the calculation of two of these terms. The $z|z + | -$ terms evaluate in the following manner.

$$\begin{aligned} z|z : & \sum_{kk', m, \sigma} c_{q\sigma}^\dagger c_{k'\sigma} |m\rangle \langle m| \frac{J_m^{\sigma 2}}{\omega - \frac{D}{2} + \frac{J}{2}\sigma S_d^z} |m\rangle \langle m| c_{k\sigma}^\dagger c_{q\sigma} = - \sum_{kk', m, \sigma} n_{q\sigma} \frac{J_m^{\sigma 2} c_{k\sigma}^\dagger c_{k'\sigma} |m\rangle \langle m|}{\omega_{m, \sigma} - \frac{D}{2} + \frac{J}{2}\sigma m}, \\ +|- : & \sum_{kk', m} c_{q\uparrow}^\dagger c_{k'\downarrow} |m\rangle \langle m+1| \frac{J_m^2}{\omega - \frac{D}{2} + \frac{J}{2}S_d^z} |m+1\rangle \langle m| c_{k\downarrow}^\dagger c_{q\uparrow} = -n_{q\uparrow} \sum_{kk', m} \frac{J_m^2 c_{k\downarrow}^\dagger c_{k'\downarrow} |m\rangle \langle m|}{\omega_{m+1, \uparrow} - \frac{D}{2} + \frac{J}{2}(m+1)}. \end{aligned} \quad (3.5)$$

We similarly compute the rest of the terms. We again define $\sum_q \hat{n}_{q\sigma} = n(D)$. To compare with the spin- $\frac{1}{2}$ RG equations, we will transform the general spin- S ω to the spin- $\frac{1}{2}$ ω , using $\omega_{m, \sigma} \rightarrow$

$\omega - \frac{J}{2} \left(m\sigma - \frac{1}{2} \right)$. The renormalisation in J_m^σ is

$$\Delta J_m^\sigma = -n(D) \frac{(J_m^\sigma)^2 + \left(J_{m-\frac{1+\sigma}{2}}^t \right)^2}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.6)$$

Here, we have defined $J_m^t = 0$ for $|m| > S$. Two relations can be obtained from this RG equation, the RG equations for the sum and difference of the couplings: $J_m^\pm = \frac{1}{2} (J_m^\uparrow \pm J_m^\downarrow)$. The RG equation for the sum of the couplings is

$$\Delta J_m^+ = -n(D) \frac{\sum_\sigma (J_m^\sigma)^2 + \sum_\sigma \left(J_{m-\frac{1+\sigma}{2}}^t \right)^2}{2 \left(\omega - \frac{D}{2} + \frac{J}{4} \right)} = -n(D) \frac{J^2}{4} \frac{S(S+1)}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.7)$$

This is an m -independent piece, so it can be summed over to produce an impurity-independent potential scattering term, which we ignore.

The second is the RG equation for the difference of the couplings:

$$\Delta J_m^- = -n(D) \frac{1}{2} \frac{\left(J_{m-1}^t \right)^2 - \left(J_m^t \right)^2}{\omega - \frac{D}{2} + \frac{J}{4}} = -\frac{1}{4} \frac{n(D) m J^2}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.8)$$

The usual J Kondo coupling is recovered through $J = 2J_m^-/m$. Substituting this gives

$$\Delta_{\text{p sector}} J = -\frac{1}{2} n(D) \frac{J^2}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.9)$$

We can also obtain the RG equation for J from the transverse renormalisation:

$$\Delta J_m^t = -\frac{n(D) J_m^t (J_m^\downarrow + J_{m+1}^\uparrow)}{\omega - \frac{D}{2} + \frac{J}{4}} = -\frac{1}{2} \frac{n(D) J_m^t J}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.10)$$

Since $J_m^t \propto J$, we have

$$\Delta J_{\text{p sector}} = -\frac{1}{2} \frac{n(D) J^2}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.11)$$

The total renormalisation from both particle and hole sectors, at this order, is

$$\Delta J^{(2)} = -\frac{n(D) J^2}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.12)$$

We now come to the third-order renormalisation. Following eq. (3.3), the next order renormalisation is

$$\Delta H_j^{(3)} = c^\dagger T \frac{1}{\omega' - H_D} H_X \frac{1}{\omega' - H_D} T^\dagger c + c^\dagger T \frac{1}{\omega' - H_D} T^\dagger c \frac{1}{\omega' - H_D} H_X. \quad (3.13)$$

The first term will be of the form

$$\sum_{q,k,l_1} c_{q\beta,l_1}^\dagger c_{k\alpha,l_1} |m_1\rangle \langle m_2| \frac{1}{\omega - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{\beta J}{4} S_d^z} |m_2\rangle c_{k_1\sigma_1,l_2}^\dagger c_{k_2\sigma_2,l_2} \langle m_3| \frac{1}{\omega - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{\beta J}{4} S_d^z} |m_3\rangle \langle m_4| c_{k\alpha,l_1}^\dagger c_{q\beta,l_1}. \quad (3.14)$$

We have not bothered to write all the summations and the couplings correctly, because we will only simplify the denominator here. Evaluating the inner products gives

$$\sum_{q,k,l_1} \frac{|m_1\rangle \langle m_4| c_{q\beta}^\dagger c_{k\alpha} c_{k_1\sigma_1,l_2}^\dagger}{\omega_{m_2,\beta} - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{\beta J m_2}{4}} \frac{c_{k_2\sigma_2,l_2} c_{k\alpha}^\dagger c_{q\beta}}{\omega_{m_3,\beta} - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{\beta J m_3}{4}}. \quad (3.15)$$

We again use $\omega_{m,\sigma} \rightarrow \omega - \frac{J}{2} \left(m\sigma - \frac{1}{2} \right)$.

$$|m_1\rangle \langle m_4| c_{k_1\sigma_1,l_2}^\dagger c_{k_2\sigma_2,l_2} \sum_{q,k,l_1} \frac{\hat{n}_{q\beta} (1 - \hat{n}_{k\alpha})}{\left(\omega - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{J}{4} \right)^2}. \quad (3.16)$$

We define $\sum_q \hat{n}_{q\beta} = n(D)$. Performing the sums over k and l_1 gives

$$-\frac{1}{2} |m_1\rangle \langle m_4| c_{k_1\sigma_1,l_2}^\dagger c_{k_2\sigma_2,l_2} \frac{\rho n(D) K}{\omega - \frac{D}{2} + \frac{J}{4}}. \quad (3.17)$$

ρ is the density of states which we have taken to be constant. Reinstating the complete summation and the couplings gives

$$-\frac{1}{2} \frac{\rho n(D) K}{\omega - \frac{D}{2} + \frac{J}{4}} \sum_{\substack{m_1, m_4, k_1, k_2, \\ l_2, \sigma_1 \sigma_2}} \lambda_1 \lambda_2 \lambda_3 |m_1\rangle \langle m_4| c_{k_1\sigma_1,l_2}^\dagger c_{k_2\sigma_2,l_2}. \quad (3.18)$$

There is no sum over m_2 and m_3 because they are constrained by m_1 and m_4 respectively. λ_i represent the couplings present at the three interaction vertices. $k_{1,2}$ sum over the momenta, $\sigma_{1,2}$ sum over the spin indices and l_2 sums over the channels.

The second term in eq. (3.13) can be evaluated in an almost identical fashion. The integral here will be negative of the first term, because of an exchange in the scattering processes.

$$\frac{1}{2} \frac{\rho n(D) K}{\omega - \frac{D}{2} + \frac{J}{4}} \sum_{\substack{m_1, m_4, k_1, k_2, \\ l_2, \sigma_1 \sigma_2}} \lambda_1 \lambda_3 \lambda_2 |m_1\rangle \langle m_4| c_{k_1\sigma_1,l_2}^\dagger c_{k_2\sigma_2,l_2}. \quad (3.19)$$

The first group of terms (those that appear in (3.18)) in the particle sector can be represented as $a|b'|c$, where $a, b, c \in \{z, +, -\}$ and represent the operator for the conduction electrons in the three connected processes. The $\prime$ on b indicates that it is the state of the electrons *not being decoupled*. The second group of terms (those that appear in (3.19)) are therefore represented as $a|b|c'$, because in this group, the interaction H_X between the electrons that are not being decoupled occur at the

very end. We will only calculate the terms in the particle sector, the ones in hole sector will be equal to these because of particle-hole symmetry. The full list of terms is:

$$\begin{array}{cccccccccccc}
 z|z'|z & z|z|z' & -|z'|+ & -|+|z' & +|z'|- & +|-|z' & z|+|z & z|-'|z & +|+|- & -|+|+ \\
 z|z|+ & z|z|- & +|-|+ & +|-|- & -|+|+ & z|z'|z & z|z|z' & -|z'|+ & -|+|z' & +|z'|- \\
 +|-|z' & z|+|z & z|-'|z & +|+|- & -|+|+ & z|z|+ & z|z|- & +|-|+ & +|-|- & -|+|+
 \end{array}$$

The total renormalisation in J_m^σ is

$$\begin{aligned}
 \Delta J_m^\sigma &= \frac{1}{2} \frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}} \left[(J_{m-1}^t)^2 J_m^\sigma + (J_m^t)^2 J_m^\sigma - (J_{m-1}^t)^2 J_{m-1}^\sigma - (J_m^t)^2 J_{m+1}^\sigma \right] \\
 &= \frac{1}{2} \frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}} J^3 m \sigma .
 \end{aligned} \tag{3.20}$$

Since we had defined $J_m^\sigma \equiv \frac{1}{2} J m \sigma$, we have $\Delta J = \frac{2}{m \sigma} \Delta J_m^\sigma$, and we get $\Delta_{\text{p sector}} J = \frac{1}{4} \frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}} J^3$. Combining with the hole sector renormalisation, we get

$$\Delta J^{(3)} = \frac{1}{2} \frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}} J^3 . \tag{3.21}$$

The total renormalisation in J after combining all orders is

$$\Delta J = -\frac{n(D)J^2}{\omega - \frac{D}{2} + \frac{J}{4}} + \frac{1}{2} \frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}} J^3 . \tag{3.22}$$

2.4 Example III: Particle in a periodic potential

We consider electrons moving in one dimension in the presence of a periodic potential $V(x)$ satisfying $V(x+a) = V(x)$. The full Hamiltonian can be represented as

$$\mathcal{H} = \int_{-\infty}^{\infty} dx \, c^\dagger(x) \left[\hat{p}^2/2m + V(x) \right] c(x) , \tag{4.1}$$

where $c(x)$ destroys a fermion at the position x .

Physically, the model represents a set of N_{well} number of wells (located at the minima of the potential) linked to each other through hopping (brought about by the kinetic energy term). Each well is described a potential $V(x)$, whose spectrum consists of N_{levs} single particle levels. These levels are localised in real space (around each minima of width a), so the corresponding momentum states are widely separated; each of these momentum states eventually give rise to an energy band, we therefore represent them using the band index: $k_n = nG$ ($n = 0, 1, \dots, N_{\text{levs}} - 1$), where G is the spacing between the momentum states arising from a single potential well, and is equal to the reciprocal lattice vector $2\pi/a$.

These are of course not the only momentum states in the problem; since each potential well is spaced far apart (at distances greater than a), the multiplicity of the wells leads to closely spaced

momentum states, N_{well} in number (the number of potential wells). Since these are closely spaced, they lead to the dense set of states within each band: $k_{n,j} = nG + \frac{2\pi j}{N_{\text{well}}}$ ($j = 0, 1, \dots, N_{\text{well}} - 1$). The spacing $2\pi/N_{\text{well}}$ between the states within the band is imposed by applying periodic boundary conditions on the system: $\psi(x=0) = \psi(x=aN_{\text{well}})$. We take the thermodynamic limit by allowing $N_{\text{well}} \rightarrow \infty$, keeping all other independent parameters unchanged. Because of the periodicity of the real space potential, it can only scatter states that are separated by a reciprocal lattice vector G ; this means that states within a band (that are within a reciprocal lattice vector of each other) do not scatter off each other: $\langle k_{n+m,j+l} | \hat{V} | k_{n,j} \rangle = \delta_{l,0} \langle k_{n+m,j} | \hat{V} | k_{n,j} \rangle$.

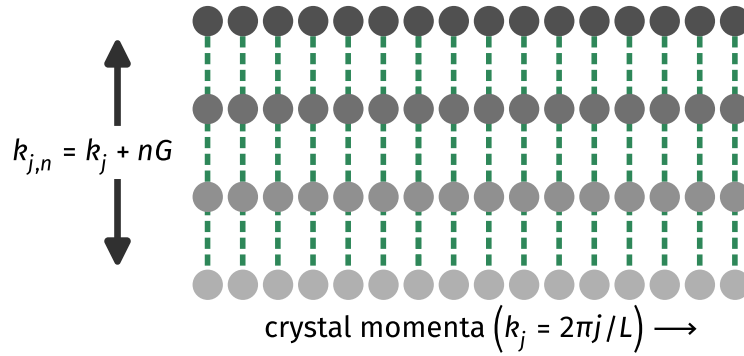


Figure 2.6: The complete set of momentum states in the system. Each row represents a band of states. Each column contains the set of momenta that are connected by a reciprocal lattice vector nG . The states in the lowest band give rise to the crystal momenta. All the momentum states in a column are connected to each other through the potential term (green dotted line). Any two states within a band, on the other hand, are decoupled from each other.

Decomposition into sum of zero-dimensional problems

We begin by writing the Hamiltonian completely in momentum space. A general momentum is represented as $k+nG$, where n is the band index goes from 0 to $N-1$, and k is the crystal momentum that spans through the lowest band. With this notation, we have

$$\begin{aligned} \mathcal{H} &= \frac{\hbar^2}{2m} \int dk \sum_n c^\dagger(k+nG) (k+nG)^2 c(k+nG) \\ &\quad + \int dk \sum_{m,n} [\mathcal{V}(nG) c^\dagger(k+nG+mG) c(k+mG) + \text{h.c.}] \\ &= \frac{2\pi}{a} \int dk \mathcal{H}(k) \end{aligned} \tag{4.2}$$

where

$$\begin{aligned}
 \mathcal{H}(k) &= \frac{a}{2\pi} \frac{\hbar^2}{2m} \sum_n c^\dagger(k+nG) (k+nG)^2 c(k+nG) \\
 &\quad + \frac{a}{2\pi} \sum_{m,n} [v(nG) c^\dagger(k+nG+mG) c(k+mG) + \text{h.c.}] \\
 &= \frac{a}{2\pi} \int dx' c^\dagger(x') \left[\frac{1}{2m} (-i\hbar\partial_{x'} + \hbar k)^2 + V(x') \right] c(x').
 \end{aligned} \tag{4.3}$$

We have used x' to symbolise the position variable conjugate to the new set of momenta $\{nG\} =$

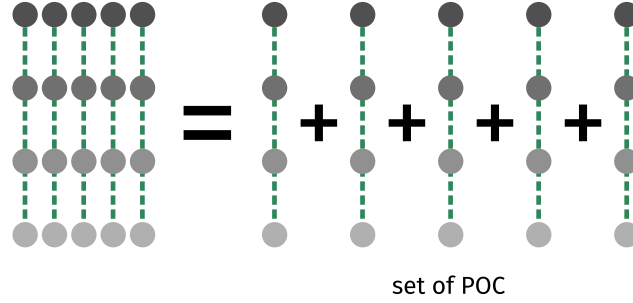


Figure 2.7: Because of the periodicity of the potential, the problem splits into a sum of particle on a circle (POC) problems, each formed by a specific crystal momentum and other momentum states that are related to it by a reciprocal vector.

$-i\partial_{x'}$ that span the restricted Hilbert space of $\mathcal{H}(k)$. Also, $\mathcal{V}(k)$ is the Fourier transform of the real space potential $V(x)$.

This decoupling of the full Hamiltonian $\mathcal{H}$ into a block-diagonal form (each block indexed by the crystal momentum k) can be attributed to the discrete translation symmetry present in the problem and could have been anticipated very generally, as we now show. The Hamiltonian satisfies $[\mathcal{H}, T(a)] = 0$, where $T(x) = e^{i\hat{p}x}$ is the translation operator. By choosing a starting location $x = 0$, applying the translation operator cycles through the set of states $|x = na\rangle$; $n = 0, 1, \dots, N_{\text{well}}$; this set is closed under the application of $T(a)$. It is easy enough to guess the eigenstates of $T(a)$. Since the application of $T(a)$ increases the position monotonically, the only way to create an eigenstate is to involve all states. With that in mind, the simplest ansatz is of the form $|\Phi\rangle = \sum_n e^{-i\Phi n} |an\rangle$. Applying the translation operator on this state gives

$$T(a) |\Phi\rangle = \sum_{n=0}^{N_{\text{well}}-2} e^{-i\Phi n} |a(n+1)\rangle + e^{-i\Phi(N_{\text{well}}-1)} |aN_{\text{well}}\rangle = e^{i\Phi} \left[e^{-i\Phi N_{\text{well}}} |aN_{\text{well}}\rangle + \sum_{n=1}^{N_{\text{well}}-1} e^{-i\Phi n} |an\rangle \right] \tag{4.4}$$

Since we have applied PBC on the system, we used $|aN_{\text{well}}\rangle$ for $|0\rangle$. If the values of Φ are chosen to be some integer multiple of $2\pi/N_{\text{well}}$, we would have $e^{-i\Phi N_{\text{well}}} = 1$, and the entire state can be resummed into the form

$$T(a) |\Phi\rangle = e^{i\Phi} |\Phi\rangle \tag{4.5}$$

This shows that $|\Phi\rangle$ is an eigenstate of the translation operator. Since the set of states has a dimension of N_{well} , the number of eigenstates and the number of possible values of Φ will also be N_{well} . This

fixes the eigenvalues to $\Phi = 2\pi m/N_{\text{well}}, m = 0, 1, \dots, N_{\text{well}} - 1$. These will also be eigenstates of the Hamiltonian, because they are indexed by different values of Φ and the Hamiltonian cannot change Φ (because it commutes with T). As will be shown later, the eigenvalues Φ are actually related to the crystal momentum, through $\Phi = ka$. The set of allowed values of Φ form the first Brillouin zone.

However, $|\Phi\rangle$ as defined above is not the only eigenstate with eigenvalue $e^{i\Phi}$. One can simply shift the starting location $x = 0$ by any distance smaller than a , and repeat the entire procedure. The eigenstates hence obtained will have the same eigenvalues as above:

$$|\Phi, \delta\rangle = \sum_n e^{-i\Phi n} |an + \delta\rangle, \quad 0 \leq \delta < a; \quad (4.6)$$

$$T(a) |\Phi, \delta\rangle = \sum_{n=0}^{N_{\text{well}}-2} e^{-i\Phi n} |a(n+1) + \delta\rangle + e^{-i\Phi(N_{\text{well}}-1)+\delta} |aN_{\text{well}} + \delta\rangle = e^{i\Phi} |\Phi, \delta\rangle \quad (4.7)$$

The translation operator is therefore able to at most classify the complete set of states into N_{well} number of blocks, each labelled by a specific value of Φ . Each block has a large number of states (N_{levs}), and the Hamiltonian will general scatter these states into each other. The Hamiltonian for each of these blocks can be represented as $\mathcal{H}(\Phi) = \mathcal{P}_\Phi \mathcal{H} \mathcal{P}_\Phi$ (where $\mathcal{P}_\Phi$ projects onto the block of states with eigenvalue $e^{i\Phi}$), such that the total Hamiltonian can be represented as $\mathcal{H} = \sum_\Phi \mathcal{H}(\Phi)$. Upon rescaling by $2\pi/a$ and taking the limit of $N_{\text{well}} \rightarrow \infty$, this is the same Hamiltonian decoupling relation as that obtained in eq. 4.2.

Since the spacing between two adjacent momentum is G , the maximum value of x' has to be $2\pi/G = a$, with the ends identified with each other: $\psi(x' = 0) = \psi(x' = a)$. We can then define a dimensionless variable $\theta = 2\pi x'/a$ that goes from 0 to 2π , and is conjugate to the angular wave number $Q = -i\partial_\theta$. This definition ensures $[\theta, Q] = i$. For reasons that will become clear very soon, we will also define a dimensionless crystal momentum $\Phi = ka$. With these modifications, the Hamiltonian takes the simpler form

$$\mathcal{H}(k) = \int_0^{2\pi} d\theta \, c^\dagger(\theta) \left[\frac{\hbar^2}{2ma^2} \left(\hat{Q} + \Phi/2\pi \right)^2 + U(\theta) \right] c(\theta). \quad (4.8)$$

By noting that we have the boundary condition $\psi(\theta = 0) = \psi(\theta = 2\pi)$, we can recognise this Hamiltonian as that describing the dynamics of a particle moving on a circle (POC) with a potential $U(\theta) = \frac{a}{2\pi} V(a\theta/2\pi)$, in the presence of an Aharonov-Bohm flux Φ . Note that $U(\theta)$ has a period of 2π .

Even when reduced to the simpler POC, the problem cannot be simply integrated to obtain the eigenstates. This is because the complete Hamiltonian does not commute with either the position or the momentum operator; it contains quantum fluctuations that need to be resolved in order to obtain the ground state energy. One way of obtaining the energy is to calculate the propagator for transition from $\theta = 0$ to $\theta = 2\pi$, using the method of path integrals [?, ?]. This is shown in Appendix ???. We, however, will attack the problem using a renormalisation group method, called the unitary renormalisation group in order to track the emergence of the tight-binding metal in the limit when the effective amplitude of the single-particle periodic potential at low-energies has become much larger than the typical kinetic energy of the electron.

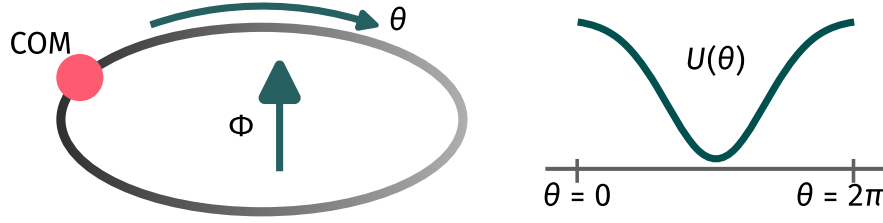


Figure 2.8: The full problem reduces to that of a particle on a circle, in the presence of potential with period 2π , and a flux Φ .

URG analysis of the particle on a circle problem

A summary of the unitary RG method is provided in Appendix ?? . For the purposes of studying the model under renormalisation group, we write the particle on a circle Hamiltonian in momentum(Q –)space. For that, we introduce the Fourier modes

$$c(\theta) = \frac{1}{\sqrt{2\pi}} \int dQ e^{iQ\theta} c(Q), \text{ and } U(\theta) = \int dQ e^{iQ\theta} \mathcal{U}(Q), \quad (4.9)$$

so that the Hamiltonian becomes

$$\mathcal{H}(k) = \int dQ \varepsilon(Q + \Phi/2\pi) \hat{n}(Q) + \int dQ_1 dQ_2 c^\dagger(Q_1) c(Q_2) \mathcal{U}(Q_1 - Q_2). \quad (4.10)$$

where we have defined the dispersion $\varepsilon(Q) = \hbar^2 Q^2 / (2ma^2)$. The second term involves scattering between momentum states Q_1 and Q_2 . The quantum fluctuations that connect the low-energy states to the high-energy states can be integrated out using the URG method [?]. This proceeds through a reduction in the momentum cutoff, ultimately leading to an IR theory. As the large momenta are integrated out, the potential term for scattering between the lower momenta undergo renormalisation. Let, at the l^{th} RG step, the momentum cutoff be Q_l . At this step, fluctuations between all momentum states more energetic than Q_l have been integrated out. The RG equations for the renormalisation of the potential $\mathcal{U}_{ij}^{(l)}$ that scatters any two lower momenta Q_i and Q_j ($Q_i, Q_j < Q_l$) is given by [?]

$$\Delta \mathcal{U}_{ij}^{(l)}(\omega) = \frac{\mathcal{U}_{il} \mathcal{U}_{lj}}{\omega - \varepsilon(Q_l + \Phi/2\pi)}, \quad (4.11)$$

where ω is an energy scale that tracks the amount of quantum fluctuations present in the system, and is characteristic of the URG method. The dispersion also undergoes renormalisation::

$$\Delta \varepsilon^{(l)}(Q_i, \omega) = \frac{|\mathcal{U}_{il}|^2}{\omega - \varepsilon(Q_l + \Phi/2\pi)}, \quad (4.12)$$

Effective description at low energies

Appearance of bands and band gaps

As the URG proceeds, the quantum fluctuations between the momentum states are resolved, leading to a simpler theory for the IR. We focus on the final two states $|Q_0 + \Phi/2\pi\rangle$, $|Q_1 + \Phi/2\pi\rangle$, for $\Phi = 0$

(Brillouin zone edge). The Hamiltonian for their dynamics, just prior to the decoupling of Q_j , is

$$H_{01}^* = \varepsilon^*(Q_0) |Q_0\rangle \langle Q_0| + \varepsilon^*(Q_1) |Q_1\rangle \langle Q_1| + (\mathcal{U}_{01}^* |Q_1\rangle \langle Q_0| + \text{h.c.}) \quad (4.13)$$

Since both the states are at the edge of the band, we can assume $\varepsilon^*(Q_0) \simeq \varepsilon^*(Q_1)$, in which case the problem becomes that of a symmetric double well with tunnelling. The eigenstates are $|Q_0\rangle \pm |Q_1\rangle$, with energies $\varepsilon^*(Q_0) \pm |\mathcal{U}_{01}^*|$. We can substitute this estimate of the energy eigenvalues into the RG equations in the form of the quantum fluctuation scale ω . At the last decoupling, the shifts in the kinetic energy are therefore

$$\Delta\varepsilon^*(Q_i, \omega^\pm) \simeq \frac{|\mathcal{U}_{01}^*|^2}{\varepsilon^*(Q_0) \pm |\mathcal{U}_{01}^*| - \varepsilon(Q_0)} = \pm |\mathcal{U}_{01}^*| = \pm |\mathcal{U}^*(G)|, \quad (4.14)$$

where we used $\mathcal{U}_{01} = \mathcal{U}(Q_1 - Q_0) = \mathcal{U}(G)$. The difference between the two shifts gives the energy difference between the two lowest states of the POC. In this way, the above RG equation shows the emergence of the single-particle band gap and the formation of the band insulator (if the chemical potential is placed within the gap). If we now reintroduce the flux degree of freedom, the lowest energy eigenstates of all the POCs can be combined to form a band of closely spaced energy levels $\{\varepsilon^*(Q_0 + \Phi_j/2\pi)\}$, $\Phi = k_{0,j}a$ (see Fig. 2.9). These levels are indexed by $k_{0,j}$ with j ranging through the first Brillouin zone; $k_{0,j}$ are the crystal momenta of the system.

The second band of states is formed similarly by considering the first excited of the POCs: $\{\varepsilon^*(Q_1 + \Phi_j/2\pi)\}$, $\Phi = k_{0,j}a$. The gap between the first and second band is set by the energy difference between the ground state and first excited state of the POC at $\Phi = 0$. This difference, and hence the gap, is equal to $2|\mathcal{U}_{01}^*|$.

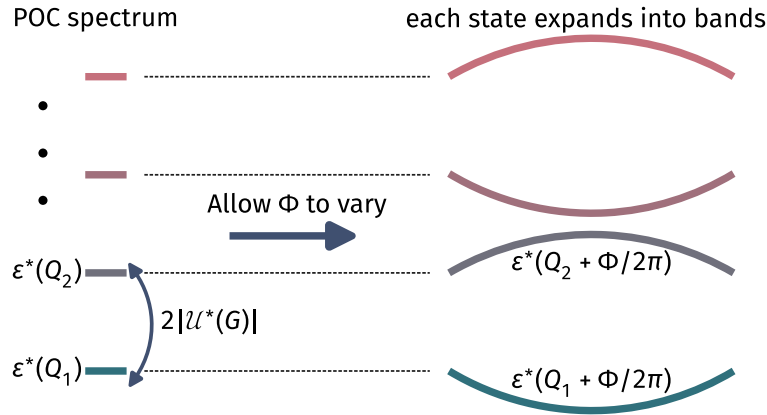


Figure 2.9: Spectral flow transformations on the POC by tuning the flux Φ generates the full band of delocalised single-particle electronic states whose crystal momenta correspond to $\hbar\Phi/a$.

Dispersion relation of the lowest band

In order to derive the dispersion for the lowest band, we note that the final fixed point Hamiltonian (obtained upon decoupling all higher momentum states) is described by the single momentum state

$|Q_0\rangle$. In terms of real space hopping, this means that all short wavelength processes have been removed, and the only process that remains is the longest hopping, that from 0 to 2π . The fixed point Hamiltonian can be written in terms of this process:

$$\begin{aligned} \varepsilon^*(Q_0) |Q_0\rangle \langle Q_0| &= \frac{1}{2} \varepsilon^*(2\pi) (|\theta = 0\rangle + |\theta = 2\pi\rangle) (\langle\theta = 0| + \langle\theta = 2\pi|) = \frac{1}{2} \varepsilon^*(2\pi) (\hat{n}(0) + \hat{n}(2\pi)) \\ &\quad + \frac{1}{2} \varepsilon^*(2\pi) (c^\dagger(0)c(2\pi) + \text{h.c.}) \end{aligned} \quad (4.15)$$

We now recall that we had mapped the spatial point $x = 0$ ($x = a$) to the angular point $\theta = 0$ ($\theta = 2\pi$) point. We now transform back to spatial coordinates to get

$$\frac{1}{2} \varepsilon^*(2\pi) (\hat{n}(0) + \hat{n}(a)) + \frac{1}{2} \varepsilon^*(a) (c^\dagger(0)c(a) + \text{h.c.}) . \quad (4.16)$$

This is computed for a specific value $\Phi = 0$ for the flux (or the crystal momentum). The POC corresponding to a particular flux is built from the states within a particular well. Changing the flux is then equivalent to studying the POC corresponding to the states within a different well. What this means is that tuning the flux by one interval amounts to translating the fixed point Hamiltonian by one period a of the potential. Adding the contributions from all the values of the flux leads to the following tight-binding Hamiltonian:

$$H_{TB} = \varepsilon^*(2\pi) \sum_{j=0}^{N_{\text{well}}-1} \hat{n}(ja) + \frac{1}{2} \varepsilon^*(a) \sum_{j=0}^{N_{\text{well}}-1} (c^\dagger(ja)c((j+1)a) + \text{h.c.}) , \quad (4.17)$$

where the first term corresponds to a constant chemical potential applied to the tight-binding metal in one dimension with a single-particle dispersion $\varepsilon(k) = -\varepsilon^*(a) \cos(ka)$ where k corresponds to the set of crystal momenta $k \in [-\frac{\pi}{a}, \frac{\pi}{a}]$. The exercise, therefore, shows the emergence of nearest-neighbour hopping in the lowest band by starting from a more microscopic model of a particle in a periodic potential. (An alternate derivation of the tight-binding metal has also been shown from path-integral methods in Appendix ??).

2.5 Comparison of renormalisation group approaches based on canonical transformations

We have considered three canonical transformations in this section: the Poor Man's scaling (PMS), the Schrieffer-Wolff transformation (SWT) and the continuous unitary transformation renormalization group (CUT-RG). PMS and SWT are more or less identical; they differ in the context in which they are used. PMS is used when there is an entire spectrum of energies in the model, ranging from a low energy limit to a high energy; it is then employed to decouple the highest energy modes in an iterative fashion. SWT is used when the Hamiltonian can be split into one high and one low energy part, and we need to decouple these two modes. It is clear when seen from this perspective that SWT is like a one-shot PMS; it decouples the UV from the IR in one step, compared to the iterative

approach of PMS. However, as has been shown in the previous subsections, both PMS and SWT can be formalized in an identical fashion, so that one can be switched for the other in both contexts.

Now that we have established that PMS and SWT are formally identical, we can relate them to URG. URG is similar in philosophy to PMS in the fact that URG also successively decouples high energy modes from the low energy ones. The difference is that URG takes care of the quantum fluctuations, at least in principle, by introducing a new set of energy scales ω . These ω then parameterise the RG flows of URG, compared to the single RG flow of PMS. Since SWT is formally the same as PMS, there is also a comparison between SWT and URG. Both PMS and SWT trivialize the quantum fluctuation scales of URG by replacing it with the bare energy values.

CUT-RG is philosophically different from the other transformations. Its goal is to gradually reduce the contributions of the off-diagonal terms by making them decay along a certain scale l . Off-diagonal terms that connect states with large energy differences decay faster. In this sense, there is no sequential dropping out of off-diagonal terms; all off-diagonal terms disappear at $l = \infty$. In this sense, it is like a continuous version of SWT. While SWT strives to remove an entire off-diagonal term in one-shot, CUT RG does this gradually by introducing a scale l . This separation of scales is absent in SWT. It does exist in URG and PMS, albeit in a different fashion. There, the separation comes in when we decouple single electron states starting from the Brillouin zone boundary Λ_N and come down to the Fermi surface Λ_0 . Each Λ provides a natural energy scale for separating the high and low energy physics.

If one integrates the continuum generator η over all the scales, one should recover something analogous to the SWT generator. This generator is however necessarily perturbative in the off-diagonal term, since by definition η will only be of first order in $\mathcal{H}_X$. This is in contrast to the non-perturbative generators in URG and, at least in principle, PMS. This non-perturbative form is encapsulated in the presence of a second completely different set of energy scales ω (or E in PMS). This second energy scale is absent in CUT RG because it takes care of at most the second order term.

Another point to note is that since SWT keeps the entire off-diagonal piece in the generator, new terms will almost certainly be generated at every step, and they have to be truncated. This is not the case with URG or PMS, because in those methods, we decouple just a single-electron at each step, and so those electrons become integrals of motion at that step, leading to their removal from the off-diagonal piece, and very often the Hamiltonian retains the same form as the bare model. This makes the philosophy of URG and PMS easier to work with. Tied to this is the fact that CUT RG often takes a certain type of interaction in the generator part, and not the entire off-diagonal piece. Hence, at the limit of the flow parameter going to ∞ , the chosen off-diagonal piece goes to zero but the remaining off-diagonal pieces still remain. As a result, the Hamiltonian is at most block diagonal at this stage. On the other hand, URG progressively decouples single electrons, meaning all scattering terms corresponding to that electron vanish at each step.

One last thing to note is that URG, being unitarily implemented with a well-defined generator, does not accommodate for spontaneous symmetry breaking (SSB). In order to see SSB, the symmetry-breaking term has to be added to the bare model explicitly; if this term grows under the RG, then the ground state will be symmetry-broken. CUT RG, however, does allow for SSB through the idea that the generator is not uniquely defined. If the generator commutes with a particular symmetry, then the family of Hamiltonians will also have the symmetry [370]. However, if the generator is replaced

by something that is normal ordered w.r.t a particular expectation value, then the CUT RG flow will usually be towards either the symmetry-preserved state or the symmetry-broken state, depending on whichever is stable [371].

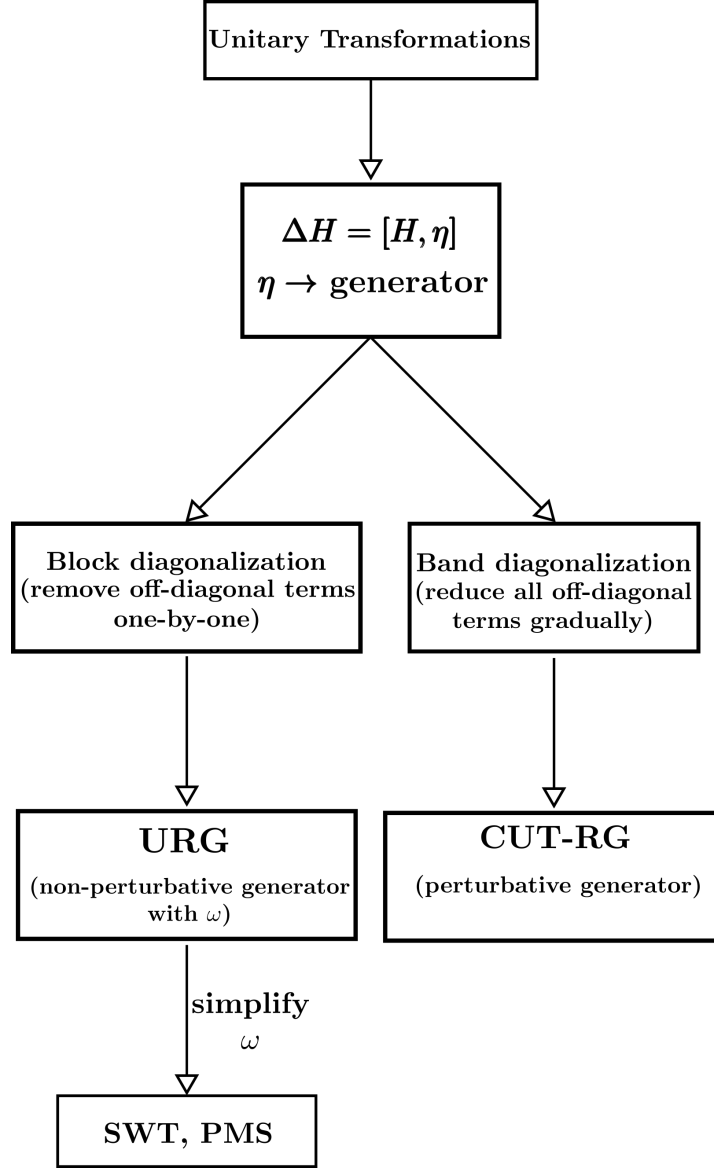


Figure 2.10: Comparison of the various unitary transformations and their relationships to each other.

2.6 Iterative Diagonalisation of Quantum Impurity Models

The broad idea is the following. We have a Hamiltonian describing an interacting set of states (in real or momentum space). We can express the total Hamiltonian in an incremental fashion: $H = \sum_i H_i$, where H_{i+1} involves more number of states than H_i . For example, a tight-binding model can be

written in that form, with the definition $H_i = c_{i+1}^\dagger c_i + \text{h.c.}$. The iterative diagonalisation method obtains the low-energy spectrum of this problem in the following manner: We first diagonalise the Hamiltonian H for a smaller value of i , small enough such that this can be done exactly. We then truncate the spectrum to a predefined size, and rotate all existing operators to this truncated basis, including the Hamiltonian H . We then consider the "bonding Hamiltonian" ΔH between the existing sites and the new sites, and rotate the same into the truncated basis. Adding the previous rotated Hamiltonian H and the rotated increment Hamiltonian ΔH gives us a truncated but effective Hamiltonian for the increased number of sites. We again diagonalise this, and again retain only a fixed number of states in the spectrum. We keep repeating this until we reach the required number of sites.

Jordan-Wigner Matrix Representation of Fermionic Operators

For a single qubit, the creation/annihilation matrices are

$$c = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, c^\dagger = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad (6.1)$$

given the convention that $|1\rangle = (0 \ 1)$ is the occupied state. For a many-body system, these must be replaced with field operators that have the canonical fermionic algebra. For a system of N 1-particle levels, this can be accomplished through a *Jordan-Wigner*-like transformation

$$c_j = \left(\otimes_1^j \sigma_z \right) \otimes c \otimes \left(\otimes_1^{N-j} \mathbb{I} \right), j \in [0, N-1], \quad (6.2)$$

where $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. For example, for spinless electrons on a three-site lattice, we can have three fermionic operators: c_1, c_2 , and c_3 . Following the above expression, their matrices are

$$c_1 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, c_2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, c_3 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

For later use, we define the *antisymmetriser matrix* $S(j)$ and identity matrix $\mathbb{I}(j)$,

$$\begin{aligned} S(j) &= \otimes_1^j \sigma_z = \sigma_z \otimes \dots j \text{ times } \dots \otimes \sigma_z, \\ \mathbb{I}(j) &= \otimes_1^{N-j} \mathbb{I} = \mathbb{I} \otimes \dots j \text{ times } \dots \otimes \mathbb{I}, \end{aligned} \quad (6.3)$$

to express the fermionic representation compactly:

$$c_j = S(j) \otimes c \otimes \mathbb{I}(N-j). \quad (6.4)$$

Structure of Hamiltonian

We consider a general impurity model, of L number of sites (excluding the impurity site):

$$H_{\text{sys}}(L) = H_{\text{imp}} + H_{\text{imp-bath}} - t \sum_{\sigma, j=1}^{L-1} \left(c_{j,\sigma}^\dagger c_{j+1,\sigma} + \text{h.c.} \right), \quad (6.5)$$

where H_{imp} is the Hamiltonian for the decoupled impurity site and $H_{\text{imp-bath}}$ is the impurity-bath coupling. Such a system has $2L + 2$ single-particle levels in the full system (2 spin levels for each of the $L + 1$ sites).

Iteration scheme (the initial number of states is L_0):

$$\begin{aligned} H_0(L_0) &= H_{\text{sys}}(L_0), \\ H_{r+1}(L_0) &= H_r(L_0) + \Delta H_r(L_0), \quad r \geq 0, \\ \Delta H_r(L_0) &= -t \sum_{\sigma} \left(c_{L_0+r,\sigma}^\dagger c_{L_0+r+1,\sigma} + \text{h.c.} \right). \end{aligned} \quad (6.6)$$

H_0 is the initial Hamiltonian, consisting of L_0 lattice sites in the bath, $H_{r+1}(L_0)$ is the Hamiltonian after $r + 1$ iterations having started with L_0 sites, and $\Delta H_r(L_0)$ is the increment term that gets added to $H_r(L_0)$ during the $(r + 1)^{\text{th}}$ iteration to give rise to $H_{r+1}(L_0)$.

Iterative Diagonalization

Let the starting Hamiltonian be $H_0(L_0)$, consisting of L_0 single-particle levels (hence a Hilbert space dimension of 2^{L_0}). We start with a single-particle computational basis and construct the L_0 fermionic operators $c_1, c_2, \dots, c_{L_0}$ in this basis, using eq. 6.4. We also keep track of a large antisymmetriser matrix $S(L_0)$ that will be used to attach new sites when we expand the system. Let M_s be the maximum number of eigenstates we retain in the spectrum at any given step. The value of M_s should be chosen so that a $M_s \times M_s$ matrix can be diagonalised in reasonable time.

- S1. Construct the complete Hamiltonian matrix H_0 in our present basis using the field operators $\{c_j\}$. Diagonalise the Hamiltonian (of size $D_0 \times D_0$) and obtain the eigenvalues E_n and eigenstates X_n . Each X_n is a column vector of size D_0 .
- S2. Retain at most M_s number of eigenstates, preferring the ones with lower energy. The reduced basis for this rotated truncated subspace is constructed by stacking the column vectors X_n horizontally:

$$R = [X_1 X_2 \dots X_{M_s}]_{D_0 \times M_s}. \quad (6.7)$$

This matrix also acts as the transformation to rotates and truncate all operators from the old basis into the new one.

- S3. We rotate our Hamiltonian H_0 , our fermionic operators $\{c_j\}$ and the large antisymmetrizer matrix $S(L_0)$ into the new reduced basis, using the transformation $O \rightarrow R^\dagger O R$.

- S4. We now need to expand our system by adding the increment Hamiltonian ΔH_0 . Let the number of new 1-particle levels in ΔH_0 be L_Δ . These new levels will be indexed as $L_0+1, \dots, L_0+L_\Delta$. We need to define antisymmetrized fermionic operators for the new sites:

$$\begin{aligned} c_{L_0+1} &= c \otimes \mathbb{I}(2^{L_\Delta-1}), \\ c_{L_0+2} &= \sigma_z \otimes c \otimes \mathbb{I}(2^{L_\Delta-2}), \\ &\dots \\ c_{L_0+L_\Delta} &= (\otimes_1^{L_\Delta-1} \sigma) \otimes c. \end{aligned} \tag{6.8}$$

These have to be calculated in the local computational basis (of size 2^{L_Δ}) of the new levels.

- S5. Combining the new sites with the old sites leads to a combined Hilbert space dimension of $(M_s + 2^{L_\Delta}) \times (M_s + 2^{L_\Delta})$. To allow all operators to act on the enlarged Hilbert space, we expand both sets of operators:

$$\begin{aligned} c_j &= S(L_0) \otimes c_j; \quad j = L_0 + 1, \dots, L_\Delta, \\ c_j &= c_j \otimes \mathbb{I}(2^{L_\Delta}); \quad j = 1, 2, \dots, L_0, \\ H_0 &\rightarrow H_0 \otimes \mathbb{I}(2^{L_\Delta}), \\ S(L_1) &= S(L_0) \otimes \mathbb{I}(2^{L_\Delta}). \end{aligned} \tag{6.9}$$

where $\mathbb{I}(2^{L_\Delta})$ is an identity matrix of dimension $2^{L_\Delta} \times 2^{L_\Delta}$. Note that the operators in the last three equations are the rotated ones (following Step 3).

- S6. Using the transformed operators $c_{L_0+1}, \dots, c_{L_0+L_\Delta}$ for the new sites, construct the difference Hamiltonian matrix ΔH_0 and hence the updated Hamiltonian $H_1 = H_0 + \Delta H_0$ for the next step. Repeat the process starting from step 2 with the new Hamiltonian H_1 , the new operators c_j and the new matrix $S(L_1)$ replacing the old counterparts.

Static correlations

We are in general interested in n -point correlations of the form $\langle O_1 O_2 \dots O_n \rangle$, where O_i are operators that act on 1-particle Hilbert spaces. Let the earliest step of the iterative diagonalisation procedure at which all these operators have entered the system be r . At this step r , construct the correlation operator $O_1 O_2 \dots O_n$ using the fermionic matrices c_j at that step (recall that these matrices will be highly rotated versions of the matrices we started with). Having constructed the operator O , we expand and rotate it after the completion of every future step: $O_{n+1} = (R_n^\dagger O_n R_n) \otimes \mathbb{I}(2^{L_\Delta})$, where R_n is the rotation matrix for the n^{th} step. The last step n^* of the iterative diagonalisation consists of only a diagonalisation and no expansion, resulting in a final set of eigenstates $\{X_i\}$. The form of the correlation operator in this basis is $O_{n^*} = R_{n^*-1}^\dagger O_{n^*-1} R_{n^*}$. The expectation value can now be calculated using the matrix O_{n^*} and the ground state of $\{X_i\}$.

Reduced density matrices

2.7 Optimising Fermi Surface Calculations

Rotated basis for simplified representation

At the renormalisation group fixed point, we have a renormalised theory for the interaction of the impurity spin S_f with the f -layer Fermi surface:

$$H^* = \sum_{\alpha,\beta} \mathbf{S}_f \cdot \boldsymbol{\sigma}_{\alpha,\beta} \sum_{\mathbf{k},\mathbf{q}} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k},\alpha}^\dagger f_{\mathbf{q},\beta} . \quad (7.1)$$

We will now obtain a more minimal representation of this interaction. Each momentum label is summed over all four quadrants Q_1 through Q_4 ; eq. ?? relates Q_1 with Q_3 and Q_2 with Q_4 :

$$\begin{aligned} \sum_{\mathbf{k},\mathbf{q}} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k},\alpha}^\dagger f_{\mathbf{q},\beta} &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k} \in Q_1, Q_2} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k},\alpha}^\dagger + \sum_{\mathbf{k} \in Q_3, Q_4} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k},\alpha}^\dagger \right] f_{\mathbf{q},\beta} \\ &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k} \in Q_1, Q_2} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k},\alpha}^\dagger + \sum_{\mathbf{k} \in Q_1} J_f^*(\mathbf{k} - \mathbf{Q}_1, \mathbf{q}) f_{\mathbf{k}-\mathbf{Q}_1,\alpha}^\dagger + \sum_{\mathbf{k} \in Q_2} J_f^*(\mathbf{k} + \mathbf{Q}_2, \mathbf{q}) f_{\mathbf{k}+\mathbf{Q}_2,\alpha}^\dagger \right] f_{\mathbf{q},\beta} \\ &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k} \in Q_1} J_f^*(\mathbf{k}, \mathbf{q}) (f_{\mathbf{k},\alpha}^\dagger - f_{\mathbf{k}-\mathbf{Q}_1,\alpha}^\dagger) + \sum_{\mathbf{k} \in Q_2} J_f^*(\mathbf{k}, \mathbf{q}) (f_{\mathbf{k},\alpha}^\dagger - f_{\mathbf{k}+\mathbf{Q}_2,\alpha}^\dagger) \right] f_{\mathbf{q},\beta} . \end{aligned} \quad (7.2)$$

For ease of notation, we define new fermionic operators $A_{\mathbf{k},\sigma}$:

$$A_{\mathbf{k},\sigma,\pm} = \begin{cases} \frac{1}{\sqrt{2}} (f_{\mathbf{k},\sigma} \pm f_{\mathbf{k}-\mathbf{Q}_1,\sigma}) , & \mathbf{k} \in Q_1 , \\ \frac{1}{\sqrt{2}} (f_{\mathbf{k},\sigma} \pm f_{\mathbf{k}+\mathbf{Q}_2,\sigma}) , & \mathbf{k} \in Q_2 , \end{cases} \quad (7.3)$$

which satisfy the appropriate algebra: $\{A_{\mathbf{k},\sigma,p}, A_{\mathbf{k}',\sigma',p'}\} = \delta_{\mathbf{k},\mathbf{k}'} \delta_{\sigma,\sigma'} \delta_{p,p'}$, with $p = \pm$ denoting the flavours of the A fields. Note that only the $p = -1$ flavour enters the Hamiltonian. Henceforth, we drop the label $\pm$ and it is implied that A refers to the $p = -1$ variant.

Decomposing the sum over $\mathbf{q}$ in a similar fashion, we get

$$\sum_{\mathbf{k},\mathbf{q}} J_f^*(\mathbf{k}, \mathbf{q}) f_{\mathbf{k},\alpha}^\dagger f_{\mathbf{q},\beta} = 2 \sum_{\mathbf{k},\mathbf{q} \in Q_1, Q_2} J_f^*(\mathbf{k}, \mathbf{q}) A_{\mathbf{k},\alpha}^\dagger A_{\mathbf{q},\beta} . \quad (7.4)$$

Correlations in truncated representation

In order to calculate equal-time correlations (such as $\langle S_d^+ f_{\mathbf{k}\downarrow}^\dagger f_{\mathbf{k}\uparrow} \rangle$), we use eq. 7.3 to express the bare fields $f_{\mathbf{k},\sigma}$ in terms of the new fields $A_{\mathbf{k},\sigma,\pm}$ and $B_{\mathbf{k},\sigma,\pm}$:

$$\begin{aligned} f_{\mathbf{k},\sigma} &= \frac{1}{\sqrt{2}} (A_{\mathbf{k},\sigma,+} + A_{\mathbf{k},\sigma,-}) , \mathbf{k} \in Q_1, Q_2 \\ f_{\mathbf{k}-Q_1,\sigma} &= \frac{1}{\sqrt{2}} (A_{\mathbf{k},\sigma,+} - A_{\mathbf{k},\sigma,-}) , \mathbf{k} \in Q_1 \\ f_{\mathbf{k}+Q_2,\sigma} &= \frac{1}{\sqrt{2}} (A_{\mathbf{k},\sigma,+} - A_{\mathbf{k},\sigma,-}) , \mathbf{k} \in Q_2 \end{aligned} \quad (7.5)$$

where the four relations act on operators in four quadrants. These relations allow us to express correlations $F_{k_i,k_j} = \langle S_d^+ c_{k_i,\uparrow}^\dagger c_{k_j,\downarrow} \rangle$ for the bare fields f in terms of the rotated fields. Using the fact that the Hamiltonian does not couple the fields B_+ and A_+ , we have

$$\begin{aligned} F_{k_1,k_1} &= F_{k_3,k_3} = -F_{k_1,k_3} = \frac{1}{2} \langle S_d^+ A_{k_1,\downarrow,-}^\dagger A_{k_1,\uparrow,-} \rangle , \\ F_{k_2,k_2} &= F_{k_4,k_4} = -F_{k_2,k_4} = \frac{1}{2} \langle S_d^+ A_{k_2,\downarrow,-}^\dagger A_{k_2,\uparrow,-} \rangle , \\ F_{k_1,k_2} &= F_{k_3,k_4} = -F_{k_1,k_4} = -F_{k_2,k_3} = \frac{1}{2} \langle S_d^+ A_{k_1,\downarrow,-}^\dagger A_{k_1,\uparrow,-} \rangle , \end{aligned} \quad (7.6)$$

where the momentum state k_i belongs to the quadrant Q_i .

Effective Star Graph Representation of Kondo Interaction

To make computation less intensive, we want to make the Hamiltonian in eq. 7.4 sparser, by going to a star graph representation where each node of the graph couples only with the impurity spin (central node) and not to the other nodes of the graph. We therefore consider a rotated basis of excitations

$$\gamma_{i,\sigma}^\dagger = \sum_k \Lambda_{i,k} A_{k,\sigma}^\dagger , \text{ where } \sum_k \Lambda_{i,k}^* \Lambda_{j,k} = \delta_{i,j} . \quad (7.7)$$

These fermions of course satisfy the appropriate algebra:

$$\{\gamma_{i,\sigma}, \gamma_{j,\sigma'}^\dagger\} = \sum_{k,q} \Lambda_{i,k}^* \Lambda_{j,q} \{A_{k,\sigma}, A_{q,\sigma'}^\dagger\} = \delta_{i,j} \delta_{\sigma,\sigma'} . \quad (7.8)$$

In order to calculate the coefficients $\Lambda_{i,k}$, we impose the star graph simplification condition described above:

$$\sum_{\mathbf{k},\mathbf{q}} J_f^*(\mathbf{k},\mathbf{q}) A_{\mathbf{k},\alpha}^\dagger A_{\mathbf{q},\beta} = \sum_i K_i \gamma_{i,\alpha}^\dagger \gamma_{i,\beta} , \quad (7.9)$$

where K_i are the eigenvalues of the Kondo coupling matrix $J_f^*(\mathbf{k},\mathbf{q})$. Substituting the definition of the γ fields (eq. 7.7) leads to the equation

$$J_f^*(\mathbf{k},\mathbf{q}) = \sum_i \Lambda_{k,i} K_i \Lambda_{i,q}^\dagger \implies \mathcal{J} = \Lambda K \Lambda^\dagger , \quad (7.10)$$

where $\mathcal{J}$ is the Kondo coupling matrix with matrix elements $J_f^*(\mathbf{k}, \mathbf{q})$, K is a diagonal matrix with the eigenvalues (K_i) of $\mathcal{J}$ along its diagonal, and Λ is the matrix $\Lambda_{i,k}$ (the rotated indices i along the row, and the earlier momentum indices along the column). From eq. 7.10, it is clear that since $\mathcal{J}$ is Hermitian, the matrix Λ is simply the unitary matrix which diagonalises the Kondo matrix $\mathcal{J}$:

$$\Lambda^\dagger \mathcal{J} \Lambda = K . \quad (7.11)$$

The prescription is therefore as follows. We compute the unitary transformation Λ that diagonalises $\mathcal{J}$ and obtain the eigenvalues $\{K_i\}$. Using the eigenvalues, we work with the sparser Hamiltonian

$$\vec{S}_f \cdot \sum_{\alpha, \beta} \vec{\sigma}_{\alpha, \beta} \sum_i K_i \gamma_{i, \alpha}^\dagger \gamma_{i, \beta} . \quad (7.12)$$

If we have to calculate a certain correlation $C_{k,q}$, we instead calculate the correlation matrix $\tilde{C}_{i,j}$, and then transform back using the unitary transformation:

$$C = \Lambda \tilde{C} \Lambda^\dagger . \quad (7.13)$$

2.8 Measures of entanglement and their computation

The present millennium has seen a rapid surge in the use of quantum entanglement towards the study of quantum and condensed matter systems [160–165]. By tracking quantum correlations among particles across both small and large distances, entanglement measures offer valuable insights into the nature of the ground state and low-energy excitations. This, for example, allows us to distinguish strongly-correlated topologically ordered states characterised by long-ranged entanglement (in the form of a sub-leading topological term in their entanglement entropy [166, 167, 170, 174]) from weakly-interacting topologically trivial phases that have short-ranged area-law entanglement [162, 187, 188].

2.8.1 Reduced density matrix elements as equal-time correlation functions

A density matrix plays the role of a probability density function (PDF) for a quantum system. A classical variable X that can take values $x \in [-X^*, X^*]$ has a PDF $F(X = x)$, which is used, for example, for calculating the average value (expectation value) of the variable:

$$\langle X \rangle = \int_{-X^*}^{X^*} x F(x) dx . \quad (8.1)$$

In quantum mechanics, the variables we are interested in are operators O , and a quantum PDF ρ will then give the expectation value of the operator, in a fashion similar to its classical counterpart:

$$\langle O \rangle = \sum_{\phi} \langle \phi | \rho O | \phi \rangle ; \quad (8.2)$$

here, the integral over the classical values x is replaced by a sum over the all possible quantum states $|\phi\rangle$ (complete orthonormal basis) of the system. However, we already know that the expectation value of a quantum operator is given by

$$\langle O \rangle = \langle \Psi | O | \Psi \rangle . \quad (8.3)$$

To bring this form closer to the one in terms of ρ (eq. 2), we insert an identity resolution: $1 = \sum_{\phi} |\phi\rangle \langle \phi|$:

$$\langle O \rangle = \sum_{\phi} \langle \Psi | O | \phi \rangle \langle \phi | \Psi \rangle = \sum_{\phi} \langle \phi | \Psi \rangle \langle \Psi | O | \phi \rangle . \quad (8.4)$$

A comparison of this equation to eq. (2) gives

$$\rho = |\Psi\rangle \langle \Psi| ; \quad (8.5)$$

a density matrix (also called density operator) is therefore formally defined as the outer product of the wavefunction with its conjugate. It encodes all information about the state; for example, the probability of measuring the system to be at a position x (hence in a state $|x\rangle$) is given by $P(x) = \langle x | \rho | x \rangle$. From its definition in eq. (5), the DM has the very important property that it has unity trace:

$$\text{Tr}[\rho] = 1 , \quad (8.6)$$

where $\text{Tr}[\cdot]$ is the trace of an operator. This is seen very easily by taking the trace in an orthonormal basis in which the state $|\Psi\rangle$ is one of the basis states. For example, consider the DM

$$\rho = \frac{1}{\sqrt{2}} [|\Psi_1\rangle \langle \Psi_1| + |\Psi_2\rangle \langle \Psi_2|] , \quad (8.7)$$

where $|\Psi_{1,2}\rangle$ are orthogonal normalised states. The trace of ρ^2 for such a state is only $1/2$; such states satisfying $\text{Tr}[\rho^2] < 1$ are termed *mixed states*; the DM describing such states cannot be written in the form $\rho = |\Psi\rangle \langle \Psi|$ for any wavefunction $|\Psi\rangle$. On the other hand, the DMs that *can* be expressed like that are termed *pure states*, and have $\text{Tr}[\rho^2] = 1$.

One way in which a pure state can be realistically converted into a mixed state is through a measurement process. Consider the zero total spin state formed by two spins:

$$|\Psi\rangle = \frac{1}{\sqrt{2}} [|\uparrow, \downarrow\rangle + |\downarrow, \uparrow\rangle] , \rho = |\Psi\rangle \langle \Psi| . \quad (8.8)$$

The state $|\uparrow, \downarrow\rangle$ has the first spin pointing up and the second one down, and this is in a superposition with its flipped counterpart $|\downarrow, \uparrow\rangle$. This is clearly a pure state.

If we now measure the state of the second spin, it will collapse into one of its two states; the other spin will collapse into the opposite state (given the antiparallel configurations of the spins). What this means is that until one checks what the result of the measurement is, the first spin remains in a statistical superposition of both outcomes (up and down). This is formally expressed through a partial trace operation: the state of the first spin after a measurement of the second spin is obtained

by taking the full density matrix ρ and computing its trace over the states $|\sigma_B\rangle$ of the second spin, leaving out the states of the first spin from the trace:

$$\rho_1 = \text{Tr}_2[\rho] = \sum_{\sigma_2} \langle \sigma_2 | \rho | \sigma_2 \rangle ; \quad (8.9)$$

where $|\sigma_2\rangle$ sums over the states $|\uparrow\rangle$ and $|\downarrow\rangle$ of the second spin. this partial tracing operation Tr_2 codifies the measurement process, and the partially traced density matrix ρ_1 is termed a reduced density matrix (RDM) of the first spin, because it only captures information about the first spin (the second spin has been traced out).

To see the connection with mixedness, let us compute the RDM for the first spin explicitly:

$$\rho_1 = \langle \uparrow_2 | \rho | \uparrow_2 \rangle + \langle \downarrow_2 | \rho | \downarrow_2 \rangle = \frac{1}{2} [|\uparrow_1\rangle \langle \uparrow_1| + |\downarrow_1\rangle \langle \downarrow_1|] , \quad (8.10)$$

where the numbered subscripts on the kets and bras indicate that they only act on the respective Hilbert spaces. To see that ρ_1 describes a mixed state, we adopt a specific representation:

$$\begin{aligned} |\uparrow_1\rangle &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |\downarrow_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \\ \Rightarrow \rho_1 &= \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \rho_1^2 = \frac{1}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned} \quad (8.11)$$

The trace of ρ_1^2 is half, hence it is a mixed state.

The deviation of trace of ρ^2 from its maximum value of unity can be thought of as a measure of the lost information. In the example above, when we traced out the states of the second spin, we lost some information about the state of the first spin as well. This is because the two spins were *entangled*; they had quantum correlations between them. The entanglement arises from the fact that the states of the two spins are anti-correlated.

If instead we had chosen a non-entangled parent state to start from, the reduced density matrix would not have been mixed. For example, consider the state

$$|\Psi\rangle = \frac{1}{2} (|\uparrow_1\rangle + |\downarrow_1\rangle) \otimes (|\uparrow_2\rangle + |\downarrow_2\rangle) . \quad (8.12)$$

It is easy to see that this state does not contain any correlations between the two spins - no matter the configuration of the second spin, the first spin is always in the state $|\uparrow_1\rangle + |\downarrow_1\rangle$. The RDM for the first spin comes out to be

$$\rho_1 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \rho_1^2 = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad (8.13)$$

revealing that ρ_1 is not mixed.

The above method of calculating an RDM works for simple cases but can fail if (i) we have fermion signs to account for, or (ii) the full basis cannot be expressed as a direct product of local basis sets. The former arises if we are dealing with fermions, and the latter arises if we are working in a

rotated basis. We now derive an alternative method for calculating the RDM, one that automatically takes care of the fermion signs and works even if the basis is rotated and complicated to write down. We start by defining the RDM more precisely. Given a state $|\Psi\rangle$ and a bipartition into subsystems S_1 and its complement S_2 , the density matrix for the state is $\rho = |\Psi\rangle \langle\Psi|$. A basis state of the full system can be written as $|A, \alpha\rangle = |A\rangle \otimes |\alpha\rangle$, where $\{A\}$ and $\{\alpha\}$ are complete sets of basis states describing S_1 and S_2 respectively. The RDM for S_2 is then defined as

$$\rho_1 = \text{Tr}_\alpha \rho = \sum_\alpha \langle \alpha | \rho | \alpha \rangle . \quad (8.14)$$

The matrix elements of ρ_1 in the basis $\{A\}$ can be expressed as

$$\langle A | \rho_1 | B \rangle = \sum_\alpha \langle A, \alpha | \rho | B, \alpha \rangle . \quad (8.15)$$

We next define an operator $O_{A,B}$ that represents scattering processes between the various states of subsystem S_1 : $O_{A,B} = |A\rangle \langle B| \otimes I_2$, but acts as identity on S_2 . The expectation value of this operator is

$$\langle O_{A,B} \rangle = \text{Tr}(\rho O_{A,B}) = \sum_{C,\beta} \langle C, \beta | \rho | A \rangle \langle B | C, \beta \rangle = \sum_\beta \langle B, \beta | \rho | A, \beta \rangle . \quad (8.16)$$

Comparing with eq. 8.15, we obtain the relation

$$\langle A | \rho_1 | B \rangle = \langle O_{B,A} \rangle . \quad (8.17)$$

This result is significant because it guarantees that in order to calculate the RDM for a subsystem, it is enough to calculate the expectation values of all inter-state off-diagonal transition operators. For example, if we take subsystem S_1 to be a two fermion system represented by the operators c_1 and c_2 , the RDM for this subsystem is determined by the following matrix:

$$\rho_1 = \begin{pmatrix} \langle h_1 h_2 \rangle & \langle h_1 c_2^\dagger \rangle & \langle c_1^\dagger h_2 \rangle & \langle c_1^\dagger c_2^\dagger \rangle \\ & \langle h_1 n_2 \rangle & \langle c_1^\dagger c_2 \rangle & \langle c_1^\dagger n_2 \rangle \\ & & \langle n_1 h_2 \rangle & \langle n_1 c_2^\dagger \rangle \\ & & & \langle n_1 n_2 \rangle \end{pmatrix} \quad (8.18)$$

Quantum Fisher information: a witness of multipartite entanglement

The quantum Fisher information (QFI) F_Q [372, 373] quantifies how sensitive a state ρ is to unitary transformations generated by an observable $\hat{O}$. For a pure state $\rho = |\psi\rangle \langle\psi|$, $F_Q(\psi, \hat{O})$ is defined in terms of the variance of the operator $\hat{O}$ (i.e., the connected correlation function):

$$F_Q(\psi, \hat{O}) = 4\Delta(\hat{O})^2 = 4 \left(\langle \psi | \hat{O}^2 | \psi \rangle - \langle \psi | \hat{O} | \psi \rangle^2 \right) , \quad (8.19)$$

and it provides an upper bound on the precision that can be achieved in measuring the parameter θ that is dual to the observable $\hat{O}$: $\mathcal{N}(\Delta\theta)^2 \geq F_Q^{-1}$, $\mathcal{N}$ being the number of independent measurements [372,373]. $F_Q(\psi, \hat{O})$ is thus a measure of the entanglement arising from quantum fluctuation content in $|\psi\rangle$ (considered with respect to an eigenstate of $\hat{O}$). This can be made more manifest by considering a traceless operator $\hat{M}$ with eigenstates $\{|m\rangle\}$. Without loss of generality, we can rescale the operator such that $\sum m^2 = 1/2$. If $\varepsilon_m(\psi) \equiv 1 - |\langle m|\psi\rangle|^2$ is the geometric entanglement between a given state $|\psi\rangle$ and the eigenstates of $\hat{M}$, the QFI corresponding to $\hat{M}$ in the state ψ can be written as

$$F_Q(\psi, \hat{M}) = 4 \left[\frac{1}{2} - \sum_m m^2 \varepsilon_m - \left(\sum_m m \varepsilon_m \right)^2 \right]. \quad (8.20)$$

The total geometric entanglement $\sum_m \varepsilon_m$ is constrained to be $d_M - 1$, where d_M is the dimension of the operator $\hat{M}$. Because each eigenvalue m has a magnitude of at most $1/\sqrt{2}$ (all m^2 must add up to $1/2$), the maximum magnitude of the last two terms in eq. (8.20) (and hence the minimum value of the QFI) is attained for the case when some of the ε_m are zero. However, because the eigenstates are orthogonal, only one of them can have perfect overlap with the state $|\psi\rangle$ and only one ε_m can be zero at a time. The case of minimum QFI therefore corresponds to $\varepsilon_{m^*} = 0, \varepsilon_{m \neq m^*} = 1$, leading to $F_Q = 0$. The opposite situation arises when all the geometric entanglement measures are non-zero and equal, $\varepsilon_m = 1/d_M$, leading to a maximum QFI value of $F_Q = 2/d_M$. A uniformly spread geometric entanglement distribution, therefore, leads to a larger QFI, while a more focused distribution leads to a smaller QFI. This sums up the relation between the quantum Fisher information and geometric entanglement.

If the operator $\hat{M}$ is such that its eigenstates are separable states from the perspective of a certain party, the vanishing QFI arises when $|\psi\rangle$ has zero geometric entanglement with respect to one of the separable states, and the QFI saturates when $|\psi\rangle$ has a finite geometric entanglement with all of the states. A large QFI can thus be associated with more entanglement. Moreover, if the eigenstates of $\hat{M}$ are symmetry-broken states, a vanishing QFI indicates that $|\psi\rangle$ is very close to such a symmetry-broken state, while a large QFI indicates that $|\psi\rangle$ is a uniform superposition of such symmetry-broken states and therefore preserves the symmetry as a whole. A larger QFI points towards the presence of more quantum fluctuations and a reduced susceptibility towards symmetry-breaking.

We will now relate the QFI to some other measures of correlation. The QFI corresponding to an observable can be directly related to the static lesser Greens functions associated with that observable:

$$F_Q(\psi, \hat{O}_2) = 4 \left(\left\langle \hat{O}_2 \hat{O}_1^\dagger \right\rangle_{\hat{O}_1^\dagger = \hat{O}_2} - \left\langle \hat{O}_2 \hat{O}_1^\dagger \right\rangle_{\hat{O}_1 = 1}^2 \right) = 4 \left[G_{O_2, O_2}^<(t \rightarrow \infty) - \left(G_{1, O_2}^<(t \rightarrow \infty) \right)^2 \right] \quad (8.21)$$

Further, the retarded Greens function defined above in eq. (??) can very generally be related to the QFI. By using the simplified forms $\frac{i}{2} G_{O_2, O_2}^R(t \rightarrow \infty) = \langle (O_2)^2 \rangle$ and $\frac{i}{2} G_{1, O_2}^R(t \rightarrow \infty) = \langle O_2 \rangle$, we get

$$F_Q(\psi, \hat{O}_2) = 2i G_{O_2, O_2}^R(t \rightarrow \infty) + \left(G_{1, O_2}^R(t \rightarrow \infty) \right)^2. \quad (8.22)$$

For the sake of completeness, we generalise the above relations to finite temperatures by adopting the approach of Hauke et al. [373]. For a mixed state at inverse temperature β characterised by the density matrix $\rho = \sum_n f_n |n\rangle \langle n|$ where $(E_n, |n\rangle)$ are the eigenvalue-eigenstate pairs and $f_n = e^{-\beta E_n} / \sum_n e^{-\beta E_n}$ are the Boltzmann weights, the FQI can be assigned a spectral representation of the form [373]

$$F_Q(\rho, \hat{O}) = 2 \sum_{m,n} \frac{(f_m - f_n)^2}{f_m + f_n} |\langle m | \hat{O} | n \rangle|^2. \quad (8.23)$$

By using the identity $\frac{f_m - f_n}{f_m + f_n} = \int_{-\infty}^{\infty} d\omega \delta(\omega + E_m - E_n) \tanh(\beta\omega/2)$, the finite temperature FQI can be written in the form

$$F_Q(\rho, \hat{O}) = 2 \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \sum_{m,n} (f_m - f_n) \delta(\omega + E_m - E_n) |\langle m | \hat{O} | n \rangle|^2. \quad (8.24)$$

In order to relate this with a correlation function, we now point out that the lesser Greens function $G^<$ and the greater Greens function $G^>$ that we will now define admit spectral representations in terms of the eigenstates $|n\rangle$ of the full problem:

$$G_{O^\dagger, O}^<(t) \equiv i \langle \hat{O}^\dagger \hat{O}(t) \rangle \implies G_{O^\dagger, O}^<(\omega) = 2\pi i \sum_{m,n} f_n \delta(\omega + E_m - E_n) |\langle m | \hat{O}^\dagger | n \rangle|^2, \quad (8.25)$$

$$G_{O^\dagger, O}^>(t) \equiv -i \langle \hat{O}(t) \hat{O}^\dagger \rangle \implies G_{O^\dagger, O}^>(\omega) = -2\pi i \sum_{m,n} f_m \delta(\omega + E_m - E_n) |\langle m | \hat{O} | n \rangle|^2. \quad (8.26)$$

If $\hat{O}$ is a Hermitian operator, eqs. (8.25) can be used to rewrite the QFI in terms of $G_{O^\dagger, O}^\lessgtr(\omega)$. Indeed, by combining eqs. (8.25) and eq. (8.24) with the assumption $\hat{O} = \hat{O}^\dagger$, we get

$$F_Q(\rho, \hat{O}) = -\frac{1}{\pi i} \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \left(G_{O, O}^>(\omega) + G_{O, O}^<(\omega) \right). \quad (8.27)$$

This constitutes the general relation between the finite temperature QFI for a general Hermitian many-particle operator $\hat{O}$ and the corresponding many-particle correlations at zero and finite frequencies.

The Kubo linear response function associated with the operator $\hat{O}(t)$ due to a perturbation from the same operator is associated with a correlation function/susceptibility $C_{\hat{O}}(t, t') = i\theta(t - t') \langle [\hat{O}(t), \hat{O}(t')] \rangle$. By going to the frequency domain, the imaginary part of such a function can be related to the sum of the lesser and greater Greens functions:

$$\begin{aligned} C_{\hat{O}}(\omega) &= i \int_0^\infty d(t - t') e^{i\omega(t-t')} \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\ \implies C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^* &= i \int_0^\infty d(t - t') \left(e^{i\omega(t-t')} - e^{-i\omega(t-t')} \right) \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\ &= i \int_{-\infty}^\infty d(t - t') e^{i\omega(t-t')} \left(\langle \hat{O}(t) \hat{O}(t') \rangle - \langle \hat{O}(t') \hat{O}(t) \rangle \right) \\ &= - \left(G_{O, O}^>(\omega) + G_{O, O}^<(\omega) \right). \end{aligned}$$

This allows us to connect the QFI with the imaginary part of the Kubo correlation function/susceptibility obtained in Ref. [373]:

$$F_Q(\rho, \hat{O}) = \frac{1}{\pi i} \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) (C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^*) . \quad (8.28)$$

2.8.2 Decomposition of entanglement entropy in terms of multipartite information I_n

We first define a partitioning of the entire system into N disjoint regions, $\mathcal{S}_N = \{\nu_1, \nu_2, \dots, \nu_N\}$. Going beyond bipartite mutual information, one can define the n -partite information I_n between the members of $\mathcal{S}_n = \{\nu_1, \nu_2, \dots, \nu_n\}$, which is itself a subset of the exhaustive set of regions $\mathcal{S}_N$:

$$I_n(\nu_1, \dots, \nu_n) = \sum_{\sigma \in \mathcal{P}(\mathcal{S}_n)} (-1)^{1+|\sigma|} S_{EE}(\sigma) , \quad (8.29)$$

where σ is a subset drawn from the power set of $\mathcal{S}_n$, and $|\sigma|$ is the number of members in the subset σ . The 4-partite information can be written as

$$\begin{aligned} I_4(\nu_1, \nu_2, \nu_3, \nu_4) &= S_{EE}(\nu_1) + S_{EE}(\nu_2) + S_{EE}(\nu_3) + S_{EE}(\nu_4) - S_{EE}(\nu_1 \cup \nu_2) - S_{EE}(\nu_1 \cup \nu_3) \\ &\quad - S_{EE}(\nu_1 \cup \nu_4) - S_{EE}(\nu_2 \cup \nu_3) - S_{EE}(\nu_2 \cup \nu_4) - S_{EE}(\nu_3 \cup \nu_4) + S_{EE}(\nu_1 \cup \nu_2 \cup \nu_3) \\ &\quad + S_{EE}(\nu_1 \cup \nu_2 \cup \nu_4) + S_{EE}(\nu_1 \cup \nu_3 \cup \nu_4) + S_{EE}(\nu_2 \cup \nu_3 \cup \nu_4) + S_{EE}(\nu_1 \cup \nu_2 \cup \nu_3 \cup \nu_4) . \end{aligned} \quad (8.30)$$

One can also define the average n -partite information $\bar{I}_n$ by considering all sets of n regions:

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \sum_{\mathcal{S}_n \in C_n(\mathcal{S}_N)} I_n(\mathcal{S}_n) = \frac{(N-n)!n!}{N!} \sum_{\mathcal{S}_n \in C_n(\mathcal{S}_N)} \sum_{\sigma \in \mathcal{P}(\mathcal{S}_n)} (-1)^{1+|\sigma|} S_{EE}(\sigma) , \quad (8.31)$$

where $C_n(\mathcal{S}_N)$ represents all ways of choosing n distinct elements out of the full set $\mathcal{S}_N$ of subsystems.

The expression for $\bar{I}_n$ can be simplified by splitting the summation over the elements of the power set $\mathcal{P}$ into sets of definite number:

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \sum_{\mathcal{S}_n \in C_n(\mathcal{S}_N)} I_n(\mathcal{S}_n) = \frac{(N-n)!n!}{N!} \sum_{\mathcal{S}_n \in C_n(\mathcal{S}_N)} \sum_{m=1}^n \sum_{\sigma_m \in \mathcal{P}_m(\mathcal{S}_n)} (-1)^{1+|\sigma|} S_{EE}(\sigma_m) , \quad (8.32)$$

where $\mathcal{P}_m$ represents all subsets of the power set that have m elements. With some combinatorial magic, it can be shown that

$$\sum_{\mathcal{S}_n \in C_n(\mathcal{S}_N)} \sum_{\sigma_m \in \mathcal{P}_m(\mathcal{S}_n)} = \binom{N-m}{n-m} \sum_{\sigma_m \in C_m(\mathcal{S}_N)} . \quad (8.33)$$

To see how this comes about, note that what this equation says is that summing over combinations of subsets of length n of the super set $\mathcal{S}_N$ and further summing over all subsets of length $m \leq n$ for

each set S_n is equivalent to summing just once over subsets of length m of the super set S_N but now each subset of length m appears with a frequency $f_m = \binom{N-m}{n-m}$. The value for the frequency f_m is simply the number of subsets S_n in which the smaller subset σ_m appears. This can be calculated by allowing m of the n slots in S_n to be filled by the members of σ_m , and the remaining $n - m$ slots then have to be filled by choosing from the remaining $N - m$ members of the super set S_N ; there's obviously $\binom{N-m}{n-m}$ ways of carrying this out.

Implementing this into the expression for $\bar{I}_n$ gives

$$\bar{I}_n = \sum_{m=1}^n (-1)^{1+m} \frac{\binom{n}{m}}{\binom{N}{m}} \sum_{\sigma_m \in C_m(S_N)} S_{EE}(\sigma_m) = \sum_{m=1}^n (-1)^{1+m} \binom{n}{m} S_{EE}^{(m)}, \quad (8.34)$$

where we have define the average entanglement entropy $S_{EE}^{(m)}$ for all m -member sets of regions:

$$S_{EE}^{(m)} = \frac{1}{\binom{N}{m}} \sum_{\sigma_m \in C_m(S_N)} S_{EE}(\sigma_m). \quad (8.35)$$

With most of the machinery in place, we would now like to express the entanglement entropy S_{EE} purely in terms multipartite measures I_n , for the reasons stated above. For a system that is invariant under real-space translation (as our lattice model is), the entanglement entropy of any real-space interval should only depend on the extent of the interval. It therefore suffices to calculate the entanglement entropy averaged over all such intervals, and the corresponding measure in our formulation is $S_{EE}^{(1)}$.

We will now express $S_{EE}^{(1)}$ in terms of $\bar{I}_n$, by eliminating $S_{EE}^{(m)}$ ($m > 1$) from the equations. We adopt the ansatz

$$S_{EE}^{(1)} = \sum_{n=2}^N \lambda_n \bar{I}_n = \sum_{n=2}^N \lambda_n \binom{n}{m} \sum_{m=1}^n (-1)^{1+m} S_{EE}^{(m)}. \quad (8.36)$$

In order to separate $S_{EE}^{(1)}$ from the rest of the $S_{EE}^{(m)}$, we exchange the order of summation:

$$S_{EE}^{(1)} = \sum_{m=1}^N (-1)^{1+m} S_{EE}^{(m)} \sum_{n=m}^N \lambda_n \binom{n}{m} - S_{EE}^{(1)} \lambda_1 = S_{EE}^{(1)} \sum_{n=2}^N n \lambda_n - \sum_{m=2}^{N-1} S_{EE}^{(m)} (-1)^m \sum_{n=m}^N \binom{n}{m} \lambda_n, \quad (8.37)$$

where the constraint of $m \leq n$ in the previous form is now enforced by letting n run from m to N . The $N - 1$ coefficients $\{\lambda_n\}$ will be determined by requiring that the $S_{EE}^{(m)}$ ($m > 1$) vanish on the right-hand side, as well as by matching the coefficients of $S_{EE}^{(1)}$ on both sides. Note that m runs only upto $N - 1$ in the second term; this is because, $S_{EE}^{(N)}$ is the entanglement entropy of the full system and hence automatically zero. The equations are

$$\sum_{n=2}^N n \lambda_n = 1, \quad (8.38)$$

$$\sum_{n=m}^N \binom{n}{m} \lambda_n = 0; \quad m = 2, \dots, N - 1. \quad (8.39)$$

We focus on the second equation first, since it is very short for m close to N . For $m = N - 1$ and $m = N - 2$, we get

$$\begin{aligned} m = N - 1 : \lambda_{N-1} + N\lambda_N &= 0 \implies \lambda_{N-1} = -N\lambda_N \\ m = N - 2 : \lambda_{N-2} + (N-1)\lambda_{N-1} + \frac{N(N-1)}{2}\lambda_N &= 0 \implies \lambda_{N-2} = \frac{N(N-1)}{2}\lambda_N. \end{aligned} \quad (8.40)$$

We extrapolate this to the general solution,

$$\lambda_m = (-1)^m \binom{N}{m} \lambda_N, \quad m < N, \quad (8.41)$$

and then prove that if this is true for λ_{N-m} , it also solves the equation for λ_{N-m-1} , thus amounting to a proof by mathematical induction. The general equation for λ_{N-m-1} is

$$\sum_{n=N-m-1}^N \binom{n}{N-m-1} \lambda_n = 0 \implies \lambda_{N-m-1} + \sum_{m'=0}^m \binom{N-m'}{N-m-1} \lambda_{N-m'} = 0. \quad (8.42)$$

Substituting the extrapolated solution gives

$$\begin{aligned} \lambda_{N-m-1} &= - \sum_{m'=0}^m (-1)^{N-m'} \binom{N-m'}{N-m-1} \binom{N}{m'} \lambda_N = \binom{N}{m+1} \lambda_N \sum_{m'=0}^m (-1)^{N-m'-1} \binom{m+1}{m'} \\ &= (-1)^{N-m-1} \binom{N}{m+1} \lambda_N, \end{aligned} \quad (8.43)$$

which is consistent with the general solution eq. 8.41. This determines all coefficients λ_n for $n < N$ in terms of λ_N . To determine λ_N itself, we use eq. 8.38. Substituting eq. 8.41 into it gives

$$\lambda_N \sum_{n=2}^N n (-1)^n \binom{N}{n} = 1 \implies \lambda_N = \frac{(-1)^N}{N}. \quad (8.44)$$

The decomposition of entanglement entropy into multipartite measures is now complete:

$$\begin{aligned} S_{\text{EE}}^{(1)} &= \frac{1}{N} \sum_{n=2}^N (-1)^n \binom{N}{n} \bar{I}_n, \\ \bar{I}_n &= \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(S_N)} I_n(S_n), \end{aligned} \quad (8.45)$$

where we have assumed N is even in order to drop the global phase factor $(-1)^N$.

2.9 Wavefunction Renormalisation Group

2.9.1 General idea

A reverse URG analysis involves studying the evolution of wavefunctions from the IR to the UV, by implementing inverse renormalisation group transformations. This is made possible by the fact

that the IR fixed point Hamiltonians are often tractable and allow the computation of wavefunctions. Such an analysis gives insight into how states change as high-energy quantum fluctuations are resolved during the Hamiltonian renormalisation from the UV to the IR. It also allows studying the evolution of entanglement measures and correlation functions.

We will start with a very simple IR ground state wavefunction, and then go back towards the UV ground state by applying the inverse unitary operator $U^\dagger$:

$$\begin{aligned}
 U : \underbrace{|1, 2, \dots, N\rangle}_{\text{UV ground state}} &\rightarrow |1, 2, \dots, N-1\rangle |N\rangle \rightarrow \dots \rightarrow \underbrace{|1, 2, \dots, N^*\rangle |N^*+1\rangle \dots |N\rangle}_{\text{IR ground state}} \\
 U^\dagger : \underbrace{|1, 2, \dots, N^*\rangle |N^*+1\rangle \dots |N\rangle}_{\text{IR ground state}} &\rightarrow |1, 2, \dots, N^*+1\rangle |N^*+2\rangle \dots |N\rangle \rightarrow \dots \rightarrow \underbrace{|1, 2, \dots, N\rangle}_{\text{UV ground state}}
 \end{aligned}$$

The first process is the forward RG which we used to obtain the scaling equations. The second process is the reverse RG which we will undertake now. The method has already been applied to study the renormalisation of entanglement in various models of strong correlation. In general, we start with an IR wavefunction that consists a certain number of momentum states n_1 still entangled with the impurity, and the remaining momentum states n_2 disentangled from the impurity. The former are said to be part of the emergent Kondo cloud, while the latter are said to be part of the integrals of motion (IOMs) and appear in direct product with the cloud+impurity system in the ground state wavefunction. Each momentum state is tagged with two conduction bath levels, one above the Fermi surface and one below. These will be represented as $k, \pm$. If we represent the emergent cloud momentum states as $\{k_i\}$ and the IOM states as $\{q_i\}$, the ground state wavefunction can be written as

$$|\Psi\rangle_0 = \left(\otimes_{i=1}^{n_2} |q_i, -, \uparrow\rangle |q_i, -, \downarrow\rangle \right) |\Phi\rangle_{\text{cloud}} \left(\otimes_{i=1}^{n_1} |q_i, +, \uparrow\rangle |q_i, +, \downarrow\rangle \right) \quad (9.1)$$

Throughout, we have assumed that the IOMs below the Fermi surface are occupied while those above are unoccupied. This means:

$$|\Psi\rangle_0 = \left(\otimes_{i=1}^{n_2} |\hat{n}_{q_i, -, \uparrow} = 1\rangle |\hat{n}_{q_i, -, \downarrow} = 1\rangle \right) |\Phi\rangle_{\text{cloud}} \left(\otimes_{i=1}^{n_1} |\hat{n}_{q_i, +, \uparrow} = 0\rangle |\hat{n}_{q_i, +, \downarrow} = 0\rangle \right) \quad (9.2)$$

$|\Phi\rangle_{\text{cloud}}$ will be obtained by diagonalising a small system of n_1 conduction bath k states. This completes the construction of the IR ground state $|\Psi\rangle_0$. The algorithm of reverse RG then involves applying unitary transformations on this IR wavefunction; the unitary transformations will be the inverse of those that were applied during the forward RG algorithm. Since the forwards RG transformations decoupled k states and transformed them into the IOMS, the inverse transformations will take the momentum states in the IOMS and re-entangle them with the impurity+cloud system, one at a time. This is shown in fig. 2.11.

The next step is to write down the unitaries that will take us from the IR ground state to the UV ground state. In the forward RG, we used the following unitary for decoupling the shell ϵ_q and spin β is

$$U_{q\beta} = \frac{1}{\sqrt{2}} \left[1 + \eta_{0\beta} + \eta_{1\beta}^\dagger \right]. \quad (9.3)$$

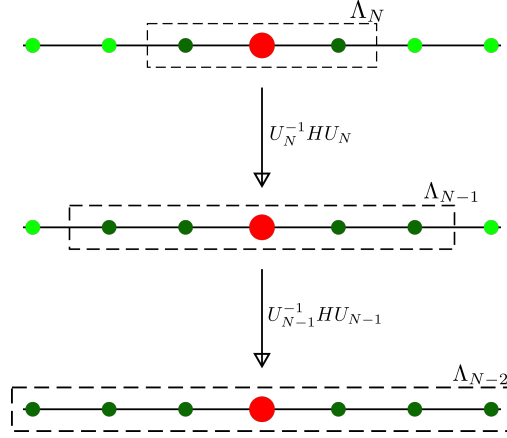


Figure 2.11: We start with a Hamiltonian with an impurity site (red) coupled with two conduction electrons (dark green), with four other decoupled electrons (bright green). The dotted rectangle represents the emergent window $(-\Lambda_j, \Lambda_j)$ at each step; the electrons inside that rectangle are still entangled with the impurity, while the ones inside have been decoupled. The next step of reverse RG involves applying the inverse transformation on the Hamiltonian, which will couple two more electrons from the IOMS (hence four dark green circles in the second step), leading to an enlargement of the emergent window. The unitary varies for each step, hence the notation U_j .

Here, the subscripts 0 and 1 indicate it decouples an electron above and below the Fermi surface respectively. The inverse transformation for re-entangling $q\beta$ is

$$U_{q\beta}^\dagger = \frac{1}{\sqrt{2}} \left[1 + \eta_{0\beta}^\dagger + \eta_{1\beta} \right]. \quad (9.4)$$

The wavefunction after reversing one step of the RG will thus be

$$|\Psi\rangle_1 = U_{q\uparrow}^\dagger U_{q\downarrow}^\dagger |\Psi\rangle_0 \quad (9.5)$$

Here, q is the first momentum state immediately outside the emergent cloud. Re-entangling further momentum states using their respective unitaries lead to the sequence of wavefunctions $|\Psi\rangle_2, |\Psi\rangle_3$ and so on. A straightforward way of calculating entanglement and correlations across the RG flow is to create a second-quantised prototypical form of the IR state $|\Psi_0\rangle$, and then apply the inverse unitary operators $\{U_i\}$ to this wavefunction in order to generate the family of states $|\Psi_i\rangle = U_i |\Psi_{i-1}\rangle, i > 0$. Quantities like entanglement entropy and correlations can then be calculated by using this family of states. If the unitary is of a specific form, there is instead a more efficient method of doing this, which is what we will describe in the next subsection.

2.9.2 Evolution equation for state tensor coefficients

In the following, we restrict ourselves to the reverse RG analysis of an impurity problem hybridising with the local degree of freedom of a conduction bath. At every step, we allow two states from the IOMS to interact with the already-entangled states (one above the Fermi energy and one below,

in order to maintain particle-hole symmetry). After the j^{th} application of the reverse RG transformation, $2j$ number of states are entangled with the impurity site, and the state can be represented as

$$|\Psi_j\rangle = \sum_{\nu_d, \nu_1, \dots, \nu_{2j}} C_{\nu_d, \nu_1, \dots, \nu_{2j}, I_{2j+1}}^{(j)} |\nu_d\rangle |\nu_1\rangle |\nu_2\rangle \dots |\nu_{2j}\rangle \otimes |I_{2j+1}\rangle, \quad (9.6)$$

where $|\nu_d\rangle$ and $|\nu_i\rangle$ ($i \in [1, 2j]$) sum over the configurations of the impurity site and the effective zeroth site configuration at the j^{th} step. $|I_{2j+1}\rangle$ is the configuration of the integrals of motion from the $(2j+1)^{\text{th}}$ site and beyond. We define the product state $|\psi_{dz}(j)\rangle = |\nu_d\rangle |\nu_1\rangle \dots |\nu_{2j}\rangle$ for future reference.

We consider a restricted form of RG evolution, where the unitary transformation from IR towards UV can be expressed at any step as

$$U_j^\dagger = 1 + \sum_{i \in [1, 2j]} \sum_{\nu_d, \mu_d, \nu_i, \mu_i} \mathcal{T}_{\nu_d}^{\mu_d} (J_{\nu_d, \nu_i}^{\mu_d, \mu_i} c_{2j+1}^{\nu_i} \mathcal{T}_{\nu_i}^{\mu_i} + K_{\nu_d, \nu_i}^{\mu_d, \mu_i} c_{2j+2}^{\dagger} \mathcal{T}_{\nu_i}^{\mu_i}), \quad (9.7)$$

where $\mathcal{T}_\alpha^\nu = |\nu\rangle \langle \alpha|$ is a transition operator and sums over all transition operators of the specific subspace (impurity or bath sites). We first obtain the action of the transformation $U_j^\dagger$ on the basis states. We have

$$\begin{aligned} & \mathcal{T}_{\mu_d}^{\nu_d} c_{2j+2}^{\dagger} \mathcal{T}_{\mu_i}^{\nu_i} |\mu_d\rangle \dots |\mu_i\rangle \dots |2_{2j+1}\rangle |0_{2j+2}\rangle \\ &= \mathcal{T}_{\mu_d}^{\nu_d} c_{2j+2}^{\dagger} (-1)^{\mu_d + \sum_{k < i} \nu_i} |\mu_d\rangle \dots |\nu_i\rangle \dots |2_{2j+1}\rangle |0_{2j+2}\rangle \\ &= \mathcal{T}_{\mu_d}^{\nu_d} (-1)^{\mu_d + \sum_{k < i} \nu_i} (-1)^{\mu_d + \sum_k \nu_i} |\mu_d\rangle \dots |\nu_i\rangle \dots |2_{2j+1}\rangle |1_{2j+2}\rangle \\ &= (-1)^{\sum_{k \geq i} \nu_i} |\nu_d\rangle \dots |\nu_i\rangle \dots |2_{2j+1}\rangle |1_{2j+2}\rangle, \end{aligned} \quad (9.8)$$

and

$$\mathcal{T}_{\mu_d}^{\nu_d} c_{2j+1} \mathcal{T}_{\mu_i}^{\nu_i} |\mu_d\rangle \dots |\mu_i\rangle \dots |2_{2j+1}\rangle |0_{2j+2}\rangle = (-1)^{\sum_{k \geq i} \nu_i} |\nu_d\rangle \dots |\nu_i\rangle \dots |1_{2j+1}\rangle |0_{2j+2}\rangle. \quad (9.9)$$

Using these expressions, we construct the state at the $(j+1)^{\text{th}}$ step:

$$\begin{aligned} |\Psi_{j+1}\rangle &= \sum_{\nu_d, \nu_1, \dots, \nu_j} C_{\nu_d, \nu_1, \dots, \nu_j}^{(j)} |\psi_{dz}(j)\rangle |2_{2j+1}\rangle |0_{2j+2}\rangle |I_{2j+3}\rangle \\ &+ \sum_i \sum_{\mu_d, \mu_i, \nu_d, \nu_i} C_{\mu_d, \dots, \mu_i, \dots, \nu_d, \nu_i}^{(j)} J_{\mu_d, \mu_i}^{\nu_d, \nu_i} (-1)^{\sum_{k \geq i} \nu_k} |\psi_{dz}(j)\rangle |1_{2j+1}\rangle |0_{2j+2}\rangle |I_{2j+3}\rangle \\ &+ \sum_i \sum_{\mu_d, \mu_i, \nu_d, \nu_i} C_{\mu_d, \dots, \mu_i, \dots, \nu_d, \nu_i}^{(j)} K_{\mu_d, \mu_i}^{\nu_d, \nu_i} (-1)^{\sum_{k \geq i} \nu_k} |\psi_{dz}(j)\rangle |2_{2j+1}\rangle |1_{2j+2}\rangle |I_{2j+3}\rangle \end{aligned} \quad (9.10)$$

Gathering similar terms and normalising the state, we can write it as

$$\begin{aligned} |\Psi_{j+1}\rangle &= \frac{1}{L_j} \sum_{\nu_d, \{\nu_i\}} |\psi_{dz}(j)\rangle \left[C_{\nu_d, \{\nu_i\}, 1_{2j+1}, 0_{2j+2}, I_{j+1}}^{(j)} |2_{2j+1}\rangle |0_{2j+2}\rangle + C_{+, \nu_d, \{\nu_i\}}^{(j)} |2_{2j+1}\rangle |1_{2j+2}\rangle \right. \\ &\quad \left. + C_{-, \nu_d, \{\nu_i\}}^{(j)} |1_{2j+1}\rangle |0_{2j+2}\rangle \right] |I_{2j+3}\rangle, \end{aligned} \quad (9.11)$$

where $L_j^2 = \sum_{v_d, \{v_i\}} \left(C_{v_d, \{v_i\}}^{(j)} \right)^2 + C_{+, v_d, \{v_i\}}^{(j)} \right)^2 + C_{-, v_d, \{v_i\}}^{(j)} \right)^2$ defines the normalisation factor, and

$$\begin{aligned} C_{+, v_d, \{v_i\}} &= \sum_{i, \mu_d, \mu_i} C_{\mu_d, \dots, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} K_{\mu_d, \mu_i}^{v_d, v_i} (-1)^{\sum_{k \geq i} v_k} \\ C_{-, v_d, \{v_i\}} &= \sum_{i, \mu_d, \mu_i} C_{\mu_d, \dots, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} J_{\mu_d, \mu_i}^{v_d, v_i} (-1)^{\sum_{k \geq i} v_k} . \end{aligned} \quad (9.12)$$

We can now read off the renormalisation of the state tensor coefficients:

$$\begin{aligned} C_{v_d, v_1, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}, I_{2j+3}}^{(j+1)} &= \frac{1}{L_j} C_{v_d, v_1, \dots, v_{2j}, 2_{2j+1}, 0_{2j+2}, I_{2j+3}}^{(j)} , \\ C_{v_d, v_1, \dots, v_{2j}, 2_{2j+1}, 1_{2j+2}, I_{2j+3}}^{(j+1)} &= \frac{1}{L_j} \sum_i (-1)^{\sum_{k \geq i} v_k} \sum_{\mu_d, \mu_i} C_{\mu_d, \dots, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} K_{\mu_d, \mu_i}^{v_d, v_i} , \\ C_{v_d, v_1, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}, I_{2j+3}}^{(j+1)} &= \frac{1}{L_j} \sum_i (-1)^{\sum_{k \geq i} v_k} \sum_{\mu_d, \mu_i} C_{\mu_d, \dots, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} J_{\mu_d, \mu_i}^{v_d, v_i} , \end{aligned} \quad (9.13)$$

We now make an important assumption: the coefficients $K_{\mu_d, \mu_i}^{v_d, v_i}$ and $J_{\mu_d, \mu_i}^{v_d, v_i}$ are uniform for the set of states $S(v_i)$ and zero for other states. That gives

$$\begin{aligned} C_{v_d, v_1, \dots, v_{2j}, 1_{2j+1}, 1_{2j+2}, I_{2j+3}}^{(j+1)} &= \frac{1}{L_j} K \sum_i (-1)^{\sum_{k \geq i} v_k} \sum_{\substack{\mu_d \in S(v_d), \\ \mu_i \in S(v_i)}} C_{\mu_d, \dots, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} , \\ C_{v_d, v_1, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}, I_{2j+3}}^{(j+1)} &= \frac{1}{L_j} J \sum_i (-1)^{\sum_{k \geq i} v_k} \sum_{\substack{\mu_d \in S(v_d), \\ \mu_i \in S(v_i)}} C_{\mu_d, \dots, \mu_i, \dots, v_{2j}, 1_{2j+1}, 0_{2j+2}}^{(j)} , \end{aligned} \quad (9.14)$$

One can also look at renormalisation of expectation values of operators; this is useful for tracking the evolution of correlations and entanglement measures. Consider an operator $O_{\mu_k}^{v_k}$ that acts on the k^{th} bath site. The expectation value of this operator at any step can be written as

$$\begin{aligned} \langle O_{v_k}^{\mu_k} \rangle_j &= \sum_{v_d, v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_{2j}} C_{v_d, \dots, \mu_k, \dots}^{(j)} C_{v_d, \dots, v_k, \dots}^{(j)} , \\ \langle O_{v_k}^{\mu_k} \rangle_{j+1} &= \sum_{v_d, v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_{2j+2}} C_{v_d, \dots, \mu_k, \dots}^{(j+1)} C_{v_d, \dots, v_k, \dots}^{(j+1)} \end{aligned} \quad (9.15)$$

2.10 Zero temperature Greens function in frequency domain

2.10.1 Spectral representation of Greens function

The impurity retarded Green's function is defined as

$$G_{d\sigma}(t) = -i\theta(t) \langle \{O_{\sigma}(t), O_{\sigma}^{\dagger}\} \rangle \quad (10.1)$$

where the average $\langle \rangle$ is over a canonical ensemble at temperature T , and O_σ is the excitation whose spectral function we are interested in. What follows is a standard calculation where we write the Green's function in the spectral representation. The ensemble average for the operator O_σ can be written in terms of the exact eigenstates of the Hamiltonian:

$$H |n\rangle = E_n |n\rangle, \quad \langle O_\sigma \rangle \equiv \frac{1}{Z} \sum_n \langle n | O_\sigma | n \rangle e^{-\beta E_n} \quad (10.2)$$

where $Z = \sum_n e^{-\beta E_n}$ is the partition function and $\{|n\rangle\}$ is the set of eigenfunctions of the Hamiltonian. We can therefore write

$$\begin{aligned} & \langle \{O_\sigma(t), O_\sigma^\dagger\} \rangle \\ &= \frac{1}{Z} \sum_m e^{-\beta E_m} \langle m | \{O_\sigma(t), O_\sigma^\dagger\} | m \rangle \\ &= \frac{1}{Z} \sum_{m,n} e^{-\beta E_m} \langle m | \left(O_\sigma(t) |n\rangle \langle n| O_\sigma^\dagger + O_\sigma^\dagger |n\rangle \langle n| O_\sigma(t) \right) | m \rangle \quad \left[\sum_n |n\rangle \langle n| = 1 \right] \\ &= \frac{1}{Z} \sum_{m,n} e^{-\beta E_m} \langle m | \left(e^{iH^*t} O_\sigma e^{-iH^*t} |n\rangle \langle n| O_\sigma^\dagger + O_\sigma^\dagger |n\rangle \langle n| e^{iH^*t} O_\sigma e^{-iH^*t} \right) | m \rangle \\ &= \frac{1}{Z} \sum_{m,n} e^{-\beta E_m} \left(e^{i(E_m - E_n)t} \langle m | O_\sigma | n \rangle \langle n | O_\sigma^\dagger | m \rangle + e^{i(E_n - E_m)t} \langle m | O_\sigma^\dagger | n \rangle \langle n | O_\sigma | m \rangle \right) \\ &= \frac{1}{Z} \sum_{m,n} e^{i(E_m - E_n)t} || \langle m | O_\sigma | n \rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \end{aligned} \quad (10.3)$$

The time-domain impurity Green's function can thus be written as (this is the so-called Lehmann-Kallen representation)

$$G_{d\sigma} = -i\theta(t) \frac{1}{Z} \sum_{m,n} e^{i(E_m - E_n)t} || \langle m | O_\sigma | n \rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \quad (10.4)$$

We are interested in the frequency domain form.

$$\begin{aligned} G_{dd}^\sigma(\omega) &= \int_{-\infty}^{\infty} dt e^{i(\omega)t} G_{dd}^\sigma(t) \\ &= \frac{1}{Z} \sum_{m,n} || \langle m | O_\sigma | n \rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) (-i) \int_{-\infty}^{\infty} dt \theta(t) e^{i(\omega + E_m - E_n)t} \end{aligned} \quad (10.5)$$

To evaluate the time-integral, we will use the integral representation of the Heaviside function:

$$\theta(t) = -\frac{1}{2\pi i} \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} \frac{1}{x + i\eta} e^{-ixt} dx \quad (10.6)$$

With this definition, the integral in $G_{d\sigma}(\omega)$ becomes

$$\begin{aligned}
 (-i) \int_{-\infty}^{\infty} dt \theta(t) e^{i(\omega+i0^+ + E_m - E_n)t} &= (-i) \frac{-1}{2\pi i} \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{1}{x + i\eta} \int_{-\infty}^{\infty} dt e^{i(\omega+i0^+ + E_m - E_n - x)t} \\
 &= \frac{1}{2\pi} \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{1}{x + i\eta} 2\pi \delta(\omega + i0^+ + E_m - E_n - x) \quad (10.7) \\
 &= \frac{1}{\omega + E_m - E_n + i0^+} .
 \end{aligned}$$

The frequency-domain Green's function is thus

$$G_{dd}^{\sigma}(\omega) = \frac{1}{Z} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \frac{1}{\omega + E_m - E_n + i0^+} \quad (10.8)$$

The infinitesimally positive imaginary part in the denominator shifts the poles onto the lower half of the complex plane, leaving it analytic in the upper half: this is necessary to make the retarded Greens function causal.

2.10.2 Real and imaginary parts - The Sokhotski-Plemelj theorem

In order to write down the real and imaginary parts of the Greens function, we first prove the *Sokhotski-Plemelj* formula:

$$\lim_{\eta \rightarrow 0^+} \frac{1}{x + i\eta} = P\left(\frac{1}{x}\right) - i\pi\delta(x) . \quad (10.9)$$

where $P(f(x))$ is the Cauchy principal value, defined as

$$P\left[\int_{-\infty}^{\infty} \frac{dx}{x}\right] = \int_{-\infty}^{0^-} \frac{dx}{x} + \int_{0^+}^{\infty} \frac{dx}{x} . \quad (10.10)$$

To prove the above identity, we integrate the left-hand side using a test function $f(x)$:

$$\begin{aligned}
 \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{f(x)}{x + i\eta} &= \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{f(x)(x - i\eta)}{x^2 + \eta^2} \\
 &= \lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx f(x) \left[\frac{x}{x^2 + \eta^2} - i\eta \frac{1}{x^2 + \eta^2} \right] \quad (10.11)
 \end{aligned}$$

The first integral can be split into three parts:

$$\lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx f(x) \frac{x}{x^2 + \eta^2} = \lim_{\eta \rightarrow 0^+} \lim_{\varepsilon \rightarrow 0^+} \left[\int_{-\infty}^{-\varepsilon} dx \frac{f(x)x}{x^2 + \eta^2} + \int_{\varepsilon}^{\infty} dx \frac{f(x)x}{x^2 + \eta^2} + \int_{-\varepsilon}^{\varepsilon} dx \frac{f(x)x}{x^2 + \eta^2} \right] \quad (10.12)$$

For the first two parts, it is safe to take the limit of η , because x is always non-vanishing there. For the third part, we can approximate $f(x)$ as $f(0)$ in the neighbourhood $x \in [-\varepsilon, \varepsilon]$, $\varepsilon \rightarrow 0$. This

leaves an odd function $x/(x^2 + \eta^2)$ as the integrand, being integrated over a symmetric range. The third term therefore vanishes. In total, we get

$$\lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx f(x) \frac{x}{x^2 + \eta^2} = \int_{-\infty}^{-\epsilon} dx \frac{f(x)}{x} + \int_{\epsilon}^{\infty} dx \frac{f(x)}{x} = P \left(\int_{-\infty}^{\infty} dx \frac{f(x)}{x} \right). \quad (10.13)$$

To evaluate the second integral of eq. 10.11, we note that the only region in which the integrand is non-zero is when $|x| \simeq 0$; there, we again approximate $f(x)$ as $f(0)$. The remaining integral can then be evaluated easily:

$$-i \lim_{\eta \rightarrow 0^+} \eta \int_{-\infty}^{\infty} dx \frac{f(x)}{x^2 + \eta^2} = -i \lim_{\eta \rightarrow 0^+} \eta f(0) \int_{-\infty}^{\infty} \frac{dx}{x^2 + \eta^2} = -i\pi f(0), \quad (10.14)$$

where we used $\int \frac{dx}{x^2 + \eta^2} = \frac{1}{\eta} \arctan(x/\eta)$. Combining eqs. 10.11, 10.13 and 10.14, we get

$$\lim_{\eta \rightarrow 0^+} \int_{-\infty}^{\infty} dx \frac{f(x)}{x + i\eta} = P \left(\int_{-\infty}^{\infty} dx \frac{f(x)}{x} \right) - i\pi \int_{-\infty}^{\infty} dx f(x) \delta(x). \quad (10.15)$$

This proves eq. 10.9.

The Sokhotski-Plemelj formula allows us to split the spectral representation into a real and an imaginary part:

$$\begin{aligned} G'_{d\sigma} &= \frac{1}{Z} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) P \left(\frac{1}{\omega + E_m - E_n} \right) \\ G''_{d\sigma} &= -\pi \frac{1}{Z} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \delta(\omega + E_m - E_n). \end{aligned} \quad (10.16)$$

The spectral function, specifically, is defined as follows:

$$A_{d\sigma}(\omega) = -\frac{1}{\pi} G'_{d\sigma} = \frac{1}{Z} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \delta(\omega + E_m - E_n) \quad (10.17)$$

2.10.3 Zero temperature spectral function

We specialise to zero temperature by taking the limit of $\beta \rightarrow \infty$. In both the partition function as well as inside the summation, the only term that will survive is the exponential of the ground state energy E_{GS} .

$$Z \equiv \sum_m e^{-\beta E_m} \implies \lim_{\beta \rightarrow \infty} Z = d_{GS} e^{-\beta E_{GS}}, \quad E_{GS} \equiv \min \{E_n\}$$

where d_{GS} is the degeneracy of the ground state. The spectral function then simplifies to

$$\begin{aligned} A_{d\sigma} &= \frac{1}{d_{GS} e^{-\beta E_{GS}}} \sum_{m,n} ||\langle m | O_{\sigma} | n \rangle||^2 \left[e^{-\beta E_m} \delta_{E_m, E_{GS}} + e^{-\beta E_n} \delta_{E_n, E_{GS}} \right] \delta(\omega + E_m - E_n) \\ &= \frac{1}{d_{GS}} \sum_{n, n_{GS}} \left[||\langle n_{GS} | O_{\sigma} | n \rangle||^2 \delta(\omega + E_{GS} - E_n) + ||\langle n | O_{\sigma} | n_{GS} \rangle||^2 \delta(\omega - E_{GS} + E_n) \right] \end{aligned} \quad (10.18)$$

The label n_{GS} sums over all states $|n_{GS}\rangle$ with energy E_{GS} . In practise, we evaluate this expression by replacing the formal Dirac delta functions with "nascent" ones, such as a Lorentzian function with a sufficiently sharp peak.

2.10.4 Reconstructing full Green function from spectral function: Kramers-Kronig relations

As mentioned earlier, the retarded Greens function $G_R(\omega + i0^+)$ has a complex pole in the lower half of the complex plane. In the following, the frequency ω is assumed to be complex. Consider the integral

$$I(\omega) = \oint_C d\omega' \frac{G_R(\omega')}{\omega' - \omega}, \quad \omega \in \mathbb{R}. \quad (10.19)$$

The contour C encloses the entirety of the upper half of the complex plane as well as almost all points of the real line but avoids the pole at $\omega' = \omega$ by forming a semicircle about that point that extends into the upper half of the plane. Since the integrand is completely analytic in the region enclosed by the contour (recall again that G_R is analytic in the upper half), the integral evaluates to zero (by Cauchy's theorem):

$$I(\omega) = 0. \quad (10.20)$$

One can also evaluate the integral along each part of the contour. The semicircular part contributes zero: this is because while the length of the semicircular arc goes as $|\omega'|$, the integrand vanishes faster than $1/|\omega'|$ as $|\omega'| \rightarrow \infty$ (this assumes that the Green function vanishes in the limit of $\omega \rightarrow \infty$). The part of the contour along the real axis can now be evaluated. The two parts of the contour on either side of the pole contribute

$$\int_{-\infty}^{\omega-0^+} d\omega' \frac{G_R(\omega')}{\omega' - \omega} + \int_{\omega+0^+}^{\infty} d\omega' \frac{G_R(\omega')}{\omega' - \omega} = P \int_{-\infty}^{\infty} d\omega' \frac{G_R(\omega')}{\omega' - \omega} \quad (10.21)$$

To evaluate the part of the integral along the small semicircular around $\omega' = \omega$ (that extends into the upper half), we argue that this contribution should be equal to that obtained by taking the semicircle that goes instead into the lower half but circulates in the opposite direction. Each of these contributions should, in turn, be equal to that obtained from joining these two contours. But joining these contours leads to a closed integral about the pole at $\omega' = \omega$, which is equal to $2\pi G_R(\omega)$.

$$\int_{\text{upper}} d\omega' \frac{G_R(\omega')}{\omega' - \omega} = \int_{\text{lower}} d\omega' \frac{G_R(\omega')}{\omega' - \omega} = \frac{1}{2} \oint_{C(\omega)} d\omega' \frac{G_R(\omega')}{\omega' - \omega} = i\pi G_R(\omega). \quad (10.22)$$

Combining the last two equations, we get

$$I(\omega) = P \int_{-\infty}^{\infty} d\omega' \frac{G_R(\omega')}{\omega' - \omega} + i\pi G_R(\omega). \quad (10.23)$$

Comparing with eq. 10.20 and separating into real part G'_R and imaginary part G''_R gives

$$P \int_{-\infty}^{\infty} d\omega' \frac{[G'_R(\omega') + iG''_R(\omega')]}{\omega' - \omega} + \pi [-G''_R(\omega) + iG'_R(\omega)] = 0. \quad (10.24)$$

Comparing the imaginary parts gives

$$G'_R(\omega) = \frac{1}{\pi} P \int_{-\infty}^{\infty} d\omega' \frac{G''_R(\omega')}{\omega' - \omega} = -\mathcal{H} [G''_R(\omega)] , \quad (10.25)$$

where $\mathcal{H} [g(\omega)] = \frac{1}{\pi} P \int_{-\infty}^{\infty} d\omega' \frac{g(\omega')}{\omega - \omega'}$ is the Hilbert transform of the function $g(\omega)$. The spectral function was defined above as $-1/\pi$ times the imaginary part of the Greens function. Exchanging G''_R for A gives

$$G'_R(\omega) = \pi \mathcal{H} [A(\omega)] . \quad (10.26)$$

This provides a method for calculating the real part of a Greens function if the spectral function is already known.

Chapter 3

Holographic entanglement renormalisation for fermionic quantum matter

3.1 Introduction

The manuscript is structured in the following manner. Sec. 3.2 describes the system we work with and defines certain physical ideas and measures of entanglement that are employed in later sections. In Sec. 3.3, we define the scaling transformations mentioned earlier, and discuss its implications on the scaling of entanglement in the system under study. In Sections 3.4, 3.5 and 3.6, we probe the holographic nature of the entanglement, unveiling an emergent spatial dimension in the process and obtain several characteristics of this emergent geometry. Section 3.7 discusses the consequences of a critical Fermi surface on the holographic space, and how the emergent curvature can be linked to a convergence parameter associated with the RG flows. In Section 3.9, we probe the topological aspects of the entanglement, and discuss its implications for the system at hand. We conclude with a discussion of our results in Sec. 3.10 and point out some open questions.

Summary of our main results

For the convenience of the reader, we present below a summary of the main results obtained by us.

1. Quantum information measures (such as entanglement entropy and multipartite information) reveal a hierarchical structure in the nature of entanglement within the system. We relate geometric parameters like distance and curvature of the holographic dimension to an RG beta function of the mass gap, providing thereby an explicit analytic manifestation of the holographic principle. We show that a change in the boundedness of the space corresponds to a change in certain topological winding numbers.

2. We go on to argue that this topological transition and the underlying critical Fermi surface coincides with the formation of a quantum wormhole geometry that connects the UV and the IR of the emergent dimension. The additional (conformal) symmetry at the transition links the emergent metric and the stress-energy tensor.

3. We show that the geometry-independent part of the entanglement is invariant under appropriate scaling transformations for a system of fixed number density, and is related to the Luttinger

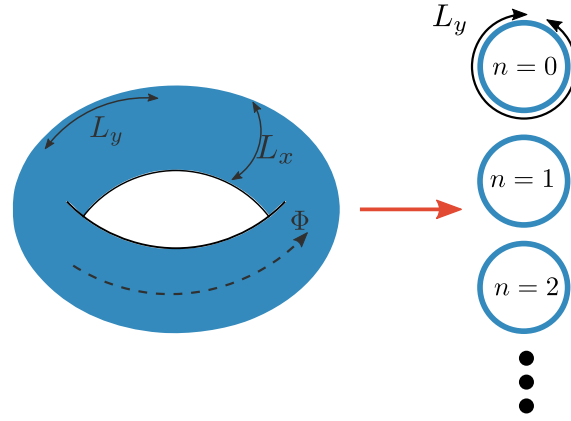


Figure 3.1: The model on the left consists of a system of non-interacting electrons placed on a torus with a flux threading through it. The model on the right consists of a sum of $1 + 1$ -dimensional systems, each consisting of massive electrons placed on a ring; the $1 + 1$ -D systems are decoupled from each other. The mass arises from the flux. These two models can be shown to be equivalent to each other [254, 255, 258, 374].

volume of the gapless fermionic system. In the presence of a strong transverse magnetic field, the topological Luttinger volume in the metallic state transforms into the Chern number C of the insulating integer quantum Hall state.

3.2 Preliminaries and definitions

3.2.1 The system

In this work, we focus on the case of Dirac fermions in two spatial dimensions described by the Lagrangian (in natural units $\hbar = c = 1$) $\mathcal{L} = \int dx dy \bar{\psi}(x, y, t) (i\gamma^\mu \partial_\mu - m) \psi(x, y, t)$. The system is placed on a 2-torus of dimensions $L_x \times L_y$ with periodic boundary conditions, such that $\psi(x, y, t) = \psi(x + L_x, y, t) = \psi(x, y + L_y, t)$. Entanglement measures will be calculated in the presence of a gauge field A that transforms the Lagrangian by shifting the momenta: $i\partial_\mu \rightarrow i\partial_\mu + eA$. We choose the vector potential A such that, in any particular setup, it couples to just one component of the momentum (say, the x -component). This results in $\phi = eAL_x/2\pi$ units of flux quantum threading the torus in the y -direction. One of our goals is to express the Luttinger volume V_L in terms of multi-partite measures of entanglement $(I_{\{\mathcal{A}_i\}}^g(\phi))$ between multiple subsystems $\{\mathcal{A}_i, i = 1, 2, \dots, g\}$ [375], in the presence of the flux ϕ . The precise definition of I^g and the method of choosing the subsystems will be given later. In order to calculate $I_{\{\mathcal{A}_i\}}^g$, however, we find it convenient to take the thin torus limit: $(L_y/L_x) \rightarrow \infty$, keeping l/L_y fixed. Importantly, this operation does not affect quantities such as V_L as long as we keep the number density unchanged.

In order to calculate entanglement measures, we will express the $2 + 1$ -dimensional system as a sum of an infinite number of massive $1 + 1$ -dimensional systems [254, 255, 258, 374]. In the

presence of a gauge field $\vec{A} = A\hat{x}$, the complete Lagrangian (in natural units) is

$$\mathcal{L} = \int dx dy \bar{\psi}(x, y) [\gamma^\mu (i\partial_\mu - eA_\mu) - m] \psi(x, y), \quad (2.1)$$

where $A_\mu = A\delta_{\mu,x}$ and $\phi = eAL_x/2\pi$. The integral ranges over the surface area of the torus, and the time dependence of the fields has been suppressed. Due to the periodic boundary conditions (PBCs) in the x -direction, the momenta k_x are quantised: $k_x^n = \frac{2\pi n}{L_x}, n \in \mathbb{Z}$. We then expand the fields $\psi(x, y)$ in these momenta:

$$\psi(x, y) = \sum_{n=-\infty}^{\infty} e^{ik_x^n x} \psi(k_x^n, y). \quad (2.2)$$

In the absence of the gauge field, the Lagrangian for 2 + 1-dimensional massive non-interacting relativistic electrons (in natural units) is

$$\mathcal{L} = \int dx dy \bar{\psi}(x, y) [\gamma^\mu (i\partial_\mu - eA_\mu) - m] \psi(x, y). \quad (2.3)$$

The integral ranges over the area of the torus on which the electrons reside, and the time dependence of the fields has been suppressed. Each component $\psi^a, a = 0, 1, 2, 3$, of the Dirac spinor $\psi = (\psi^0 \ \psi^1 \ \psi^2 \ \psi^3)$ satisfies the Dirac equation

$$[\gamma^\mu (i\partial_\mu - eA_\mu) - m] \psi^a(x, y) = 0, \quad (2.4)$$

which is equivalent to the form

$$\int dx \bar{\psi}^a(x, y) [\gamma^\nu (i\partial_\nu + eA_\nu) + m] [\gamma^\mu (i\partial_\mu - eA_\mu) - m] \psi^a(x, y) = 0. \quad (2.5)$$

Due to the periodic boundary conditions (PBCs) in the x -direction, the momenta k_x are quantised: $k_x^n = \frac{2\pi n}{L_x}, n \in \mathbb{Z}$. We expand the fields $\psi^a(x, y)$ in these momenta:

$$\psi^a(x, y) = \sum_{n=-\infty}^{\infty} e^{ik_x^n x} \psi^a(k_x^n, y). \quad (2.6)$$

Writing the Dirac equation in terms of the dual fields $\psi^a(k_x^n, y)$ gives

$$\sum_{m,n} \left(\bar{\psi}^a(k_x^m, y) e^{-ik_x^m x} \right) [\gamma^\nu (i\partial_\nu + eA_\nu) + m] [\gamma^\mu (i\partial_\mu - eA_\mu) - m] \left(e^{ik_x^n x} \psi^a(k_x^n, y) \right) = 0. \quad (2.7)$$

Evaluating the right operation gives

$$[\gamma^\mu (i\partial_\mu - eA_\mu) - m] \left(e^{ik_x^n x} \psi^a(k_x^n, y) \right) = e^{ik_x^n x} [\gamma^\mu (i\partial_\mu - eA_\mu) - (\gamma^x(k_x^n + eA_x) + m)] \psi^a. \quad (2.8)$$

Similarly, the left operation gives

$$\begin{aligned} \left(\bar{\psi}^a(k_x^m, y) e^{-ik_x^m x} \right) [\gamma^\nu (i\partial_\nu + eA_\nu) + m] &= \left[[\gamma^\nu (i\partial_\nu - eA_\nu) + m] \left(e^{ik_x^m x} \gamma_0 \psi^a \right) \right]^\dagger \\ &= \bar{\psi}^a e^{-ik_x^m x} [\gamma_\nu (i\partial^\nu + eA_\nu) - (\gamma^x (k_x^n + eA_x) - m)] . \end{aligned} \quad (2.9)$$

Substituting these into the equation eq.(2.7) gives

$$\sum_n \int dy \bar{\psi}^a(k_x^n, y) [-(i\partial_\mu + eA_\mu)(i\partial_\mu - eA_\mu) + (k_x^n + eA_x)^2 + m^2] \psi^a(k_x^n, y) = 0 , \quad (2.10)$$

where we have used $\int dx e^{ix(k_x^m - k_x^n)} = \delta_{m,n}$. In terms of these dual fields, the Lagrangian can be shown to be equivalent to a tower of massive $1 + 1$ -dimensional fermions (see Fig. 3.1):

$$\mathcal{L} = \sum_{k_x^n} \int dy \bar{\psi}(k_x^n, y) (\gamma^\mu (i\partial_\mu - eA_\mu) - M(n, \phi)) \psi(k_x^n, y) , \quad (2.11)$$

with an effective mass $M_{n,\phi}$ given by

$$M_{n,\phi} = \sqrt{m^2 + (k_x^n + eA)^2} = \sqrt{m^2 + \frac{4\pi^2}{L_x^2} (n + \phi)^2} . \quad (2.12)$$

This decoupling ensures that the various modes $n \in \mathbb{Z}$ are disentangled from one another, such that the total density matrix ρ_{2D} of the $2 + 1$ -dimensional system can be written as a product of the density matrices ρ_{1D}^n for each of the modes:

$$\rho_{2D} = \prod_{n=-\infty}^{\infty} \rho_{1D}^n . \quad (2.13)$$

We will focus primarily on the case of massless $2 + 1$ -dimensional electrons by setting $m = 0$, such that the effective mass is given by $M_{n,\phi} = \frac{2\pi}{L_x} |n + \phi|$. Certain results will, however, be generalised to the case of massive fermions ($m \neq 0$). While the massless case corresponds to the gapless excitations of a Fermi liquid or those at the surface of a topological insulator, the massive case corresponds to gapped or insulating states of matter, e.g., the Bogoliubov-de Gennes quasiparticles above an s -wave superconducting gap. Massive Dirac fermions have also been observed experimentally in several materials like ultrathin films [376], topological insulating crystals [377] and van der Waals heterostructures [378]. They are also emergent at low-energies, for instance, in fermionic nonlinear σ -models [379], and can also be realised near the IR fixed point of the kagome Heisenberg quantum antiferromagnets placed in an external magnetic field along the z -direction [380].

3.2.2 Entanglement measures

We will work with two measures of entanglement discussed below. Given the density matrix $\rho = |\Psi\rangle \langle \Psi|$ for the ground state $|\Psi\rangle$ of the complete system, the entanglement entropy (EE) of a set of subsystems $\{\mathcal{A}_i, i = 1, 2, \dots, g\}$ with the rest of the system is given by:

$$S_{\{\mathcal{A}_i\}} = -\text{Tr} [\rho_{\{\mathcal{A}_i\}} \ln \rho_{\{\mathcal{A}_i\}}] ; \quad \rho_{\{\mathcal{A}_i\}} = \text{Tr}_{\{\mathcal{A}_i\}} [\rho] . \quad (2.14)$$

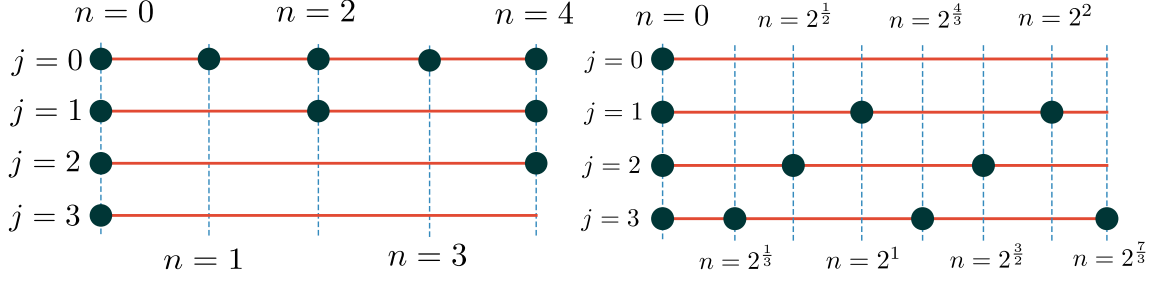


Figure 3.2: Visual depiction of the subsets $\mathcal{A}_z(j)$ employed by us in this work, for $z = 1$ and $z = -1$. As j is changed by 1, the spacing between the modes in the subset increases by a factor of 2^{j^z} . Top: $z = 1$. The initial subset $\mathcal{A}_z(0)$ has k -states at the modes $n = 0, 1, 2, 3, 4$. Upon changing j by 1, the new subset obtained has double the spacing between the modes ($2^{j^z} = 1/2^j$) of the previous number of modes. That is, if $j = 0$ has $0, 1, 2, 3, 4$, then $j = 1$ has $0, 2, 4$, $j = 2$ has $0, 4$ and $j = 3$ has 0 . Bottom: $z = -1$. In this case, the number of modes get double at each state ($2^{-j^z} = 2^j$). We start with k -modes present only at $n = 0$, but the scaling transformations lead to $j = 1$ having the modes $0, 4$, $j = 2$ having $0, 2, 4$, and so on.

$\text{Tr}[\cdot]$ is the trace operation over all degrees of freedom, while $\text{Tr}_{\{\mathcal{A}_i\}}[\cdot]$ is the partial trace over only the states corresponding to the set of subsystems $\{\mathcal{A}_i\}$. $\rho_{\{\mathcal{A}_i\}}$ is referred to as the reduced density matrix for the rest of the system (complement of the set $\{\mathcal{A}_i\}$). Using EE, one can define the mutual information $I(\{\mathcal{A}_i\} : \{\mathcal{B}_i\})$ between two subsystems A and B :

$$I_2(\{\mathcal{A}_i\} : \{\mathcal{B}_i\}) = S(\{\mathcal{A}_i\}) + S(\{\mathcal{B}_i\}) - S(\{\mathcal{A}_i\} \cup \{\mathcal{B}_i\}), \quad (2.15)$$

where $S(\{\mathcal{A}_i\} \cup \{\mathcal{B}_i\})$ is the EE of $\{\mathcal{A}_i\} \cup \{\mathcal{B}_i\}$ with the rest. Higher order measures of information among N subsystems can also be defined similarly (see, e.g., [375]). Such multi-partite measures of quantum information convey the degree of correlations present within the system - the presence of N -particle correlations ensures all information measures up to the N -partite information will be non-zero.

One final object of interest in this context is the entanglement Hamiltonian $\mathcal{H}(\{\mathcal{A}_i\})$ for a subsystem $\{\mathcal{A}_i\}$, defined through the relation [176, 181]

$$\rho_{\{\mathcal{A}_i\}} = \frac{e^{-\mathcal{H}(\{\mathcal{A}_i\})}}{\text{Tr}(e^{-\mathcal{H}(\{\mathcal{A}_i\})})}. \quad (2.16)$$

The entanglement Hamiltonian is essentially the spectrum of the reduced density matrix, and contains additional information on the nature of correlations beyond the entanglement entropy. For the system we are considering (described in the previous subsection), the total reduced density matrix can be decomposed into decoupled 1D modes (eq. (2.13)). This implies that the total entanglement Hamiltonian is simply a sum of the individual Hamiltonians for the 1D modes:

$$\rho_{2D} = \prod_{n=-\infty}^{\infty} \rho_{1D}^n \sim \prod_{n=-\infty}^{\infty} e^{-\mathcal{H}_{1D}^n(\{\mathcal{A}_i\})} = e^{-\mathcal{H}_{2D}(\{\mathcal{A}_i\})}, \quad (2.17)$$

where $\mathcal{H}_{2D} = \sum_n \mathcal{H}_{1D}^n$.

3.2.3 Luttinger volume

The Luttinger volume V_L is defined as the number of $\vec{k}$ -space points inside the Fermi surface [97,98] :

$$V_L = \int_{G_{\vec{k}}(\omega=0)>0} \frac{1}{(2\pi)^d} d\vec{k} = \int_0^{k_F} \frac{1}{(2\pi)^d} d\vec{k} . \quad (2.18)$$

The integral (referred to as Luttinger's integral) runs over all $\vec{k}$ -space points where the $\omega = 0$ Greens function is positive, and the Fermi momentum k_F is defined as the set of points in $\vec{k}$ -space where $G_{\vec{k}}(\omega = 0)$ changes sign. Luttinger's integral can be shown to be topological in nature [99, 100], promoting Luttinger's volume to the status of a topological invariant that characterises the system. For systems with long-lived excitations close to the Fermi surface, V_L is equal to the average number density of electrons in the system, leading to Luttinger's theorem [97, 98, 263]. For systems with gapped excitations in the momentum window $k \leq k^*$ above the ground state, the violation ΔV_L of Luttinger's theorem can be quantified through a count of the bound states [264]:

$$\Delta V_L = \int_0^{k_F} \frac{1}{(2\pi)^d} d\vec{k} - \int_{k^*}^{k_F} \frac{1}{(2\pi)^d} d\vec{k} = \int_0^{k^*} \frac{1}{(2\pi)^d} d\vec{k} \quad (2.19)$$

3.3 Entanglement hierarchy in mixed momentum and real space

Before beginning our calculations of entanglement measures, we define the subsystem(s) whose entanglement we want to obtain. The subsystems we will use in this work are of the following kind:

$$\mathcal{A}_z(i) = \tilde{A}_z(i) \times A_z(i) , \quad (3.1)$$

where i takes integral values starting from zero, z can take all integral values apart from 0 and $\times$ represents the Cartesian product between the set of real space points $\tilde{A}_z(i) = \{0 \leq y \leq l\}$ and the set of momentum space points

$$A_z(i) = \left\{ k_x^n \right\}; \quad k_x^n = \frac{2\pi}{L_x} n; \quad \max(\{k_x^n\}) = \frac{2\pi N}{L_x}; \\ n \in \{-N, -N + t_z(i), \dots, -t_z(i), 0, t_z(i), N\}; \quad t_z(i) = 2^{i^z} , \quad (3.2)$$

and where $t_z(i) = 2^{i^z}$ is the difference between consecutive values of n . The subset $\mathcal{A}_z(i)$ is therefore a region in mixed momentum and real spaces, spanned by the variables k_x and y respectively. In y -space, the region is of length l and bounded by the points $y = 0$ and $y = l$. In k_x -space, $\mathcal{A}_z(i)$ comprises the points $\left\{0, \pm 2^{i^z} \frac{2\pi}{L_x}, \pm 2^{i^z} \frac{4\pi}{L_x}, \dots\right\}$ such that $|k_x| \leq \frac{2\pi N}{L_x}$. For example, for the case of $z = 1$ and N being a multiple of 4, we have

$$A_1(0) = \left[-\frac{2\pi}{L_x} N, -\frac{2\pi}{L_x} (N-1), \dots, -\frac{2\pi}{L_x} 2, -\frac{2\pi}{L_x} 1, 0, \frac{2\pi}{L_x} 1, \frac{2\pi}{L_x} 2, \dots, \frac{2\pi}{L_x} (N-1), \frac{2\pi}{L_x} N \right] , \\ A_1(1) = \left[-\frac{2\pi}{L_x} N, -\frac{2\pi}{L_x} (N-2), \dots, -\frac{2\pi}{L_x} 4, -\frac{2\pi}{L_x} 2, 0, \frac{2\pi}{L_x} 2, \frac{2\pi}{L_x} 4, \dots, \frac{2\pi}{L_x} (N-2), \frac{2\pi}{L_x} N \right] , \quad (3.3) \\ A_1(2) = \left[-\frac{2\pi}{L_x} N, -\frac{2\pi}{L_x} (N-4), \dots, -\frac{2\pi}{L_x} 8, -\frac{2\pi}{L_x} 4, 0, \frac{2\pi}{L_x} 4, \frac{2\pi}{L_x} 8, \dots, \frac{2\pi}{L_x} (N-4), \frac{2\pi}{L_x} N \right] .$$

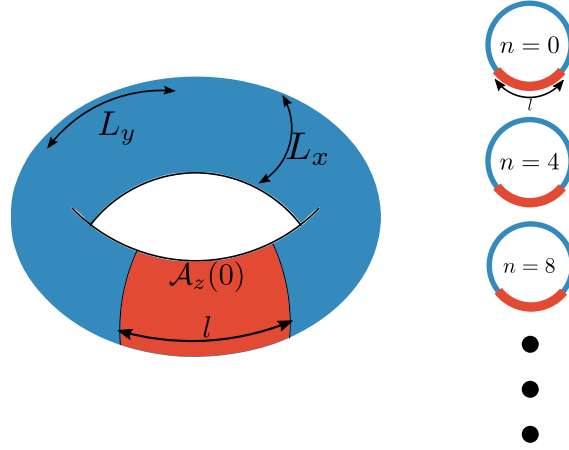


Figure 3.3: Top: Depiction of the subset $\mathcal{A}_z(i)$ for the choice $i = 0$. Setting $i = 0$ leads to unchanged length L_x along x and the constraint $0 \leq y \leq l$ in the y -direction. This creates an annulus (red region) of height l and circumference L_x .

Bottom: Typical k -modes present in the subset $\mathcal{A}_z(i)$ for $z = 1$ and $i = 2$. This is because, if $i = 0$ has the modes $n = 0, 1, 2, 3, 4, \dots$, increasing i by 2 increases the spacing between the modes to $2^{2^z} = 4$, so that the mode just after $n = 0$ is $n = 4$, and the one following that is $n = 8$, and so on.

Note that N represents the edge of the Brillouin zone, and will be sent to infinity when we calculate quantities in the continuum limit. In Fig. 3.2, we have shown two such families of subsets $\mathcal{A}_z(0), \mathcal{A}_z(1), \mathcal{A}_z(2), \mathcal{A}_z(3), \dots$, for $z = 1$ and $z = -1$.

We note here that the transformations for $z < 0$ correspond to increasing the density (fine-graining) of k_x -states by adding more momentum states while keeping the bounds fixed. This involves reducing the interval in k_x -space, and hence equivalent to increasing the system size L_x . This is similar to methods such as numerical renormalisation group treatment of the Kondo problem [362] where an increasing number of states are added to the system at increasingly larger distances from the impurity site, such that the system asymptotically goes to the thermodynamic limit. The other case of $z > 0$ is, on the contrary, a decimation process in k_x -space, and equivalent to reducing the system size. Indeed, it will be shown later that the sequence of Hamiltonians generated in the process (for both positive and negative z) are related to one other by renormalisation group transformations.

Within a given set $\mathcal{A}_z(i)$, the distance between adjacent k -space points is $\frac{2\pi}{L_x} t_z(i)$ where $t_z(i) = 2^{i^z}$. Choosing $i = 0$ keeps the set of k_x -space points unchanged, such that we recover the entire Lagrangian of eq. (2.1) in the x -direction for system size L_x , while the y -direction is constrained to length l . This means that $\mathcal{A}_z(0)$ is just an annulus that wraps around the x -direction of the torus and extends to a length l longitudinally in the y -direction (shown in Fig. 3.3). Along with the period $t_z(i)$, one can also define the fraction of maximum points present in the set $\mathcal{A}_z(i)$ as

$$f_z(i) = 1/t_z(i) = 2^{-i^z}. \quad (3.4)$$

For example, for $z > 0$, $f_z(i) = 2^{-i^z}$ expresses the fact that in going from the set at $j = 0$ to the set at $j = i$, the number of elements in the set is scaled down by a factor of 2^{i^z} (see Fig. 3.4).

We now define the shorthand notation $S_z(i)(\phi) \equiv S_{\mathcal{A}_z(i)}(\phi)$ to denote the EE of the subsystem $\mathcal{A}_z(i)$ in the presence of the Aharonov-Bohm flux ϕ . In order to calculate $S_z(i)$, we start with $i = 0$. As mentioned earlier, this is essentially the EE between an annulus of length l and the rest of the 2-torus. Since the density matrices are in product, the total EE is just a sum of the EE arising from each mode:

$$S_{\mathcal{A}_z(0)}(\phi) \equiv S_z(0)(\phi) = \lim_{N \rightarrow \infty} \sum_{n=-N}^N S_{\tilde{A}_z(0)}^n, \quad (3.5)$$

where $\tilde{A}_z(0)$ acts as a projection of the annulus on the $k_x - y$ plane, and S^n indicates that the EE is calculated for the n^{th} mode. Each massive fermionic mode has a correlation length-scale $\xi_n = 1/M_{n,\phi}$. We choose l to be much larger than the largest correlation length: $l \gg \max(\{\xi_n, \forall n\})$. Using the expression for ϕ and the quantisation of the momenta k_x^n , this condition can be expressed as

$$l \gg \frac{L_x}{2\pi} \times \max\left(\left\{\frac{1}{|n + \phi|}\right\}\right) = \frac{L_x}{2\pi \{\phi\}}, \quad (3.6)$$

where $\{\phi\} = \phi - \lfloor \phi \rfloor$ is the fractional part of the flux ϕ in units of the flux quantum. Since we are in the thin torus limit, L_x must remain finite, and the maximum correlation can diverge only if $\{\phi\} = 0$. We ensure that l is larger than the correlation lengths by choosing ϕ to be necessarily non-integral, as this guarantees that the correlation lengths remain finite. Then, the EE of the annulus is given by [255, 258]

$$S_z(0)(\phi) = c \left[\alpha \frac{L_x}{\epsilon} - \ln |2 \sin(\pi \phi)| \right], \quad (3.7)$$

where c is the conformal central charge and ϵ is a UV cutoff; c is $1/3$ for a Dirac fermion. The computation of $S_z(0)(\phi)$ uses the EE $S_{\tilde{A}_z(0)}^n$ of a 1-D chain of fermions with a mass $M_{n,\phi}$ [232, 240]:

$S_{\tilde{A}_z(0)}^n = c \ln \frac{L_x}{2\pi\epsilon} - c \ln |n + \phi|$. On summing over n and regularising the UV cutoff, the first term of eq.(3.7) produces the leading area-law term of $S_0(\phi)$ [255]:

$$\sum_{n=-\infty}^{\infty} c \ln \frac{L_x}{2\pi\epsilon} \rightarrow c\alpha \frac{L_x}{\epsilon}, \quad (3.8)$$

where α is a cutoff-dependent non-universal constant. The flux-dependent term in $S_{\tilde{A}_z(0)}^n$ leads to the sub-leading term of $S_z(0)(\phi)$, and requires a regularisation using the Hurwitz zeta function [255]:

$$\sum_{n=-\infty}^{\infty} \ln |n + \phi| = \sum_{n=-\infty}^{\infty} [\ln |n + \phi| + \ln |n + (1 - \phi)|] = -\frac{\partial \zeta(\phi)}{\partial \phi} - \frac{\partial \zeta(1 - \phi)}{\partial \phi}, \quad (3.9)$$

where $\zeta(\phi) = \sum_{n=-\infty}^{\infty} (n + \phi)^{-1}$ is the Hurwitz zeta function. The derivative converges to $\frac{\partial \zeta(\phi)}{\partial \phi} = \ln \Gamma(\phi) - \frac{1}{2} \ln 2\pi$, as long as the zero mode $n + \phi = 0$ is excluded from the summation. The latter condition is guaranteed as we have chosen ϕ to be a non-integer. Γ is the Gamma function, satisfying $\Gamma(\phi)\Gamma(1 - \phi) = \pi/\sin(\pi\phi)$. Substituting the derivative gives [255, 258, 262]

$$\sum_{n=-\infty}^{\infty} \ln |n + \phi| = \ln 2\pi - \ln [\Gamma(\phi)\Gamma(1 - \phi)] = \ln |2 \sin(\pi \phi)|. \quad (3.10)$$

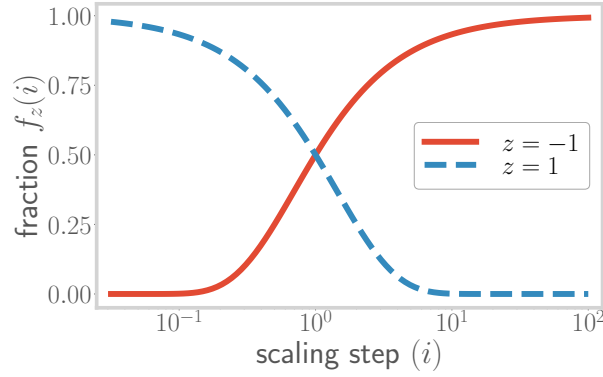


Figure 3.4: A plot of the fraction of states remaining upon performing i number of transformations on the starting Hamiltonian. Obtained by plotting the expression $f = 2^{-i^z}$, for a range of i , and for two values of z (red and blue). This fraction decreases for $z > 0$ (blue curve), and increases for $z < 0$ (red curve).

For a general $\mathcal{A}_z(i)$, there are certain modifications to the expression in eq. (3.7). For the first term in eq.(3.7), we have

$$\lim_{N \rightarrow \infty} \sum_{n=-N, \dots, -N+t_z(i), \dots}^N c \ln \frac{L_x}{2\pi\epsilon} = \lim_{N \rightarrow \infty} \sum_{n'=-\frac{N}{t_z(i)}, \dots, -\frac{N}{t_z(i)}+1, \dots}^{\frac{N}{t_z(i)}} c \ln \frac{L_x}{2\pi\epsilon} = t_z^{-1}(i) \lim_{N \rightarrow \infty} \sum_{n=-N}^N c \ln \frac{L_x}{2\pi\epsilon} \rightarrow c f_z(i) \frac{\alpha L_x}{\epsilon}. \quad (3.11)$$

Similarly, for the second term, we have

$$\begin{aligned} \lim_{N \rightarrow \infty} \sum_{n=-N, \dots, -N+t_z(i), \dots}^N \ln |n + \phi| &= \lim_{N \rightarrow \infty} \sum_{n'=-\frac{N}{t_z(i)}, \dots, -\frac{N}{t_z(i)}+1, \dots}^{\frac{N}{t_z(i)}} \ln |n' t_z(i) + \phi| = \lim_{N \rightarrow \infty} \sum_{n'=-\frac{N}{t_z(i)}, \dots, -\frac{N}{t_z(i)}+1, \dots}^{\frac{N}{t_z(i)}} \ln (f_z(i) |n' + f_z(i)\phi|) \\ &= \sum_{n'=-\infty}^{\infty} \ln |n' + f_z(i)\phi|, \end{aligned} \quad (3.12)$$

where we have assumed that $t_z(i)$ is finite, and dropped the constant part $\ln f_z(i)$ (which can be absorbed into the UV cutoff ϵ in eq. (3.11)). Following eq. (3.10), this gives

$$\lim_{N \rightarrow \infty} \sum_{n=-N, \dots, -N+t_z(i), \dots}^N \ln |n + \phi| = \ln |2 \sin(\pi f_z(i)\phi)|. \quad (3.13)$$

Combining both parts, we obtain the expression for the EE that will be employed in what lies ahead:

$$S_z(i)(\phi) = c f_z(i) \frac{\alpha L_x}{\epsilon} - c \ln |2 \sin(\pi f_z(i)\phi)|. \quad (3.14)$$

It is evident from the expression in eq. (3.14) that, as modes are added into the subsystem, the L_x -dependent geometric part of the EE increases, leading to a *hierarchical structure* in the entanglement. A similar result was previously shown to hold for geometric measures of entanglement of quantum-mechanical systems with eigenstates such as the W and GHZ states and their symmetric superpositions [381]. The variation of the geometric part with the subsystem index j is shown in Fig. 3.4.

The above-mentioned hierarchy in entanglement measures leads to an interesting property for the entanglement: the combined entanglement entropy of any number of systems $\{\mathcal{A}_z(i)\}$ is equal to the entanglement entropy of the most dense system among them. For $z > 0$, the sets become smaller as the RG index increases, while for $z < 0$, the sets become larger as the RG index increases. Thus, we may write

$$S_{\{\mathcal{A}_z(i)\}} = \theta(z)S_{\min\{i\},z} + \theta(-z)S_{\max\{i\},z}, \quad (3.15)$$

The mutual information $I^2(i, j)$ between the systems at two RG steps i and j can then be computed:

$$I^2(i, j) \equiv S_z(i) + S_{j,z} - S_{i \cup j, z} = \theta(-z)S_{\min(i,j),z} + \theta(z)S_{\max(i,j),z}. \quad (3.16)$$

In light of this subsystem-based hierarchical structure, we can also consider the g -partite information $I_{\{\mathcal{A}_z(i)\}}^g$ among g sets $\{\mathcal{A}_z(i)\}$. I^g is defined as [375]

$$I_{\{\mathcal{A}_z(i)\}}^g = \sum_{r=1}^g (-1)^{r+1} \sum_{\substack{\beta \in \mathcal{P}(\{\mathcal{A}_z(i)\}) \\ |\beta|=r}} S_{\beta}, \quad (3.17)$$

where $\mathcal{P}(\{\mathcal{A}_z(i)\})$ is the power set of $\{\mathcal{A}_z(i)\}$, and $|\beta| = r$ indicates that the sum is carried out only over those sets β that have r number of elements in them. For example, for $g = 3$, we have

$$I^3 = S_{\mathcal{A}_z(0)} + S_{\mathcal{A}_z(1)} + S_{\mathcal{A}_z(2)} - S_{\mathcal{A}_z(0) \cup \mathcal{A}_z(1)} - S_{\mathcal{A}_z(1) \cup \mathcal{A}_z(2)} - S_{\mathcal{A}_z(2) \cup \mathcal{A}_z(0)} + S_{\mathcal{A}_z(0) \cup \mathcal{A}_z(1) \cup \mathcal{A}_z(2)}. \quad (3.18)$$

For our choice of subsystems, the g -partite information can be shown to be equal to the EE of the *intersection set* of the concerned g -sets (details in Appendix A). For the case of $z > 0$, this takes the form

$$I_{\{\mathcal{A}_z(i)\}}^g(\phi) = S_g(\phi) = c f_{g,z} \alpha \frac{L_x}{\epsilon} - c \ln |2 \sin(\pi f_{g,z} \phi)|. \quad (3.19)$$

Since the g -partite information is simply the EE of the g^{th} subsystem, the hierarchy in the EE can thus be seen to lead to multiple levels of correlation in the system, corresponding to an increasing number of parties entering the set.

Finally, we note that the geometric (L_x -dependent) part of the EE in eq. (3.14), as well as any of the information measures I^g (eq. (3.19)), has the same form even if we start with a non-zero mass $m > 0$ [255]. This ensures that the above observations regarding the hierarchical structure of the entanglement hold even for gapped electronic systems.

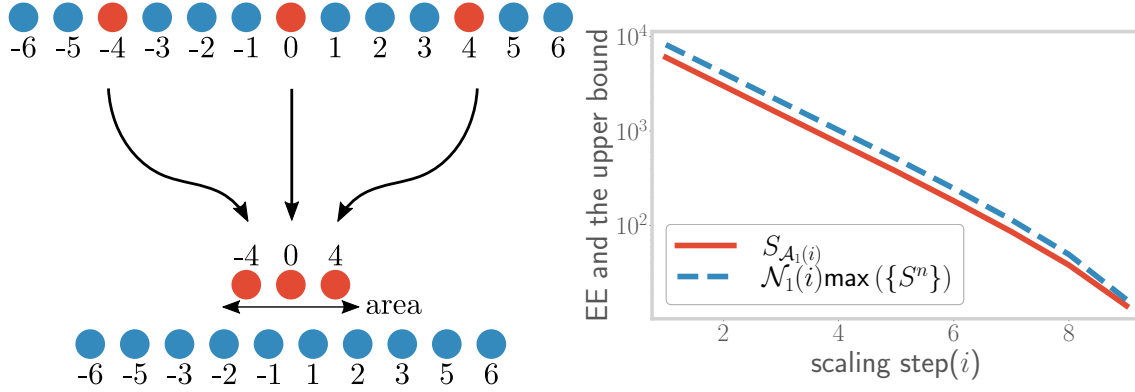


Figure 3.5: Top: Modes forming the subsystem $\mathcal{A}_z(i)$ (red circles) form the boundary between $\mathcal{A}_z(i)$ and its complement (blue circles). Bottom: Scaling of the subsystem entanglement entropy (red), and its upper bound (blue), as the RG transformations are performed (denoted by increasing index i). The bound is saturated at large i .

3.4 EHM, the Ryu-Takayanagi bound and holographic inequalities

As we will now show, the approach laid out above to the creation of subsystems amounts essentially to performing a sequence of renormalisation group (RG) transformations on the momentum modes of the Hamiltonian in the x -direction. We consider here the general case of massive Dirac electrons $M_{n,\phi} = \sqrt{m^2 + \frac{\pi^2}{L_x^2} (n + \phi)^2}$. Further, we define the superset $\mathcal{A}_z^{(0)}$ of k_x -states: $\mathcal{A}_z^{(0)} = \bigcup_{i=0}^{\infty} \mathcal{A}_z(i)$, such that all k_x -states that can appear at any point of the sequence $\{\mathcal{A}_z(i); i = 0, 1, 2, \dots\}$ are present within it. This superset can now be used to define a parent Hamiltonian $H(0)$ that consists of all the modes in the x -direction:

$$H(0) = \sum_{k_x \in \mathcal{A}_z^{(0)}} H(k_x), \quad (4.1)$$

where $H(k_x)$ is the Hamiltonian for a massive Dirac fermion that extends over a length L_y in the single spatial dimension, and having mass $M_{n,\phi}$. The Hamiltonian at any given step of the RG is then obtained by projecting onto appropriate sets of momenta. The set of momentum at step j is given by $\mathcal{A}_{j,z} = \frac{2\pi}{L_x} \times \{0, \pm t_z(j), \pm, 2t_z(j), \dots\}$ with $t_z(j) = 2^{j_z}$. The projector onto this set is given by

$$P_{j,z} = \sum_{k_x^1=0,1} \sum_{k_x^2=0,1} \dots \sum_{k_x^N=0,1} |\hat{n}_{k_x^1} \hat{n}_{k_x^2} \dots\rangle \langle \hat{n}_{k_x^1} \hat{n}_{k_x^2} \dots|, \quad k_x^j \in \mathcal{A}_{j,z}, \quad (4.2)$$

and the Hamiltonian at the corresponding step is then obtained as

$$H_z(i) = P_z(i) H(0) P_z(i). \quad (4.3)$$

In this formulation of the RG, the index i represents the RG step, while the parameter z indicates the *strength of the scaling* mechanism: for $z > 0$, a higher value of z means that the states of the Hamiltonian become rarer as the index i increases, while for $z < 0$, the states become denser with increasing i . Importantly, we will see subsequently that the parameter z corresponds also to the anomalous dimension of the coupling that leads to a spectral gap in the electronic dispersion, dictating its growth or decay under RG evolution (eq.(5.6)).

Since this is a free theory, the k -modes are decoupled, and the projection operation preserves the spectrum of the remaining degrees of freedom. The renormalised Hamiltonian (eq. (4.3)) can then be said to constitute an exact holographic mapping (EHM) [271, 272] in the direction of the RG transformations. It is also pertinent to note that the sequence of transformations in eq. (4.3) can be thought of as a simple and specific example of a unitary renormalisation group (URG) procedure [264, 265] that first disentangles and then projects out a certain set of degrees of freedom. Because the operation is unitary, the URG preserves the spectrum in the process. In the present case, the disentanglement operation is unity. The URG has already been shown to lead to a holographic renormalisation of the Hilbert space geometry which can be seen through the evolution of the Fubini-Study metric under the URG transformations [264]. Our approach differs from MERA implementations like those in refs. [382, 383] in the fact that while those involve a real space renormalisation scheme, we perform our scaling transformations in momentum space.

More evidence for the holographic nature of the construction comes from the fact that the entanglement entropy for both the massless and massive cases satisfy the bound provided by the Ryu-Takayanagi proposal that relates the entanglement of a conformal field theory (CFT) with the geometry of its gravity dual [302, 303]:

$$S_{\mathcal{A}_z(i)} \leq \mathcal{N}_z(i) \max(\{S^n\}) . \quad (4.4)$$

In the above constraint, $\max(\{S^n\})$ is the maximum single entropy of all the modes n residing in $\mathcal{A}_z(i)$, and $\mathcal{N}$ is the area of the minimal surface that separates the subsystem $\mathcal{A}_z(i)$ from the rest. In order to see how this is satisfied in the presence case, we note that since the subsystem $\mathcal{A}_z(i)$ is interspersed among the rest of the system (see left panel of Fig. 3.5), the minimal surface is as large as the subsystem itself. The area of this surface is therefore given by the number of modes in $\mathcal{A}_z(i)$. Using this information, the right hand side of eq. (4.4) can be written as

$$\mathcal{N}_z(i) \max(\{S^n\}) = \max(\{S^n\}) \mathcal{N}_{f_z(i)} = \max(\{S^n\}) \sum_{n \in \mathcal{A}_z(i)} . \quad (4.5)$$

Following eq. (2.13), the left hand side of eq. (4.4) can, on the other hand, be written as

$$S_{\mathcal{A}_z(i)} = \sum_{n' \in \mathcal{A}_z(i)} S^{n'} \leq \sum_{n' \in \mathcal{A}_z(i)} \max(\{S^n\}) . \quad (4.6)$$

Comparing eqs. (4.5) and (4.6), we recover the Ryu-Takayanagi condition of eq. (4.4). The scaling of both the left and right hand sides of eq. (4.4) has been shown in the right panel of Fig. 3.5.

Entanglement measures like entanglement entropy and mutual information (associated with disjoint intervals) have been shown to satisfy a number of inequalities in quantum field theories with gravity duals. We now consider some of them here, and show that the theory considered by us

here also satisfies these inequalities. One such constraint is the monogamy of mutual information, which says that for three disjoint regions A_1, A_2 and A_3 , we must have $I_2(A_1 : A_2) + I_2(A_1 : A_3) \leq I_2(A_1 : A_2 \cup A_3)$, where $I_2(A_1 : A_2) = S(A_1) + S(A_2) - S(A_1 \cup A_2)$ [384]. For this, we consider three disjoint regions, all of the kind depicted in the left panel of fig. 3.3, but at different positions $y_i \in [l_i, l_i + L]$, $i = 1, 2, 3$. We assume that the distance $l_i - l_{i+1}$ between adjacent regions is much larger than the correlation length. Calabrese and Cardy have shown that the total entanglement entropy of the union of such disjoint regions is obtained by multiplying the entanglement entropy of any one region with the number of disjoint intervals. Putting all this together, we get

$$\begin{aligned} S(A_1) &= S(A_2) = S(A_3), \\ S(A_1 \cup A_2) &= S(A_1 \cup A_3) = 2S(A_1), \\ S(A_1 \cup A_2 \cup A_3) &= 3S(A_1), \end{aligned} \quad (4.7)$$

leading to the above inequality being satisfied:

$$I_2(A_1 : A_2) + I_2(A_1 : A_3) - I_2(A_1 : A_2 \cup A_3) = 0. \quad (4.8)$$

We next consider a cyclic family of inequalities in terms of conditional entropies, introduced and proved in Ref. [385]: $\sum_{i=1}^{2k+1} S(A_i | A_{i+1} \dots A_{i+k}) \geq S(A_1 \dots A_{2k+1})$, where $S(A_i | A_j) = S(A_i \cup A_j) - S(A_j)$. Using the results of eq. (4.7), we get $\sum_{i=1}^{2k+1} S(A_i | A_{i+1} \dots A_{i+k}) = \sum_{i=1}^{2k+1} S(A_i) = (2k + 1)S(A_1)$ and $S(A_1 \dots A_{2k+1}) = (2k + 1)S(A_i)$, leading to $\sum_{i=1}^{2k+1} S(A_i | A_{i+1} \dots A_{i+k}) = S(A_1 \dots A_{2k+1})$, which also satisfies the inequality. In both cases, we find that the inequality is satisfied by saturating the bound; this is a consequence of the fact that the intervals are separated by distances larger than the correlation length and the total entanglement entropies are simply a sum of the individual entropies.

3.5 Holographic geometry of the RG flow

3.5.1 RG evolution as the emergent dimension

To obtain a more specific description of the RG flow as the emergence of a geometric space, we introduce a measure of distance. For this, we will employ another bipartite measure of entanglement - the mutual information I^2 , as defined in eq. (2.15), due to its non-negative nature. We focus on the massless case $m = 0$ for the time-being. Inspired by Refs. [269, 272, 365, 386], we adopt the following definition for distance:

$$d_z(i, j) \equiv \ln \frac{I_{\max}^2}{I_z^2(i : j)}, \quad (5.1)$$

where $I_z^2(i : j) \equiv I^2(\mathcal{A}_z(i), \mathcal{A}_{j,z})$ is the mutual information between the two mentioned sets, and $I_{\max}^2$ is the mutual information between two maximally entangled systems. Accordingly, $d_z(j)$ is zero (i.e., the mutual information is maximum) at $j = 0$ if $z > 0$ and $j = \infty$ if $z < 0$, and this distance increases (decreases) with the decrease (increase) in the mutual information as the scaling transformations are performed.

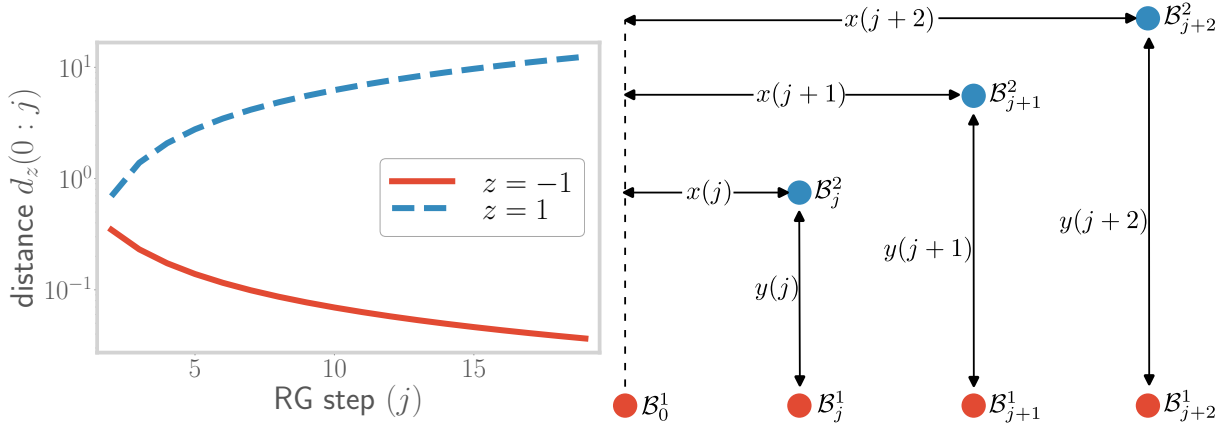


Figure 3.6: Top: Variation of the information distance $d_z(0 : j)$ (defined along the RG direction) under the scaling transformations. Obtained by plotting the expression for the distance, as a function of the number of transformations j , for two values of z . For $z = 1$ (blue curve), the distance between two extreme points increases, showing a decrease in entanglement. For $z = -1$ (red curve), the distance decreases, showing an increase in entanglement. Bottom: Geometry of subsystems used for calculating the emergent curvature. The red blobs represent the subsystems generated along the emergent direction, through the scaling transformations: $\mathcal{A}_{\in j} \rightarrow \mathcal{B}_{j+1}^1$. The blue blobs represent subsystems generated within the 1+1-dimensional field theory, transverse to the emergent direction.

We compute the distance between the points i and j to obtain the scaling of the distance $d_z(j)$. We assume $i < j$ without loss of generality. At each step, we tune the flux ϕ so as to remove the flux-dependent part from the EE. Following eq. (3.16), the mutual information is of the form $\theta(-z)S_i + \theta(z)S_j$, such that the distance is given by

$$d_z(i, j) = \theta(-z) \ln \frac{S_{\max}}{S_z(i)} + \theta(z) \ln \frac{S_{\max}}{S_{j,z}} = \theta(z)j^z \ln 2 + \theta(-z)i^z \ln 2. \quad (5.2)$$

For $z > 0$, the distance increases as j increases (keeping i fixed), as the k -space points are made more coarse-grained under the RG transformations. On the other hand, for $z < 0$, the distance decreases as i is increased (keeping j fixed). The rate of increase or decrease depends on the value of z . This scaling of the distance is shown in the left panel of Fig. 3.6.

At any given RG step j , we define two subsystems $\mathcal{B}_j^1$ and $\mathcal{B}_j^2$. The first subsystem $\mathcal{B}_j^1$ is the entire set of states $\mathcal{A}_{j,z}$ available at the j^{th} RG step, while the second subsystem $\mathcal{B}_j^2$ is the set comprising those modes that remain to be considered at the next RG step if $z > 0$, and the set comprising only the modes that were present in the previous step for $z < 0$. This scheme is shown in the right panel of Fig. 3.6. The mutual information between these two sets defines a measure of distance y_j at the j^{th} RG step:

$$y_z(j) \equiv \ln \frac{I_{\max}^2}{I_z^2(\mathcal{B}_j^1 : \mathcal{B}_j^2)} = \begin{cases} (j+1)^z \ln 2, & z > 0, \\ (j-1)^z \ln 2, & z < 0. \end{cases} \quad (5.3)$$

The distance along the RG direction ($x_z(j)$) is, on the other hand, defined as the distance between

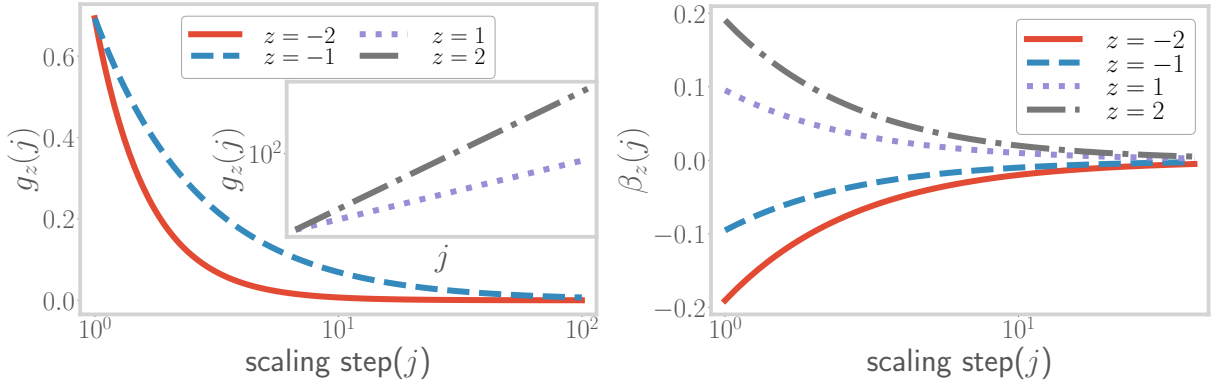


Figure 3.7: Top: Depiction of the RG flow of the coupling $g_z(j)$. For $z < 0$ (blue and red curves), the coupling is RG-irrelevant and shows a decrease as j is increased. For $z > 0$ (gray and violet curves), the coupling is RG-relevant and shows an increase as j is increased. Bottom: Variation of RG beta function $\beta_z(j)$ for the coupling $g_z(j)$, as a function of step index j . For all values of z , the beta function converges to 0 as $j \rightarrow \infty$, indicating a fixed point of the coupling g .

the current RG step and the one with the maximum entanglement entropy:

$$x_z(j) \equiv \theta(z)d(0, j) + \theta(-z)d(j, \infty) = j^z \ln 2, \quad (5.4)$$

where $d(i, j)$ was defined in eq. (5.1).

The connection of the x - and y -distances to the RG can be made manifest by relating them to the RG beta function. As we have seen above, even though we are working with massless Dirac fermions ($m = 0$), the Aharonov-Bohm flux and the dimensional reduction into $1 + 1$ -dimensional modes generate an effective mass $M_{n,\phi} = \frac{2\pi}{L_x}(n + \phi)$ for $n > 0$. At a specific RG step j , the n^{th} mode is given by $t_z(j) \times n$ on account of the RG transformations in k_x -space. With these points in mind, we define a coupling $g_z(j)$ that acts as a measure of the gap in the single-particle spectrum

$$g_z(j) \equiv \ln \frac{M_{n+1,\phi}(j) - M_{n,\phi}(j)}{2\pi/L_x} = \ln t_z(j) = j^z \ln 2. \quad (5.5)$$

The RG beta-function for the coupling $g_z(j)$ can then be written as

$$\beta_z(j) \equiv \frac{\Delta \log g_z}{\Delta j} = \log \frac{g_z(j+1)}{g_z(j)} = z \ln \left(1 + \frac{1}{j} \right), \quad (5.6)$$

where we have substituted $\Delta j = 1$. The beta function is positive for $z > 0$, indicating that the coupling $g_z(j)$ is RG-relevant in this regime (left panel of Fig. 3.7). On the other hand, the beta function is negative for $z < 0$, leading to irrelevant RG flows in this regime. The RG beta-function $\beta_z(j)$ itself decreases monotonically as j increases under the RG transformations (see Fig. 3.7): for a finite z , a fixed point is reached at $j \rightarrow \infty$ where the beta function goes to zero.

As we will now see, the flow of the RG beta functions can be related to the extremisation of a potential defined in the space of Hamiltonians. This potential is a function of the mutual information

$I_z^2(0 : j)$:

$$\beta_z(j) = -\Delta \left[\ln \left(\ln \frac{I_z^2(0 : j)}{I_{\max}^2} \right) \right]. \quad (5.7)$$

The potential given within the square brackets above corresponds to an RG monotone whose fixed point corresponds with the fixed point of the RG. For relevant flows, the above equation ensures that the function $\ln \left(\ln \frac{I_z^2(0:j)}{I_{\max}^2} \right)$ is always minimised under the RG flow, with the consequence that the trajectories ultimately lead to minimisation of the mutual information. This indicates that the relevance of the mass coupling leads in turn to the decrease of the correlation length and hence a lowering of entanglement between points that are separated in real space. On the other hand, for irrelevant flows, the above-mentioned function is maximised, implying that the RG trajectories now maximise the entanglement. This reflects the fact that the decreasing mass leads to increased correlation lengths and an approach towards criticality.

Given eqs.(5.3), (5.4) and (5.7), we can relate the distances x and y to the beta function β_z :

$$x_z = \left(e^{\frac{\beta_z}{z}} - 1 \right)^{-z} \ln 2, \quad (5.8)$$

$$y_z = \begin{cases} x_z e^{\beta} & z > 0, \\ x_z \left(2 - e^{\frac{\beta}{z}} \right)^z & z < 0. \end{cases} \quad (5.9)$$

The contrasting behaviour of the beta function depending on the sign of z indicates that it tracks the effective number of degrees of freedom that contribute to the entanglement (eq. (3.14)) at any given RG step. Indeed, we find that the effective number of degrees of freedom is the effective c -function (or central charge) $\tilde{c}_z(j)$ of the theory, taking the form $\tilde{c}_z(j) = f_z(i)c$. The beta function is thus related to the c -function as

$$\beta_z = \Delta \left[\ln \left(\ln \frac{c}{\tilde{c}_z(j)} \right) \right]. \quad (5.10)$$

For RG relevant flows, the c -function is found to diminish from the free fermion central charge $c = 1/3$, reaching the value of zero at the fixed point (where only the central $n = 0$ mode remains and the rest have been removed because of the large mass gap). The non-negative, non-increasing nature of the c -function, and its flow towards its minimum at the fixed point, leads to speculate that eq.(5.10) can be thought of as a holographic counterpart of Zamalodchikov's c -theorem [287, 387–389]. For RG irrelevant flows, the behaviour is reversed. The flow of the c -function is shown in the left panel of Fig. 3.8.

3.5.2 Embedding the emergent dimension on a metric space

We have shown above that the renormalisation group flow of the Hamiltonian leads to the emergence of an additional direction, quantified by the parameter x_j . This will now be used to construct a metric, in the following way. We already know that the scaling transformations lead to a graph of theories that are connected via edges that represent the mutual information between any two theories. As a result, this “information graph” can be mapped on to a new “distance graph” where the edges

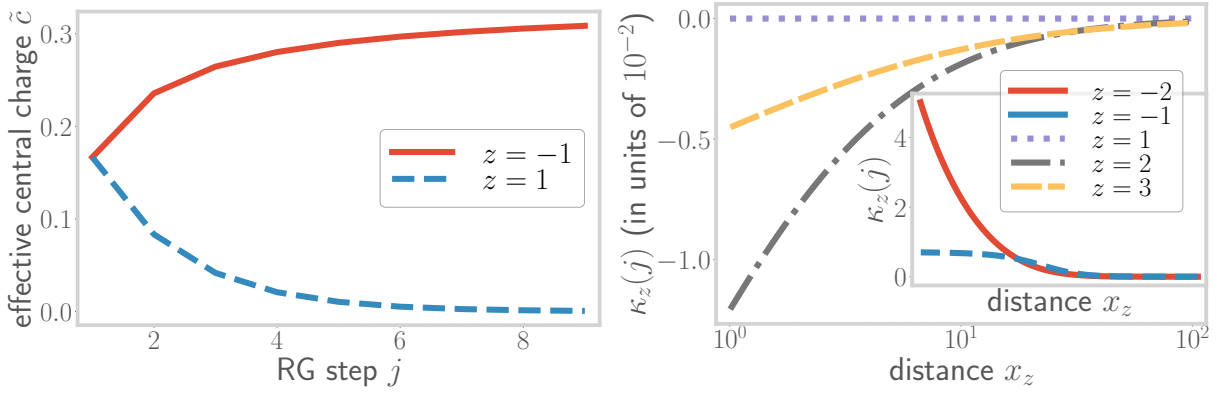


Figure 3.8: Top: Variation of the effective central charge defined in eq. (5.10), under the RG transformations for relevant ($z = 1$, blue curve) and irrelevant ($z = -1$, red curve) flows. In the former case, the central charge starts from the fermionic gapless value of $1/3$ and vanishes towards the fixed point, indicating the flow to a gapped massive phase. In the latter case, it starts from the gapped value of 0, and increases to $1/3$ at the fixed point, indicating the flow to a gapless phase. Bottom: Variation of the curvature $\kappa_z(j)$ along the RG for multiple values of z . $z = 1$ leads to a flat space with zero curvature, while $z > 1$ and $z < 1$ (inset) lead to negative and positive curvature respectively.

now represent the distance between the theories at the nodes. Finally, these distances can be used to define a metric for the distance graph: the metric between any two nodes will be the distance along the shortest path between these two nodes. This approach as well as some of the terminology are inherited from reference [390].

The RG flow can be represented as an “information graph” $G(V, E)$, where the systems $\mathcal{B}_j^1$ (defined above eq. (5.3)) constitute the set of vertices $V = \{V_j\}$ and the mutual information $I_z^2(j : j')$ between the systems $\mathcal{B}_j^1$ and $\mathcal{B}_{j'}^1$, constitute the edges E_{ij} between the vertices. This information graph is characterised by an adjacency matrix whose elements are defined by the mutual information edge weights $I_z^2(j : j')$ [391]. This graph structure is shown in Fig. (3.6), with the red spheres indicating the vertices. In order to define a metric space, we will map the graph G to another graph $\mathcal{G}$ that has the same vertices as G , but whose edges are weighted according to a well-defined notion of distance.

A path $\mathcal{P}$ in $\mathcal{G}$ is defined as a sequence of m vertices: $\mathcal{P}(V_1, V_2, \dots V_m)$. The distance along the path is defined as the sum of distances along every pair of consecutive vertices: $l(\mathcal{P}) = \sum_{i=1}^{m-1} d(V_i, V_{i+1})$, where $d(V_i, V_{i+1}) = d(i, i+1)$ (see eq. (5.1)). The *geodesic* between two points V_i and V_j is the path $\mathcal{P}$ which minimises the distance $l(\mathcal{P})$, and the distance along the geodesic then defines the *metric* $\tilde{d}(V_i, V_j)$ between two vertices V_i and V_j [390, 391]:

$$\tilde{d}(V_i, V_j) = \min_{\mathcal{P}} \{l(\mathcal{P})\} = \min_{\mathcal{P}} \left\{ \sum_{i=1}^{m-1} d(V_i, V_{i+1}) \right\}, \quad (5.11)$$

where the minimisation is carried out over all possible connected paths $\{\mathcal{P}\}$ between V_i and V_j , and m is the total number of vertices in a particular connected path $\mathcal{P}_m$. Following the expression for

the distance $d(i, j)$ obtained in eq. (5.2), the metric takes the form

$$\tilde{d}(V_i, V_j) = \theta(z)j^z \ln 2 + \theta(-z)i^z \ln 2, \quad (i < j), \quad (5.12)$$

where we have assumed $i < j$, without loss of generality.

Finally, the metric dimension of the emergent space can be determined by finding the set with the lowest possible cardinality that can resolve the metric space [390, 392]. In order for a set to resolve the metric space, we must have $d(i, a) = d(j, a) \implies i = j$ for all points a in the set and for any pair of points i and j in the metric space. In our case, the singleton set $\mathcal{R} = \{0\}$ can resolve the metric space for $z > 0$, as $d(i, 0) = d(j, 0) \implies i^z = j^z \implies i = j$. Similarly, the singleton set $\mathcal{R} = \{\infty\}$ resolves the metric space for $z < 0$. As a result, we conclude that our emergent metric space is one-dimensional in nature.

3.6 Curvature of the emergent space

Further insight on the spatial geometry of the RG flows can be obtained by looking at the curvature generated by the RG transformations. In the plane of the directions x and y generated by the RG flow, the curvature can be written using the ideas of analytic geometry as

$$\kappa_z(j) = \frac{v'_z(j)}{\left[1 + v_z(j)^2\right]^{\frac{3}{2}}}, \quad (6.1)$$

where

$$\begin{aligned} v_z(j) &\equiv \frac{\Delta y_z(j)}{\Delta x_z(j)} = \frac{y_z(j+1) - y_z(j)}{x_z(j+1) - x_z(j)} \\ &= \begin{cases} \frac{(j+2)^z - (j+1)^z}{(j+1)^z - j^z} & z > 0, \\ \frac{(j)^z - (j-1)^z}{(j+1)^z - j^z} & z < 0, \end{cases} \end{aligned} \quad (6.2)$$

and

$$v'_z(j) \equiv \frac{v_z(j+1) - v_z(j)}{x_z(j+1) - x_z(j)}. \quad (6.3)$$

The curvature $\kappa_z(j)$ has been plotted against the RG step index j in right panel of Fig. 3.8. For $z = 1$, the first derivative $v_z(j)$ becomes unity, so the second derivative $v'_z(j)$ (and hence the curvature) vanishes, leading to a flat (Minkowski) space. For $z > 1$, the curvature becomes negative, while it is positive for $z < 0$. All the flows lead to an asymptotically flat space (i.e., the curvature vanishes) as $j \rightarrow \infty$.

The different signatures of the emergent curvatures can be attributed to the different ways in which the mass of the theory scales under the decimation procedure. For $z > 0$, the curvature defined in eq. (6.1) can be written in terms of the beta function:

$$\kappa_z(j) = \frac{\left[e^{\beta(j)} - 1\right]^{\frac{1}{2}} j^{-z}}{\left[e^{\beta(j)+\beta(j+1)} - 1\right]^{\frac{3}{2}}} \left[\frac{e^{\beta(j+2)} - 1}{1 - e^{-\beta(j+1)}} - \frac{e^{\beta(j+1)} - 1}{1 - e^{-\beta(j)}} \right]. \quad (6.4)$$

This equation relates directly a geometric quantity (i.e., the curvature) of the bulk emergent space with the RG beta function for the field theory on the boundary. From Fig. 3.7, as the beta functions flow towards zero, we see from eq. (6.4) that the first fraction inside the square brackets is the first to vanish, followed by the second fraction. The sequential vanishing of these two fractions prior to the vanishing of the denominator $e^{\beta(j)} - 1$ in eq.(6.4) leads to the asymptotic vanishing of the curvature towards the fixed point. Different choices of $z > 0$ can be thought to represent different relevant perturbations that drive the system towards *gapped fixed points*, leading to distinct curvatures of the bulk. By employing the relation $v_z(j) = \Delta S_{j+1,z} / \Delta S_{j,z}$, eq. (6.1) can also be cast in terms of the entanglement entropy $S_{j,z}$, $S_{j+1,z}$ and $S_{j+2,z}$. The resulting non-linear relationship between $\kappa_z(j)$ and $S_{j,z}$ obtained is, however, in contrast to the linear dependence obtained by Cao et al. [390] between entanglement perturbations on the Hilbert space and the curvature of the resultant holographic space.

3.6.1 Topological nature of the sign of curvature

We will now argue that the signature of the curvature κ corresponds to a topological quantum number, and that a change in the signature as z is tuned through 1 amounts to a topological translation. For this, we first point out (following eq. (6.1)) that the sign of the curvature is determined purely by the sign of $\gamma_z(j) \equiv 1 - v_z(j+1)/v_z(j)$ and $\alpha_z(j) = y_z(j+1) - y_z(j)$:

$$\begin{aligned} \kappa_z(j) &= \frac{v'_z(j)}{[1 + v_z(j)^2]^{\frac{3}{2}}} = -\frac{\alpha_z(j) \gamma_z(j)}{(\Delta x_z(j))^2 [1 + v_z(j)^2]^{\frac{3}{2}}} \\ \implies \text{sign} [\kappa_z(j)] &= -\text{sign} [\alpha_z(j)] \text{sign} [\gamma_z(j)] . \end{aligned} \quad (6.5)$$

We now promote j to a continuous complex variable with the real part being greater than or equal to 1: $\text{Re} [j] \geq 1$. The quantities $\alpha_z(j)$ and $\gamma_z(j)$ can be written as

$$\alpha_z(j) = \begin{cases} [(j+2)^z - (j+1)^z] \log 2, & z > 0 \\ [j^z - (j-1)^z] \log 2, & z < 0 \end{cases} , \quad (6.6)$$

$$\gamma_z(j) = \begin{cases} 1 - \frac{[(j+3)^z - (j+2)^z][(j+1)^z - j^z]}{[(j+2)^z - (j+1)^z]^2}, & z > 0 \\ 1 - \frac{[(j+1)^z - j^z]^2}{[j^z - (j-1)^z][(j+2)^z - (j+1)^z]}, & z < 0 . \end{cases} \quad (6.7)$$

As shown in Fig.3.9(left panel), the function γ_z is positive and monotonically decreasing for $z > 1$ or $z \leq -1$, and vanishes identically for $z = 1$. On the other hand, $\alpha_z(j)$ is positive for $z > 1$, zero at $z = 1$ and negative for $z \leq -1$ (see Fig.3.9(right panel)). Combining this, we find that the sign of the curvature can be written as:

$$\text{sign} [\kappa_z] = \begin{cases} -1, & z \geq 1 \\ 1, & z \leq -1 \end{cases} = \begin{cases} -\text{sign} [\gamma_z(j)], & z \geq 1 \\ -\text{sign} [\alpha_z(j)], & z \leq -1 . \end{cases} \quad (6.8)$$

We start by considering the region $z \geq 1$. Since $z = 0$ is excluded by definition, the denominator is always non-zero. This indicates that $\gamma_z(j)$ is continuous and monotonically decreasing in this

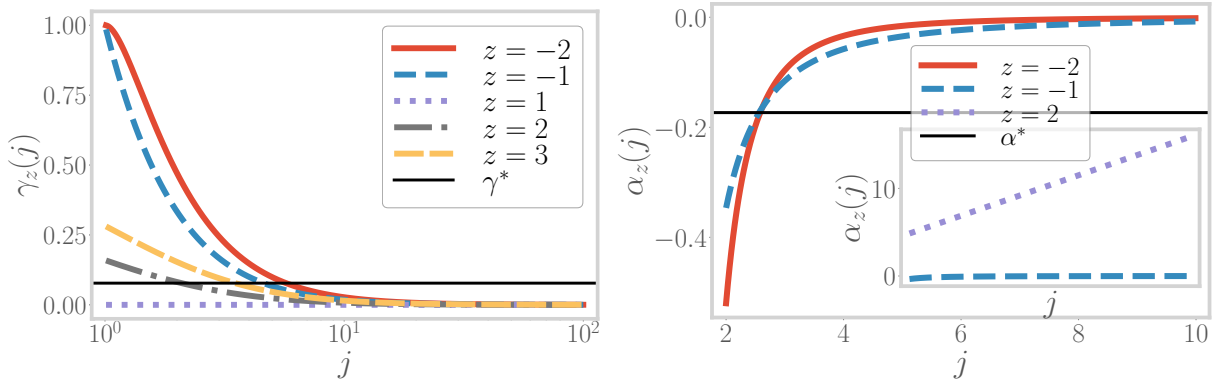


Figure 3.9: Top: Variation of γ_z along the RG for multiple values of $z > 0$. All trajectories for $z > 1$ intersect the line $y = \gamma^*$. Bottom: Variation of α_z along the RG for multiple values of $z < 0$. All trajectories intersect the line $y = \alpha^*$

range: the maximum value of $\{\gamma_z(j), j \in [1, \infty]\}$ occurs at $j = 1$. Further, the maximum value $\gamma_z(1)$ increases with the value of z , ranging between $\gamma_2(2)$ at $z = 2$ and unity at $z \rightarrow \infty$. These two features are demonstrated in the left panel of Fig. 3.9, and imply that all curves $\{\gamma_z(j) : j\}$ will take values both greater than and less than the value $\gamma^* \equiv \gamma_2(2)$ (dotted line in left panel of Fig. 3.9). Thus, the plot shows that there always exists a real value j_z^* for any given integer value $z > 1$ such that $\gamma_z(j_z^*) = \gamma^* = \gamma_2(2)$. This is precisely what ceases to happen for $z = 1$, as that is the only curve in the left panel of Fig. 3.9 that does not pass through γ^* .

Indeed, the difference between the curvature for the $z = 1$ and $z > 1$ classes can be captured by expressing the presence of the solution j_z^* in the form of a winding number. For this, we use the fact that the sign of $\gamma_j(z)$ is given by the following integral:

$$\text{sign} [\gamma_z(j)] \Big|_{z \geq 1} = \frac{1}{2\pi i} \oint_C dj \frac{\partial}{\partial j} \ln [\gamma_z(j) - \gamma^*] , \quad (6.9)$$

where the contour C extends from $j = 1$ to $j = \infty$, and is shown in the left panel of Fig. 3.12. As the value of z is tuned from 2 to 1, the pole at j_z^* disappears and the integral reduces to zero. As shown in Fig. 3.10, the integral can thus be expressed as the winding number of the curve $\gamma_z(C(j))$ that counts the number of times the contour $\gamma_z(C)$ winds around the singularity at γ^* :

$$\text{sign} [\gamma_z(j)] = \frac{1}{2\pi i} \oint_C dj \frac{\partial(\gamma_z(j) - \gamma^*)}{\gamma_z(j) - \gamma^*} = \frac{1}{2\pi i} \oint_{\gamma_z(C)} \frac{d\gamma_z}{\gamma_z - \gamma^*} . \quad (6.10)$$

This is easily seen from the following argument. If we write a general complex variable y in the form $y = r e^{i\theta}$ and consider a general contour C that starts from $r = r_0, \theta = 0$ and ends at $r = r_0, \theta = 2w\pi$ (such that the winding number of the curve is w), integrals of the form given above simplify and return the winding number itself:

$$\frac{1}{2\pi i} \oint_C \frac{dy}{y} = \frac{1}{2\pi i} \int_{r_0}^{r_0} \frac{dr}{r} + \frac{1}{2\pi} \int_0^{2w\pi} d\theta = w , \quad (6.11)$$

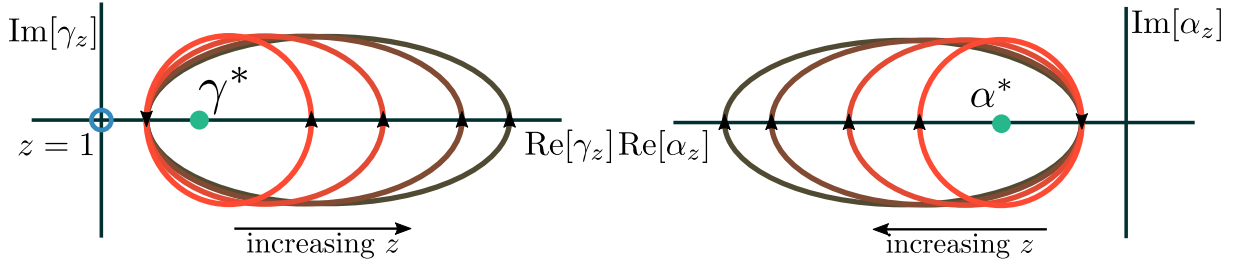


Figure 3.10: Top: Contours $\gamma_z(C)$ for multiple values of $z > 0$. As z increases, the contours extend in the positive $\text{Re}[\gamma_z]$ direction. Bottom: Contours $\alpha_z(C)$ for multiple values of $z > 0$. As z increases, the contours extend in the negative $\text{Re}[\gamma_z]$ direction.

where we used $dy = y \left(\frac{dr}{r} + d\theta \right)$.

The winding numbers take integer values, and are topological in nature: they are robust against geometric deformations, and change only when the curve crosses the singularity at $\gamma = \gamma^*$. Thus, for $z > 1$, the contour $\gamma_z(C)$ encloses the singularity at γ^* exactly once, leading to a winding number of unity. As z is lowered towards $z = 2$, the contours becomes smaller such that, in order to go from $z = 2$ to $z = 1$, the contour has to be moved across the singularity. This constituting a change in the topology of the contour: for $z = 1$, the contour no longer encloses the pole and the winding number is zero. In fact, at $z = 1$, the contour itself (depicted in blue) is of vanishing radius because $\gamma_1(j)$ has a constant value of zero. To complete the connection, we use eq. (6.5) to relate this topological winding number to the sign of the curvature:

$$\text{sign}[\kappa_z] \Big|_{z \geq 1} = -\mathcal{W}_z(\gamma^*), \quad (6.12)$$

where $\mathcal{W}_z(\gamma^*)$ is the winding number for $\gamma_z(C)$ around γ^* .

One can make a very similar argument for the case of $z \leq -1$. Here, as shown in the right panel of Fig. 3.10, the relevant quantity is $\alpha_z(j)$. For each curve α_z , the minimum value is at $j = 1$. For all z , this minimum starting value is largest for $z = -1$, and all subsequent curves $\alpha_z(j)$ start from even lower values. If we define a value $\alpha^* \equiv \alpha_z(j = 1)/2$, it is then clear that all curves $\alpha_z(j)$ will pass through the value α^* (dotted line in right panel of Fig. 3.10) at some value of j . This is very similar to the situation discussed just above for the case of $z \geq 1$ (with γ^* playing the role of α^*); we, therefore, employ the same technique here. Taking the same contour C as defined above, we write the sign of α_z as the winding number of the curve $\alpha_z(C)$ around the point $\alpha_z = \alpha^*$

$$\text{sign}[\alpha_z(j)] \Big|_{z \leq -1} = -\mathcal{W}'_z(\alpha^*) = \frac{1}{2\pi i} \oint_{\alpha_z(C)} \frac{d\alpha_z}{\alpha_z - \alpha^*}. \quad (6.13)$$

For $z > 0$, α_z is positive, and the curve $\alpha_z(C)$ encloses points in the positive half of the real axis such that the winding number in the right panel of Fig. 3.10 is then zero. This indicates that the function $1 - 2\mathcal{W}'_z(\alpha^*)$ is unity for $z > 0$, leaving any other function in product with it unchanged. On the other hand, it is -1 for $z < 0$, returning the correct sign of $\alpha_z(j)$ in that range. Combining with eq. (6.12), the sign in the entire range can be expressed as

$$\text{sign}[\kappa_z] = \mathcal{W}_z(\gamma^*) \times [2\mathcal{W}'_z(\alpha^*) - 1]. \quad (6.14)$$

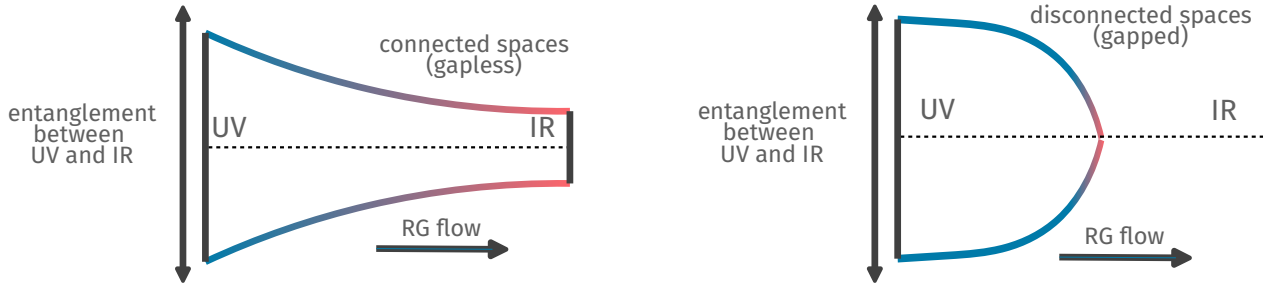


Figure 3.11: Schematic depiction of the emergent geometries in the cases of gapless (top) and gapped (bottom) RG flows. While the gapless RG flow involves a growth in the entanglement between the UV and IR spaces, the latter leads to the vanishing of UV-IR entanglement. This results in the emergence of connected and disconnected geometries, respectively.

We can now summarise the topological nature of the sign of the curvature. For $z > 0$, the second winding number $\mathcal{W}$ is 0. As z varies from 2 to 1, the other winding number $\mathcal{W}$ changes from 1 to 0, constituting the first topological change. In contrast, upon varying z from 1 to -1 , both winding numbers change: $\mathcal{W}$ from 0 to 1, and $\mathcal{W}$ from 0 to 1. This constitutes the second topological change. Together, the two changes show that going from an open geometry (negative curvature) to a closed geometry (positive curvature) involves two topological transformations.

3.7 Entanglement Holography and Fermionic Criticality

3.7.1 Equivalence between a critical Fermi surface and a holographic worm-hole

As mentioned previously under eq. (4.3), the quantity z corresponds to the anomalous dimension of the spectral gap g_z in the effective field theory: a change in the sign of z corresponds to a change in the nature of the RG flow from relevance to irrelevance and vice-versa. The holographic nature of the RG flow dictates that such a change arises by a variation in some quantity in the parent interacting theory from which the effective field theory of non-interacting Dirac fermions can be thought to have originated. Indeed, such a change in the RG relevance has drastic consequences: a relevant spectral distance g_z implies a gap in the excitations at the fixed point, while an irrelevant g_z indicates the presence of gapless excitations. The presence or absence of a gap in the low-energy excitations determines the topology of the electronic Fermi surface through the change of a topological invariant, the Luttinger volume of the system [97, 99, 100, 264, 265] (also see eq. (2.19)).

At the transition between the gapless (metallic) and gapped (insulating) states, the Fermi surface becomes critical, signalled by the vanishing lifetime of the gapless excitations close to it. Our formulation shows that such a gap-inducing transition and its associated quantum critical Fermi surface has consequences for the space generated along the RG flow. For the gapless case ($z < 0$), the degrees of freedom in the UV and the IR have finite entanglement, indicating that they are part of the same connected holographic space (see Figs. 3.6 and 3.7), while for the gapped case ($z > 0$), the entanglement vanishes into the IR, leading to a decoupling of these two regions and

their separation into disconnected spaces [269]. These two scenarios are shown schematically in Fig. 3.11. It is therefore tempting to conclude that at the transition between the gapped and gapless states, a *wormhole geometry* connects the UV and the IR via some minimal entanglement [269, 390]. Our results also suggest that the presence of such a wormhole would then coincide with two topological transitions, the first being the change in the sign of the curvature (as shown in the previous subsection), and the second being the change in the topological Luttinger volume of the underlying quantum field theory (as discussed in the previous paragraph). We end by noting that we have provided, in a later section of this work, additional support for the topological nature of Luttinger's volume and its connection to the holographically generated space by relating it to multipartite entanglement (see eq. (9.17)).

3.7.2 Connecting modular energy with the metric

The notion of an entanglement Hamiltonian was introduced in eq. (2.16). At the transition between the gapless (metallic) and gapped (insulating) states, the system becomes critical, and the Dirac electrons are described by a CFT. It turns out that for such a CFT, the entanglement Hamiltonian takes a particularly simple form. In this subsection, we will relate the emergent metric to the entanglement Hamiltonian obtained at the IR fixed point, focusing on the theory at the transition.

For a 1D CFT, the entanglement Hamiltonian corresponding to a partition of length l is given by [393, 394]

$$\mathcal{H}_{1D} = \frac{2\pi L}{c} \int_0^l dx \frac{L^2 - x^2}{2L^2} T_{00}(x), \quad (7.1)$$

where $T_{\mu\nu}$ is the stress-energy tensor of the CFT and c is the speed of light. We have already shown in eq. (2.17) that the entanglement Hamiltonian for the 2D theory is simply a sum of the 1D counterparts, owing to the decoupling of 1D modes in the Lagrangian. Working in a region where T_{00} is approximately uniform, the 2D entanglement Hamiltonian can be expressed as

$$\mathcal{H}_{2D} \sim \sum_n \mathcal{H}_{1D} = \frac{2\pi L^2}{3c} \sum_n T_{00}^n, \quad (7.2)$$

where T_{00}^n is the average value of T_{00} within the partition for the CFT associated with the n^{th} mode.

In order to make the connection with the emergent geometry, we use the entanglement first law [395], which states that at first order, small variations ΔS of the entanglement entropy can be equated to the associated change in the expectation value of the entanglement Hamiltonian:

$$\Delta S(A) = \Delta I_2(A_0 : A) = \langle \Delta \mathcal{H} \rangle = \frac{2\pi L^2}{3c} \sum_n \langle T_{00}^n \rangle, \quad (7.3)$$

where $\langle \mathcal{H} \rangle = \text{Tr}(\rho \mathcal{H})$ is often referred to as the modular energy, and we have equated $\Delta S(A)$ and $\Delta I_2(A_0 : A)$ following eq. (3.16). Further, since the entanglement at the critical point is expected to be large, we can approximate the expression for the metric between points (i, j) obtained in subsection 3.5.2 as $\tilde{d}(i, j) = -\ln I^2(i : j)/I_{\text{max}}^2 \simeq |I^2(i : j)/I_{\text{max}}^2 - 1|$. Combining this equation

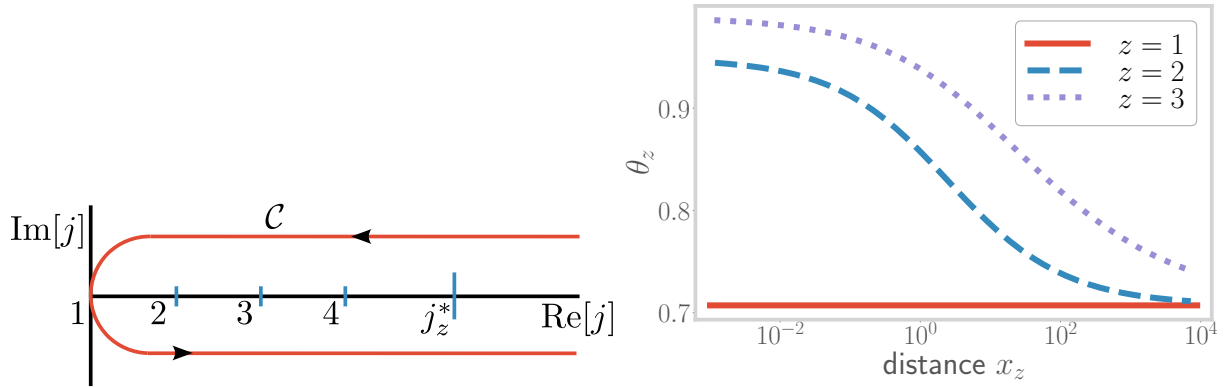


Figure 3.12: Top: The contour C used in eq. (6.9). It extends across all values of j , and the integral counts the number of poles of the function $1/(\gamma - \gamma^*)$ encircled by the contour. Bottom: Evolution of expansion parameter θ_z with distance x_z , leading to the fixed point at $\theta_z^* = 1/\sqrt{2}$.

with eq. (7.3) leads to the following interesting relation:

$$\Delta \tilde{d}(i, j) \simeq \frac{2\pi L^2}{3c} \sum_n \langle T_{00}^n \rangle . \quad (7.4)$$

The equation relates the change in a geometric measure (the emergent metric) with a mass/energy term. An analogous relation was obtained for states for which the entropy obeys an area law by Cao et al [390], connecting the spatial curvature and the stress tensor, ultimately leading to a semi-classical Einstein's field equation for gravity. Our formulation does not however lead to a relation involving the curvature, and we speculate that this is due to the violation of the area law in entanglement of fermionic critical systems. Instead, we show in the next section that the curvature in our formulation is related to the rate of convergence of the RG flows.

3.7.3 Kinematics of the RG flows

Further statements on the RG flow and the emergent holographic dimension can be made by studying quantities that can serve as candidates for the convergence parameter θ for the RG flows of g_z [396, 397]. We consider the following function as a candidate for the convergence parameter for the case of $z > 0$:

$$\theta_z(j) = \frac{1}{\sqrt{v_z^{-2} + 1}} = \left[1 + \left\{ \frac{1 - e^{-\beta_z(j)}}{e^{\beta_z(j+1)-1}} \right\}^2 \right]^{-\frac{1}{2}} . \quad (7.5)$$

The parameter θ_z has been plotted in Fig. 3.12, and the behaviour is observed to be very similar to that of the beta function shown in Fig. 3.7. Indeed, this similarity describes the holographic nature of the RG flow: the emergent geometry (as seen through θ_z) is dictated by the RG flow described by β .

The quantity θ can be interpreted as an estimate of the rate of change of area enclosed by a fixed number of RG trajectories. To see that this is the case, we note that by using $v_j = \Delta y / \Delta x$,

the expansion parameter can be written as $\theta = \Delta y(j)/\sqrt{\Delta x^2 + \Delta y^2} \sim \Delta g(j+1)/\sqrt{\Delta x^2 + \Delta y^2}$. Considering RG flows for $g_z(j)$ corresponding to multiple z , the area enclosed by the curves is given by the maximum g_z , and Δg_z is then the change in the area under the RG transformation. The denominator $\sqrt{\Delta x^2 + \Delta y^2}$ is the distance covered in the emergent spacetime, rendering θ as the rate of change of the area.

The equation of motion for this expansion parameter can then be obtained as:

$$\frac{d\theta_z}{dx} = \frac{v_z^{-3}}{(1 + v_z^{-2})^{\frac{3}{2}}} \frac{dv_z}{dx_j} = \kappa_z. \quad (7.6)$$

The simple form of this relation makes it clear that for relevant flows (corresponding to a negative curvature $\kappa_z < 0$), θ_z flows to a minimum. Following Fig. 3.8, this is accompanied by the minimisation of the c -function to zero. It is, therefore, tempting to conclude that eq. (7.6) for the spatial evolution of the convergence parameter θ is dual to eq. (5.10) that controls the flow of the c -function of the quantum theory along the RG flow.

3.8 Extension to the massive case

We will now generalise several of the main results obtained above to the case of two-dimensional electrons with a single-particle mass-gap $M_{n,\phi} = \sqrt{m^2 + \frac{4\pi^2}{L_x^2}(n + \phi)^2}$. Given that they depend only on the geometric part of the entanglement (which remains unchanged on introducing the mass $M_{n,\phi}$), the following set of key qualitative results and observations discussed earlier continue to hold for the case of a dispersion with a mass-gap:

- the hierarchical structure of the entanglement content (e.g., the area-law term in eqns.(3.14) and (3.17)) and the fact that the entanglement satisfies the Ryu-Takayanagi bound (right panel of Fig.3.5),
- the dependence of the emergent distances x and y on the RG flow (Fig.3.6, eqns.(5.2) and (5.3)), and the fact that the RG beta function tracks the central charge (Fig.3.8),
- various qualitative and quantitative properties of the curvature (e.g., eq.(6.1)), including the topological nature of its sign (eq.(6.14)), and
- the existence of a convergence parameter that is related to the RG flow, and the equation that governs the evolution of the convergence parameter (eq.(7.6)).

The functional forms of certain other results do, however, undergo changes. For instance, the coupling $g_z(j)$ is given by the (more general) expression

$$\begin{aligned} g_z(n, j) &= \ln \frac{M_{n+1,\phi}(j) - M_{n,\phi}(j)}{2\pi/L_x} \\ &= \ln \left[\sqrt{\tilde{m}^2 + (t_z(j)(n+1) + \phi)^2} - \sqrt{\tilde{m}^2 + (t_z(j)n + \phi)^2} \right], \end{aligned} \quad (8.1)$$

where $\tilde{m} = mL_x/2\pi$ is a dimensionless parameter. We will work with the coupling for the mode $n = 0$:

$$g_z(0, j) = \ln \left[\sqrt{\tilde{m}^2 + (t_z(j) + \phi)^2} - \sqrt{\tilde{m}^2 + \phi^2} \right]. \quad (8.2)$$

In order to simplify the analysis, we work in the large mass limit $\frac{t_z + \phi}{\tilde{m}} \rightarrow 0$. In this limit, upon expanding the square root and retaining only the lowest order term, we obtain

$$g_z(0, j) = \ln \frac{[t_z(j) + \phi]^2 - \phi^2}{2\tilde{m}}. \quad (8.3)$$

The RG beta function $\beta = \ln(g_z(j+1)/g_z(j))$ can, in the large mass limit, then be related to the emergent distances x and y in terms of the coupling $g_z(j)$:

$$x_z(j) = \ln t_z(j) = \ln \left[\sqrt{2\tilde{m} e^{g_z(0,j)} + \phi^2} - \phi \right], \quad (8.4)$$

$$y_z(j) = \begin{cases} \ln \left[\sqrt{2\tilde{m} e^{g_z(0,j+1)} + \phi^2} - \phi \right], & z > 0, \\ \ln \left[\sqrt{2\tilde{m} e^{g_z(0,j-1)} + \phi^2} - \phi \right], & z < 0. \end{cases} \quad (8.5)$$

The quantitative dependence of the c -function on the RG beta function can also be expressed more generally as

$$e^{\beta_z(j)} = \left[\ln \frac{\left(\frac{\tilde{c}(j+1)}{c} + \phi \right)^2 - \phi^2}{2\tilde{m}} \right] \left[\ln \frac{\left(\frac{\tilde{c}(j)}{c} + \phi \right)^2 - \phi^2}{2\tilde{m}} \right]^{-1}. \quad (8.6)$$

Quantities such as the curvature κ and the convergence parameter θ can similarly be expressed in terms of the coupling $g_z(j)$.

3.9 Topological content of the entanglement spectrum

3.9.1 Luttinger's volume as a winding number: the massless case

We will now show how to obtain Luttinger's volume (eq.(2.18)) from the g -partite information entanglement measure I^g of a system of massless 2D electrons. The flux-dependent part of the g -partite information I^g can be used to probe the k -space volume V_L within the Fermi surface. We define a quantity $Q_g(\phi)$ that is periodic in the flux ϕ and independent of the geometry:

$$Q_g(\phi) = f_{g,z} \left[\frac{1}{\sqrt{2}} - \frac{1}{2} e^{-\frac{1}{c} (I_{\{\mathcal{A}_z(i)\}}^g(\phi) - I_{\{\mathcal{A}_z(i)\}}^g(\phi^{(0)} f_{g,z}))} \right] = f_{g,z} \left[\frac{1}{\sqrt{2}} - |\sin(\pi \phi f_{g,z})| \right], \quad (9.1)$$

where $\phi^{(0)} = \frac{1}{6}$ is the value at which the flux-dependent part of $S_0(\phi)$ vanishes. Further, from eq.(3.19), we have

$$I_{\{\mathcal{A}_z(i)\}}^g(\phi_g) - I_{\{\mathcal{A}_z(i)\}}^g(\phi_g^{(0)} f_{g,z}) = -c \ln(2 |\sin(\pi \phi f_{g,z})|). \quad (9.2)$$

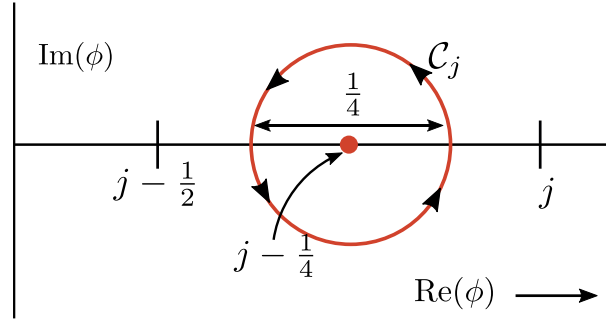


Figure 3.13: Contour C_j used to integrate eq. (9.3). The contour is chosen so as to encircle the simple pole of $1/Q$ at $\phi = j - 1/4$.

Note that the non-differentiability of $Q_g(\phi)$ at $\phi f_{g,z} = 0, 1, 2, \dots$ is not a concern as ϕ is chosen to be non-integral, ensuring that $\phi f_{g,z}$ is also non-integral.

In order to express the number N_L^x of positive k_x -states available inside the Fermi surface in the x -direction in terms of $Q_g(\phi)$, we need to establish a relation between the flux ϕ and the Fermi surface. The k_x -states of interest to us are given by $k_x^n = \frac{2\pi n}{L_x}$, $n \in \mathbb{Z}$, $0 \leq k_x^n \leq k_x^F$, where k_x^F is the x -component of the Fermi momentum $\vec{k}_F$. Assuming the flux-free Hamiltonian has time-reversal symmetry, there should be an equal number of states with the opposite momenta. Let n^* be the integer for which $k_x^{n^*} \leq k_x^F$ and $k_x^{n^*+1} > k_x^F$. Since the flux couples to the k_x -states as $k_x^n \rightarrow k_x^n + eA = \frac{2\pi n + 2\pi\phi}{L_x}$, tuning ϕ by 1 leads to the entire spectrum of k_x -states being shifted by one quantum number: $k_x^n \rightarrow \frac{2\pi(n+1)}{L_x} = k_x^{n+1}$. This means that the state $k_x^{n^*}$ moves to $k_x^{n^*+1}$, and hence out of the Fermi volume by crossing k_x^F . This flow of the spectrum is shown in Fig. 3.15.

If we now connect the system to a particle reservoir with the chemical potential set to zero, any k_x -state that moves out of the Fermi volume becomes unoccupied. We continue tuning the flux to a value ϕ^* , where the system becomes entirely devoid of any particles. The number of k_x -states that are below k_x^F can then be said to be given by the number of times ϕ goes through an integer value while changing from 0^+ to $\phi^* + 0^+$. Since we do not set the flux to an integer value, we can equivalently count the number of times ϕ goes through, say, three-quarters of an integer - $j - \frac{1}{4} : j = 1, 2, \dots$ - in changing from 0^+ to $\phi^* - 0^+$. We will now show that this count can be expressed in terms of Q_g . We begin by considering $Q_0(\phi) = \frac{1}{\sqrt{2}} - |\sin \pi \phi|$. At every three-quarter value of the flux, the quantity $\phi_j = j - \frac{1}{4} : j = 1, 2, \dots$, $|\sin \pi \phi_j|$ attains the value of $\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$, such that $Q_0^{-1}(\phi)$ goes through a simple pole. Each such pole ϕ_j can be picked up via a complex integral around a contour C_j that encloses the pole:

$$-f_j \frac{1}{2\sqrt{2}i} \oint_{C_j} \frac{d\phi}{Q_0(\phi)} = 1, \quad (9.3)$$

where f_j is either +1 or -1 and is defined through the equation $|\sin(\pi(j - \frac{1}{4}))| = f_j \sin(\pi(j - \frac{1}{4}))$, and C_j is a circle of diameter 1/4 centered at $\phi = j - \frac{1}{4}$ (shown in Fig. 3.13). The evaluation of the integral eq.(9.3) is shown in Appendix B. In order to drop the sign prefactor f_j , we reorient

those contours for which f_j is +1, and such that they wind around in the clockwise direction. The contours are shown in the left panel of Fig. 3.14, and defined as

$$C_j(\phi) = \left\{ \phi : \phi = j - \frac{1}{4} + \frac{1}{8}e^{i\theta}, \theta \in [0, -2\pi f_j] \right\}, \quad (9.4)$$

such that they host a singularity at their center. With this redefinition of the contours, we have

$$\frac{1}{2\sqrt{2}i} \oint_{C_j} \frac{d\phi}{Q_0(\phi)} = 1. \quad (9.5)$$

Counting the number of three-quarter values then reduces to counting the number of poles of Q_0^{-1} in the above-mentioned range $\phi \in (0, \phi^*)$. In this way, we can now express the count N_L^x in terms of Q_0 :

$$N_L^x = \frac{1}{2} \sum_{j=1}^{\phi^*} \frac{1}{2\sqrt{2}i} \oint_{C_j(\phi)} \frac{d\phi}{Q_0(\phi)} = \frac{1}{2\sqrt{2}} \sum_{j=1}^{\phi^*} \frac{1}{2i} \oint_{C_j(\phi)} \frac{d\phi}{Q_0(\phi)}. \quad (9.6)$$

The factor of half in eq.(9.6) is required as we want to count only the number of positive momentum states (and which is equal to the number of negative momentum states).

In going from $Q_0(\phi)$ to a general $Q_g(\phi)$, the only change required is to redefine $\phi \rightarrow \phi_g = f_{g,z}\phi$. This implies that while the states in k -space have become more coarse-grained, the flux ϕ has become more fine-tuned by the same scaling factor. This is a reflection of the overall conservation of particle number density: keeping the chemical potential fixed, the increased spacing between the k_x -states implies that a smaller number of particles are now occupying an equally smaller volume. As the number density is conserved, so is the Luttinger volume (eq.(2.18)), and this shows that our approach is consistent with the holographic Luttinger theorem [323]. Thus, given that $\phi_g^* = \phi^*$, the expression for N_L^x remains practically unchanged:

$$N_L^x = \frac{1}{2\sqrt{2}} \sum_{j=1}^{\phi^*} \frac{1}{2i} \oint_{C_j(\phi_g)} \frac{d\phi_g}{\frac{1}{\sqrt{2}} - |\sin \pi \phi_g|}. \quad (9.7)$$

In order to rewrite N_L^x in terms of the flux ϕ , we start by making the substitution $\phi_g \rightarrow f_{g,z}\phi$:

$$N_L^x = \frac{1}{2\sqrt{2}} \sum_{j=1}^{\phi^*} \frac{1}{2i} \oint_{C_j(f_{g,z}\phi)} \frac{f_{g,z}d\phi}{\frac{1}{\sqrt{2}} - |\sin(\pi f_{g,z}\phi)|}. \quad (9.8)$$

In the above equation, we can recognise $\frac{f_{g,z}}{\frac{1}{\sqrt{2}} - |\sin(\pi f_{g,z}\phi)|}$ as $Q_g^{-1}(\phi)$. Also note that the contour can be written as

$$C_j(f_{g,z}\phi) \equiv C_j^g(\phi) = \left\{ \phi : \phi = t_{g,z} \left(j - \frac{1}{4} + \frac{e^{i\theta}}{8} \right), \theta \in [0, -2\pi f_j] \right\}. \quad (9.9)$$

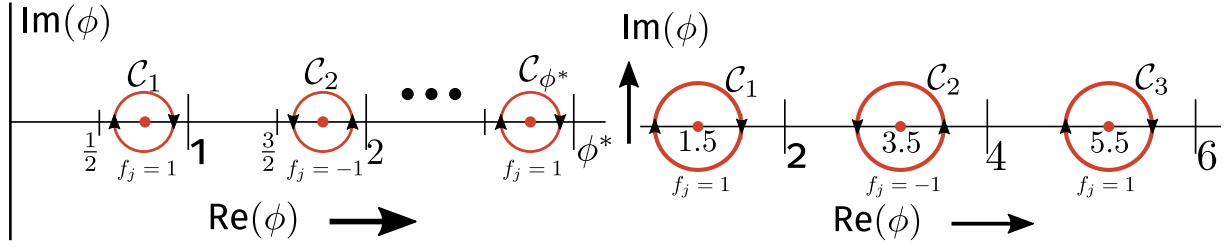


Figure 3.14: Top: Contours C_j used to integrate over Q_0^{-1} in eq. (9.6). Tall vertical lines represent integer values of $\text{Re}(\phi)$. All contours start and end within two adjacent integer values. The final contour C_{ϕ^*} ends just short of $\text{Re}(\phi) = \phi^*$. Note the alternating direction of the contours brought about by the alternation of f_j in going from $j = 1$ to $j = 2$, and so on. Bottom: Contours C_j^g for $t_{g,z} = 2$, centered at $2\left(1 - \frac{1}{4}\right)$, $2\left(2 - \frac{1}{4}\right)$ and $2\left(3 - \frac{1}{4}\right)$. Note the alternating directions of the contours.

These curves are again circles of radius $1/8$, but centered at $t_{g,z} \times \left(1 - \frac{1}{4}\right)$, $t_{g,z} \times \left(2 - \frac{1}{4}\right)$, $t_{g,z} \times \left(3 - \frac{1}{4}\right)$ and so on. These are simply a generalisation of the curves defined in eq. (9.4); the curves for $t_{g,z} = 2$ are shown in the right panel of Fig. 3.14. Substituting this in eq. (9.8) gives

$$N_L^x = \frac{1}{2\sqrt{2}} \sum_{j=1}^{\phi^*} \frac{1}{2i} \oint_{C_j^g(\phi)} \frac{d\phi}{Q_g}. \quad (9.10)$$

For a spatially isotropic system, the Fermi volume is given by

$$V_L = v_2 \times \pi (N_L^x)^2 = \frac{1}{2} \frac{\pi^3}{V} \left[\sum_{j=1}^{\phi^*} \frac{1}{2i} \oint_{C_j^g(\phi)} \frac{d\phi}{Q_g(\phi)} \right]^2, \quad (9.11)$$

where V is the real space volume of the system, and $v_d = (2\pi)^d / V$ is the volume of the smallest unit in k -space. For a generally anisotropic Fermi volume, each direction in the d -dimensional k -space can be completely specified by an angular coordinate θ with respect to some predefined coordinate system. For each direction $\hat{\theta}$, we can insert a gauge field $\vec{A} = A\hat{\theta}$ and take the thin torus limit in this direction, obtaining thereby the g -partite information $I^g(\phi_\theta)$ for length l in the direction perpendicular to $\hat{\theta}$. With the θ -dependent I^g , one can then define the quantity $Q_g(\phi_\theta)$ which counts the number of states N_L^θ in the direction $\hat{\theta}$ inside the Fermi volume. In this way, the total Fermi volume can be obtained by integrating over θ :

$$V_L = \frac{\pi^2}{4V} \int_0^{2\pi} d\theta \left[\sum_{j=1}^{\phi_\theta^*} \frac{1}{2\pi i} \oint_{C_j^g(\phi)} \frac{d\phi}{Q_g(\phi)} \right]^2. \quad (9.12)$$

It is important to note that the quantity V_L is independent of the RG index g , and is thus identical in value for all g members of the hierarchy at the g th step of the coarse-graining process.

The topological nature of Luttinger volume can be made further manifest by expressing it in terms of winding numbers, following Ref. [100] for the case of interacting electrons. To do so, we point out that the integrand $Q_g^{-1}(\phi)$ can be written as a derivative:

$$\frac{1}{Q_g(\phi)} = \frac{\sqrt{2}}{\pi} \frac{\partial}{\partial \phi} \ln Y_j^g = \frac{\sqrt{2}}{\pi} \frac{1}{Y_j^g} \frac{\partial Y_j^g}{\partial \phi}, \quad (9.13)$$

where

$$Y_j^g(\phi) \equiv \left[\frac{f_j \tan 3\pi/8 - \tan(\pi f_{g,z}\phi/2)}{\tan(\pi f_{g,z}\phi/2) - \tan(\pi/8)} \right]^{f_j} \\ = \begin{cases} \frac{\tan 3\pi/8 - \tan(\pi f_{g,z}\phi/2)}{\tan(\pi f_{g,z}\phi/2) - \tan(\pi/8)}, & \text{when } j = \text{odd, and} \\ \frac{\tan(\pi f_{g,z}\phi/2) + \tan(\pi/8)}{-\tan 3\pi/8 - \tan(\pi f_{g,z}\phi/2)}, & \text{when } j = \text{even.} \end{cases} \quad (9.14)$$

Substituting this into the expression for 2-dimensional isotropic Luttinger volume in eq. (9.11), we obtain

$$V_L = \frac{\pi^3}{V} \left[\sum_{j=1}^{\phi^*} \frac{1}{2\pi i} \oint_{Y_j^g(C_j^g)} \frac{dY_j^g}{Y_j^g} \right]^2. \quad (9.15)$$

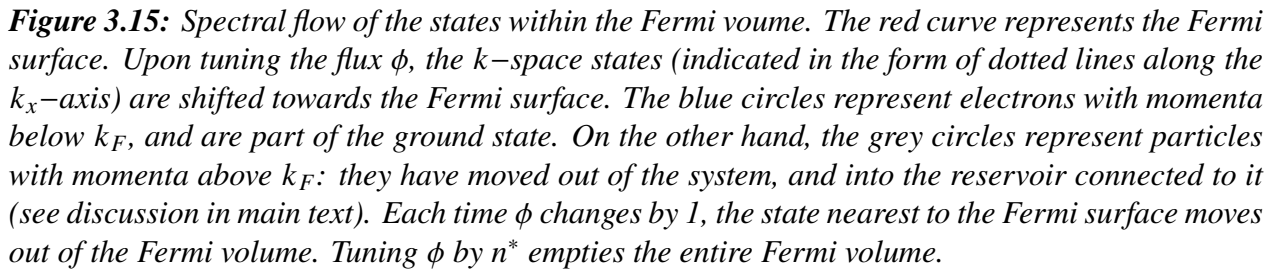
As before, the specific integral in the above expression is a winding number that counts the number of times the contour $Y_j^g(C_j^g)$ winds around the singularity at $Y_j^g = 0$. Note that these are the same singularities that exist at the centers of the contours C_j^g and that are picked up in, for example, eq. (9.3). To demonstrate this, we point out that the centers of the contours are given by $\phi = t_{g,z} \left(j - \frac{1}{4} \right)$, $j = 1, 2, \dots, \phi^*$. At these values, we have

$$\tan \left(\frac{\pi f_{g,z}\phi}{2} \right) = \tan \left(\frac{\pi j}{2} - \frac{\pi}{8} \right) = \begin{cases} \tan \frac{3\pi}{8}, & j = 1, 3, \dots \\ -\tan \frac{\pi}{8}, & j = 2, 4, \dots \end{cases} \quad (9.16)$$

Comparing with eq. (9.14), we can see that the numerators of Y_j^g are zero at these values of ϕ for both even and odd values of j . This leads to $Y_j^g = 0$, and hence a singularity in $\frac{dY_j^g}{Y_j^g}$. These values of the flux are therefore the common singularities that are picked up by the poles of $Q_g^{-1}(\phi)$ as well as the winding numbers.

We now define w_j^g as the winding number for the curve $Y_j^g(C_j^g)$ around $Y_j^g = 0$, and $\mathcal{W}(\phi^*) = \sum_{j=1}^{\phi^*} w_j^g$ as the total winding number for all the curves $\{Y_j^g(C_j^g), j = 1, 2, \dots\}$. Then, the isotropic Fermi volume takes the compact form

$$V_L = \frac{\pi^3}{V} \mathcal{W}(\phi^*)^2. \quad (9.17)$$



Interestingly, the above method employed in using entanglement measures to extract the Luttinger volume can also be applied (with suitable modifications) to an integer quantum hall system (generated upon placing the system in a strong transverse magnetic field) to obtain the number of filled Landau levels (a Chern number). To begin, we imagine a system of free non-relativistic electrons in a cylinder geometry (fig. 3.16), and placed in a magnetic field $\nabla \times \vec{A} = \vec{B}$ that points radially outwards. The field $\vec{B}$ generates Φ flux quanta through the surface of the cylinder. An Aharonov-Bohm flux $\vec{A}' = -2\pi\phi/L_x\hat{x}$ is also placed in the x -direction (i.e., along the circumference of the cylinder). We divide the cylinder into two subsystems A and B (fig. 3.16), such that tuning the dimensionless flux ϕ by 1 transfers a charge equal to the first Chern number, from A to B . The translation invariance in the x -direction ensures that the momentum k_x are good quantum numbers; for each value of k_x , we can define the matrix elements of the single-particle correlation matrix $G(k_y)$ for the subsystem A as [398, 399]

where $|\psi_{\text{gs}}\rangle$ is the ground state and i, j represent the degrees of freedom other than k_x . The total matrix G^A is constructed by joining the smaller blocks $G^A(k_x)$. As ϕ is varied, the variation of the trace of the flux-dependent matrix $G^A(\phi)$ is linear in the flux ϕ : $\text{Tr}[G^A(\phi)] - \text{Tr}[G^A(0)] = -\frac{1}{e}\sigma_H\phi$. This implies that as long as there are extended states at sufficiently small energies, we will have spectral flow that can be captured through the integral:

where C_H is a circle of radius $1/2$ in the complex plane centered at $\phi = 0$. This integral is zero if there is no spectral flow, i.e., the effect of the gauge field can be removed by a gauge transformation and

the trace is insensitive to the flux. In order to count the number of filled Landau levels (essentially, the filling factor ν), we can now vary the flux quanta Φ (due to $\vec{B}$) so as to change the filling from ν to 0:

$$\nu = -\frac{\sigma_H}{2\pi i e} \sum_{\Phi'=\Phi}^{\Phi^{\max}} \int_{C_H} \frac{d\phi}{\text{Tr}[G_{\Phi'}^A(\phi)] - \text{Tr}[G_{\Phi'}^A(0)]} . \quad (9.20)$$

The subscript Φ' on G^A indicates that the correlations depend on the variable magnetic field $\vec{B}'$. The upper limit $\Phi^{\max}$ is given by the lowest value of the magnetic field $\vec{B}'$ for which only one Landau level is filled. The magnetic field is varied such that exactly an integer number of Landau levels are filled at any given value. Importantly, the filling factor ν is known to be a Chern number $C(\Phi)$ [174,400].

We can also relate eq. (9.20) to the entanglement of the system by introducing the entanglement spectrum [163, 176], which refers to the set of eigenvalues $\{\lambda_\chi^A(k_x)\}$ of the reduced density matrix $\rho^A(k_x)$ for subsystem A corresponding to the momentum mode k_x in the x -direction and χ is a particular configuration of the degrees of freedom $\{i\}$ in the y -direction. We have the relation [399]

$$\lambda_\chi^A(k_x) = \text{Det} [P_\chi G^A(k_x) (1 - P_\chi) G^A(k_x)] , \quad (9.21)$$

where P_χ is an operator that projects on to the subset of states $\{j\}$ in $\{i\}$ that are occupied in the configuration χ . This relation between G^A and the eigenvalues λ then leads to the same conclusion as in the case without the magnetic field: the pole structure of the entanglement spectrum encodes the topological information of the system revealed in eq.(9.20). In the case of zero magnetic field, the topological quantity revealed from the pole structure was a winding number ($\mathcal{W}$), while in the present case it is a Chern number.

Finally, we reveal the connection between the winding numbers $\mathcal{W}$ in the metallic case and the Chern numbers in the integer quantum Hall system. The degeneracy of each Landau level is equal to the number of flux quanta Φ generated by the magnetic field B . Since the total number of particles in the system is fixed, the electronic states of the filled Fermi sea that form the Luttinger volume (in the absence of the external B-field) are redistributed into the Landau levels with the appropriate degeneracy as a strong transverse magnetic field is turned on. Thus, we have the constraint that the total number of occupied states $C \times \Phi$ in the quantum hall system (C being the Chern number and equal to the number of filled Landau levels) must be equal to $\pi (N_L^x)^2$, the total number of occupied states in the metallic system. Combining eqs. 9.10 and 9.17 with this discussion then allows us to relate C to the winding number $\mathcal{W}(\phi^*)$:

$$C \times \Phi = \left\lfloor \frac{\pi}{4} \mathcal{W}(\phi^*)^2 \right\rfloor = \frac{V_L}{v_2} , \quad (9.22)$$

where $v_2 = 4\pi^2/V$ is the volume of the k -space unit cell (defined below eq. (9.10)) and $\lfloor \cdot \rfloor$ is the floor function and returns the largest integer less than or equal to the argument.

3.10 Conclusions and Outlook

In summary, we have shown that considering the renormalisation group flows of the spectral gap coupling of a system of non-interacting electrons in $2 + 1$ -dimensional theory leads to hierarchical

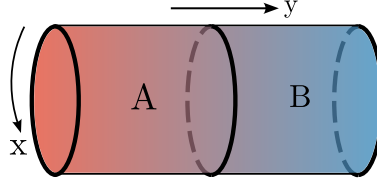


Figure 3.16: Cylindrical geometry for the integer quantum Hall system, with a magnetic field $\vec{B}$ pointing radially outwards through the surface of the cylinder. An Aharonov-Bohm vector potential acts in the x -direction (i.e., along the circumference). The entanglement is calculated between the subsystem A and its complement B .

arrangement of the entanglement with scale, and that the geometrical part of the entanglement can be described as the emergence of an additional spatial dimension, leading to holographic expressions for geometric parameters like distance and curvature. We show that changing the boundedness of the holographic space involves a topological transition with the Fermi surface of the boundary theory turning critical, and involves the generation of a geometric wormhole between the UV and IR degrees of freedom. For the massless case, the presence of an Aharonov-Bohm flux generates a geometry-independent term in the entanglement entropy which can be linked to the Luttinger volume of the gapless fermionic system. By placing the system of $2 + 1$ dimensional electrons in a strong magnetic field, we argue that the Chern number topological invariant for the ground states of the integer quantum Hall ground system are constrained by the value of the Luttinger volume in the underlying metallic system (i.e., prior to the insertion of the field).

How robust are our conclusions towards the inclusion of electronic correlations? Given the adiabatic continuity that links Fermi liquids to the non-interacting Fermi gas, i.e., the preservation of the k -space quantum numbers under adiabatic inclusion of (screened) inter-particle interactions, we expect our results to be equally valid for the many-particle entanglement of the excitations described by the fixed point effective theory for the Fermi liquid. This also suggests that our results for the wormhole geometry and the stress-energy tensor relation at the critical Fermi surface apply very generally to phase transitions that destabilise and gap out the Fermi surface. Specifically, this has profound implications for metal-insulator transitions driven by strong correlations. Non-Fermi liquids are, however, not adiabatically continuous to the Fermi gas. Thus, it should be interesting to adapt our approach to the study of interacting models of electrons that lead to gapless non-Fermi liquids (including those that are known to appear at quantum critical points related to the breakdown of a Fermi liquid). For instance, Ref. [401] reports that the second Renyi entropy of a thin k -space shell proximate to the Fermi surface is related to the quasiparticle residue; how would this change in a non-Fermi liquid? It also appears interesting to identify the effects of electronic correlations on various aspects of the emergent space. The renormalisation group techniques applied in Refs. [401–405] could also be worth investigating towards answering these questions.

It is interesting to note that our results point towards a broader similarity between the gapless fermionic systems considered here and gapped topologically ordered quantum liquids. Recall that the latter are known to possess a ground state degeneracy that is dependent on the topological nature of the interacting electronic system, leading to a subsystem entanglement entropy that has a geometry-independent subleading piece (S_{topo}) referred to as the topological entanglement entropy.

It was shown by some of us recently [375] that for a collection of N subsystems (CSS) that form a closed annular structure, the n -partite quantum entanglement information measures I^n ($3 \leq n \leq N$) are a topological invariant equal to the product of S_{topo} and the Euler characteristic of the planar manifold on which the CSS is embedded. This appears to be analogous to our finding of the hierarchical sequence of quantities $\{Q_g\}$ defined at a particular RG step (and arising out of the topological flux-dependent piece of the entanglement entropy), all of whom encode the same Luttinger volume for the gapless systems we have studied (eq. (9.12)). Further, the robustness of the n -partite information I^n against small deformations of the CSS in a topologically ordered system can be compared to the invariance of N_L^x to small deformations of the contours $C_1, C_2, \dots$ (see right panel of Fig. 3.14). It appears tempting to speculate on a possible universal origin underlying these similarities (see also [405] for a recent investigation of the entanglement properties of various quantum liquids).

Our results lead to a number of interesting questions. Firstly, the convergence parameter θ is reminiscent of the expansion rate of congruence flows on Riemannian manifolds in the context of the Raychaudhuri equation [396]. While the evolution equation for θ (eq. (7.6)) obtained by us is not of the same form (as there is no time variable, neither is there a θ^2 term in the expansion rate), we do expect such an equation to hold for an appropriately defined expansion parameter [397, 406]. This likely requires the introduction of temporal dynamics of the emergent spatial dimension, and lies well beyond the purview of the present work. Secondly, it remains to be seen how the geometry-independent part of the entanglement behaves in the presence of a mass ($m > 0$) for the Dirac fermions: a spectral gap is expected to have consequences for Luttinger's theorem [264] (see also eq. (2.19)), as well as for the many-particle entanglement. For instance, Ref. [401] reports that the second Renyi entropy shows the signatures of the condensation of Cooper pairs in the BCS problem.

Finally, we would like to point out an interesting conclusion that emerges from our analysis. As observed by us, the system considered here shows a hierarchy of multipartite entanglement measures beyond just the mutual information. Further, we have shown that the mutual information leads to the emergence of the holographic dimension. However, whether the higher-party measures also have a holographic interpretation remains an open question. We note that studies of the time-variation of tripartite information during an equilibrium process after the injection of energy suggest that it remains negative throughout the process [407–409], indicating that care must be taken in defining geometric measures using it. Nevertheless, our study raises the possibility of obtaining a generalisation of the gauge-gravity duality from the perspective of many-particle entanglement. We speculate that the higher order information measures will lead to a modification of the universal entanglement Hamiltonian obtained at criticality. Further, that these modifications can be visualised as a *hypergraph*: just as a graph contains edges that connect any two nodes, a hypergraph contains *hyperedges* that can connect more than two nodes. The eigenvalues $\epsilon^{(n)}$ (with $n > 2$) of the entanglement Hamiltonian would act as hyperedges connecting the degrees of freedom $\{i, j, \dots\}$. We leave an investigation of these aspects to future works.

Appendix A: Evaluation of multi-partite information

We define the multipartite information among g sets $\{\mathcal{A}_z(i)\}$ as [375]

$$I_{\{\mathcal{A}_z(i)\}}^g = \sum_{r=1}^g (-1)^{r-1} \sum_{\beta \in \mathcal{P}(\{\mathcal{A}_z(i)\})}^{| \beta | = r} S_{\beta}, \quad (10.1)$$

where $\mathcal{P}(\{\mathcal{A}_z(i)\})$ is the power set of $\{\mathcal{A}_z(i)\}$, and $| \beta | = r$ implies that we sum only over those sets β that have r number of elements in them. We will first evaluate the internal sum σ_r over the set β for a general r :

$$\sigma_r = \sum_{\beta \in \mathcal{P}(\{\mathcal{A}_z(i)\})}^{| \beta | = r} S_{\beta}. \quad (10.2)$$

For this, we first note that following eq. (3.15), the entanglement entropy (EE) of the union of a given sequence of sets $\{i\}$ is given by the EE of the set with the highest (lowest) index for a positive (negative) z . As a result, the EE S_{β} for the set β is simply $S_{\beta} = \theta(z)S_{\min(\beta)} + \theta(-z)S_{\max(\beta)}$. If $n_j^{(r,+)} (n_j^{(r,-)})$ is the number of sets of cardinality r that have the element $\mathcal{A}_{j,z}$ as the element with the smallest (largest) index j , the internal sum can be written as

$$\sigma_r = \sum_{j=1}^{g-(r-1)} \theta(z) n_j^{(r,+)} S_{j,z} + \sum_{j=r}^g \theta(-z) n_j^{(r,-)} S_{j,z}. \quad (10.3)$$

The number $n_j^{(r,+)}$ corresponds to the number of ways one can pick $r-1$ elements out of a set of $g-j$ elements: $n_j^{(r,+)} = \binom{g-j}{r-1}$. Similarly, $n_j^{(r,-)}$ corresponds to the number of ways one can pick $r-1$ elements out of a set of j elements: $n_j^{(r,-)} = \binom{j}{r-1}$. We can also extend the limits on the two summations by defining $\binom{n}{m} = 0$ for $n < m$. With these considerations, eq. (10.3) takes the form

$$\sigma_r = \sum_{j=1}^g \left[\theta(z) \binom{g-j}{r-1} + \theta(-z) \binom{j}{r-1} \right] S_{j,z}. \quad (10.4)$$

This can now be substituted into the full expression for the multipartite equation, giving

$$\begin{aligned} I_{\{\mathcal{A}_z(i)\}}^g &= \sum_{r=1}^g (-1)^{r-1} \sum_{j=1}^g \left[\theta(z) \binom{g-j}{r-1} + \theta(-z) \binom{j}{r-1} \right] S_{j,z} \\ &= \sum_{j=1}^g S_{j,z} \left[\theta(z) \sum_{r=0}^{g-j} (-1)^r \binom{g-j}{r} + \theta(-z) \sum_{r=0}^j (-1)^r \binom{j}{r} \right]. \end{aligned} \quad (10.5)$$

From the binomial theorem, we can now recognise each internal sum as over r as a polynomial expansion. That is, upon comparing with $(x-1)^n = \sum_{m=0}^n \binom{n}{m} (-1)^m x^{n-m}$, we obtain $\sum_{r=0}^{g-j} (-1)^r \binom{g-j}{r} =$

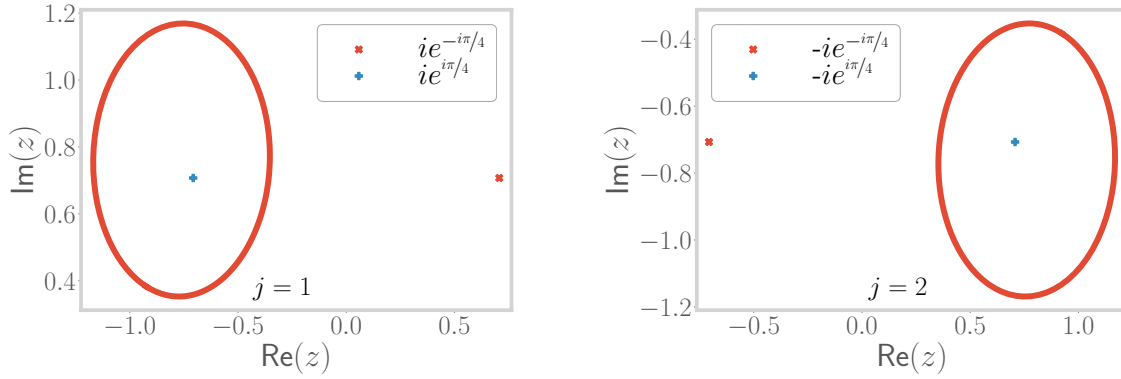


Figure 3.17: Contours $z(C_j)$ for two values of j employed for evaluating the integral eq.(10.8), and the locations of the poles $z_{\pm}^{\pm}$. For each j , only one pole (shown in blue) contributes, as the other (shown in red) is not enclosed by the contour.

$\lim_{x \rightarrow 1} (x-1)^{g-j} = \delta_{g,j}$. Similarly, the other sum evaluates to $\delta_{j,0}$. Substituting these into the expression for $I_{\{\mathcal{A}_z(i)\}}^g$ gives

$$\begin{aligned} I_{\{\mathcal{A}_z(i)\}}^g &= \sum_{j=1}^g S_{j,z} [\theta(z)\delta_{g,j} + \theta(-z)\delta_{j,0}] \\ &= \theta(z)S_g + \theta(-z)S_0. \end{aligned} \quad (10.6)$$

By noting that S_g and S_0 are, respectively, the EE of the intersection set of $\{S_z(i)\}$, one can combine the two terms in the previous equation into

$$I_{\{\mathcal{A}_z(i)\}}^g = \theta(z)S_g + \theta(-z)S_0 = S_{\cap\{\mathcal{A}_z(i)\}}. \quad (10.7)$$

Appendix B: Evaluation of the contour integral that counts poles

We prove eq. (9.3) in this appendix. The integral in question is given by

$$\mathcal{I} = \oint_{C_j} \frac{d\phi}{Q_0(\phi)} = \oint_{C_j} \frac{d\phi}{\frac{1}{\sqrt{2}} - |\sin \pi \phi|}, \quad (10.8)$$

where C_j is the contour shown in Fig. 3.13 and the variable ϕ is given by $\phi = \{\phi_j + \frac{1}{8}e^{i\theta} : \theta \in [0, 2\pi]\}$, where $\phi_j = j - \frac{1}{4}$ is the center of the contour. In order to evaluate the integral, we make a variable change: $z = e^{i\pi\phi}$, giving $dz = i\pi z d\phi$ and $\sin \pi\phi = \frac{1}{2i} \left(z - \frac{1}{z}\right)$. Depending on the value of j , $\sin \pi\phi$ can be either positive or negative. We denote either case by a subscript $\mathcal{I}_{\pm}$ corresponding to the contour $C_{\pm}$, giving $|\sin(\pi\phi)| = \pm \sin(\pi\phi)$. Casting the integral in terms of z , we obtain

$$\mathcal{I}_{\pm} = \frac{1}{i\pi} \oint_{z(C_{\pm})} \frac{dz/z}{\frac{1}{\sqrt{2}} \mp \frac{1}{2i} \left(z - \frac{1}{z}\right)} = \oint_{z(C_{\pm})} \frac{\mp \frac{2}{\pi} dz}{\left(z \mp \frac{i}{\sqrt{2}}\right)^2 - \frac{1}{2}}. \quad (10.9)$$

The integrand in $\mathcal{I}_+$ has two simple poles at $z_{\pm}^{\pm} = \pm ie^{\mp i\pi/4}$. Similarly, the integrand in $\mathcal{I}_-$ has two simple poles at $z_{\pm}^{\pm} = \mp z_{\pm}^{\pm}$. We plot the contour and the two poles for two contours of opposite signature in Fig. 3.17; it is clear that only the pole at $z_{\pm}^+$ contributes to the integral. Using the residue theorem, the integral evaluates to

$$\begin{aligned} \mathcal{I}_{\pm} &= \oint_{z(C_{\pm})} \frac{\mp \frac{2}{\pi} dz}{(z - z_{\pm}^+) (z - z_{\pm}^-)} \\ &= \mp \frac{2}{\pi} 2\pi i \lim_{z \rightarrow z_{\pm}^+} \frac{1}{z - z_{\pm}^-} \\ &= \mp 2\sqrt{2}i . \end{aligned} \tag{10.10}$$

This proves eq. (9.3).

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